Electrostatics

Electro ≡ related to charge Statics ≡ at rest

Electrostatics is branch of Physics which deals with properties of charge at rest.

→ Charge: It is property of a body or particle due to which it can create, electric and magnetic effect in the region surrounding it and it can experience electric and magnetic forces.

(Q)

Unit: SI = Coulombs
CGS = esu {electrostatic unit}

Dimension = [IT] = [AT]

3×10^9 esu = 1 C

Properties of Charge:

→ It is scalar quantity.

→ It is of two types ⇒ ① Positive → ex a) Proton
Charge on proton = $+1.6 \times 10^{-19}$ C

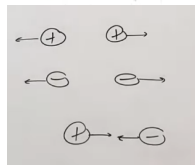
b) α particle (He^{+2})
Charge on α-particle = $2 \times q_p$
$= 3.2 \times 10^{-19}$ C

② Negative → ex Electron
Charge on electron = -1.6×10^{-19} C
Mass of electron = 9.1×10^{-31} kg

→ If a body is -ve charged then it has abundance/excess of $e^{\ominus}$.

→ If a body is +ve charged then it has deficiency of $e^{\ominus}$

→ Charge is quantized. $\boxed{Q = ne}$ $n \in Z$



→ Like point charges will repel.

→ Unlike point charges will attract.

→ Charge does not depend on its velocity and independent of reference frame

→ Charge of an isolated system is always conserved.

★ Net charge can neither be created nor destroyed.

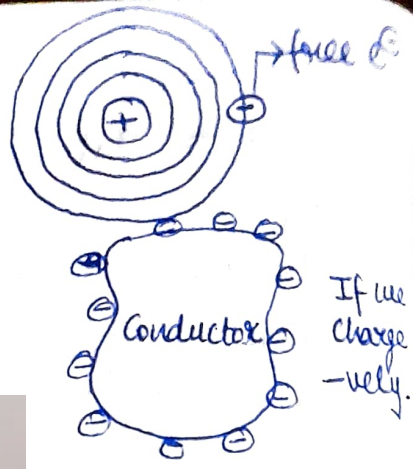
→ Charge at rest can create electric effects only and can experience electric forces only.

→ Charge moving with constant velocity will create electric as well as magnetic effects and it can experience electric as well as magnetic forces.

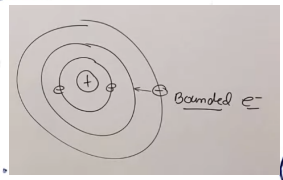
→ Charge moving with variable velocity will emit electromagnetic radiation. (i.e acceleration)

Conductors:

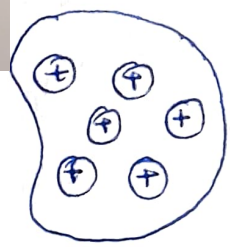
→ They have very large no. of free $e^{\ominus}$.
→ These free $e^{\ominus}$ are responsible for transfer of charge and energy from one point to another.
→ Generally, metals are conductors.
→ In a conductor charge resides on surface.
(This is because in conductor charges repel each other and try to stay as far as they can).



Non-Conductors/Insulators:

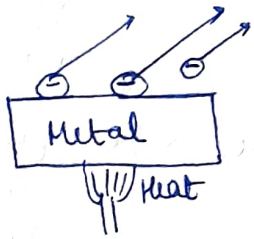


→ Negligible free $e^{\ominus}$.
→ Poor Conductor of heat and electricity.
→ It's outermost $e^{\ominus}$ is tightly bounded
→ High Resistance.
→ Charge can stay inside the bulk of insulator



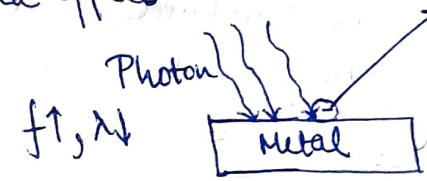
Charging:

① Thermionic effect:



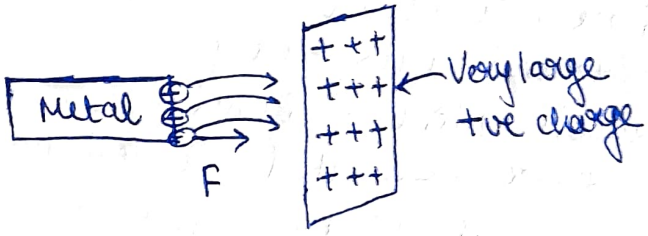
When we heat, KE of free $e^{\ominus}$ ↑ and they escape the metal surface and hence the metal becomes +ve charged.

② Photo-electric effect:



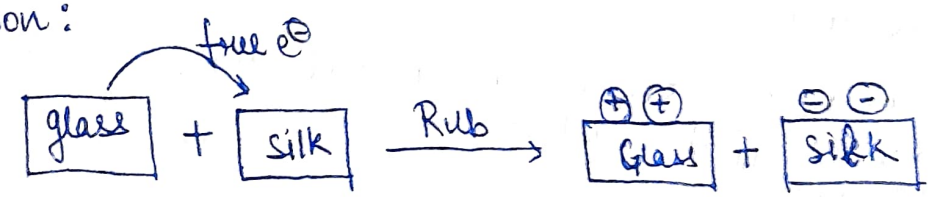
Metal becomes +ve charged.

③ Field emission:

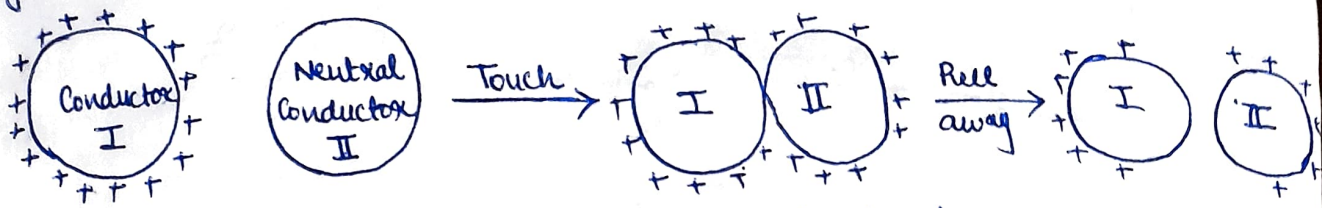


Metal becomes +vely charged.

④ Friction:



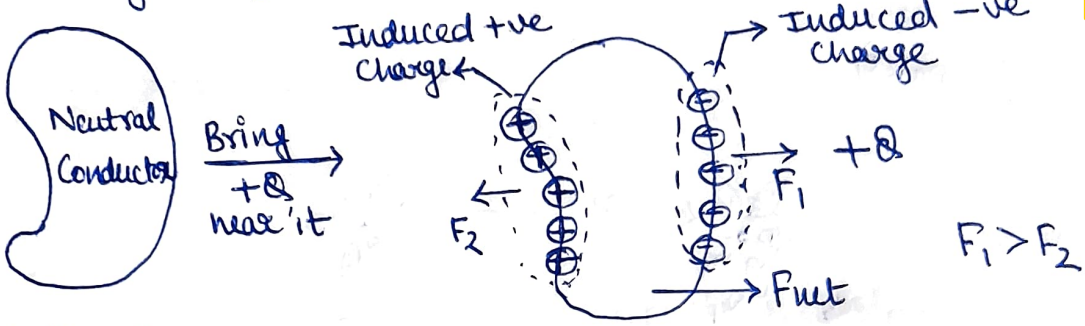
⑤ By Conduction:



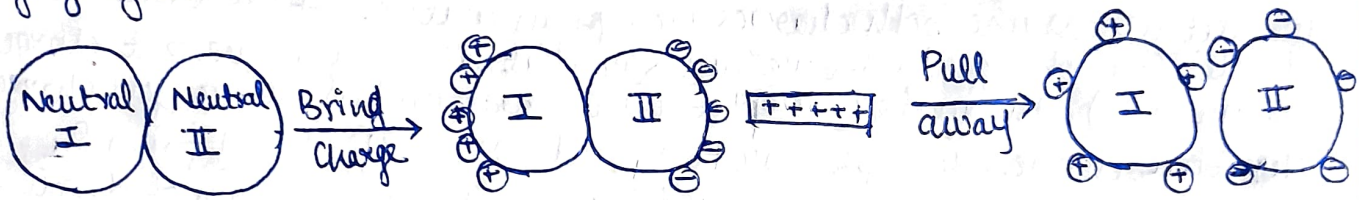
* If both conductor are identical, they will share charge equally.

Charge gets shared as now charge got more surface to reside

Induction: In a conductor charge distribution may change due to presence of nearby charges, this phenomenon is know as induction



Charging By Induction:



Coulomb's Law:

→ It is only for point charges, we will integrate for larger charge distributions

→ It is experimental law

$F \propto Q_1 Q_2$

$F \propto \frac{1}{r^2}$ (inverse square law)

$$F = \frac{K Q_1 Q_2}{r^2}$$

= Coulomb's const.

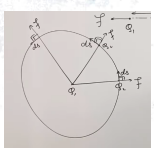
K → electrostatic const.

$K = 9 \times 10^9 \ Nm^2/C^2$

(in vacuum/air/free space)

→ Force is always along the line joining the charges. ie. F_e is a central force.

⇒ F_e is a conservative force ⇒ Potential energy is defined for electrostatic forces.



Air/vacuum

$$F = \frac{kQ_1Q_2}{r^2}$$

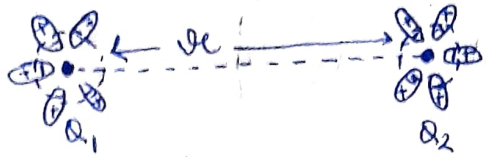
$$k = \frac{1}{4\pi\varepsilon_0}$$

$\varepsilon_0 \rightarrow$ Permitivity of free space

$$\varepsilon_0 = \frac{1}{4\pi k} = \frac{1}{4\times\pi\times 9\times 10^9} = 8.85\times10^{-12}\ C^2/N\text{-}m^2$$

Dimension of $\varepsilon_0 \Rightarrow \frac{C^2}{N\text{-}m^2} \Rightarrow \frac{I^2T^2}{MLT^{-2}L^2} = [M^{-1}L^{-3}T^4I^2]$

In any other medium (Ex-water)



Here let us assume both charges are kept in water. If we take one water droplet for observation, we can say that as soon as we put charge Q_1 (assume +ve), the water droplet which was neutral before gets charged. Negative charge develops near to Q_1 and +ve charge away from Q_1. This is known as polarisation and thus results in formation of dipole. Now many such dipoles surrounds Q_1 and Q_2 and hence reduces the effect of the charges alone. Similarly for other mediums as well.

and thus, the force reduces

$$F = \frac{k'Q_1Q_2}{r^2} = \frac{Q_1Q_2}{4\pi\varepsilon r^2}$$

$$k' = \frac{1}{4\pi\varepsilon}$$

$\varepsilon \rightarrow$ Permitivity of medium

$\varepsilon > \varepsilon_0$ unit of $\varepsilon = C^2/Nm^2$

So, $\varepsilon\uparrow, k'\downarrow, F\downarrow$

ε is property of medium

* It is more for the medium which can be easily polarised

Relative Permitivity:

$$\varepsilon_r = \frac{\varepsilon}{\varepsilon_0} = \frac{\text{Permitivity of medium}}{\text{Permitivity of vacuum}}$$

→ Unitless → Dimensionless

→ $\varepsilon_r \geq 1$ → For water: $\varepsilon_r = 81$

For metals: $\varepsilon_r \rightarrow \infty$

Force in any other medium:

$$F = \frac{k'Q_1Q_2}{r^2} = \frac{Q_1Q_2}{4\pi\varepsilon r^2} = \frac{Q_1Q_2}{4\pi\varepsilon_0\varepsilon_r r^2}$$

$$\Rightarrow k' = \frac{1}{4\pi\varepsilon} = \frac{1}{4\pi\varepsilon_0\varepsilon_r} \quad \{\varepsilon_0\varepsilon_r = \varepsilon\}$$

$$k' = \frac{k}{\varepsilon_r} = \frac{9\times10^9}{\varepsilon_r}$$

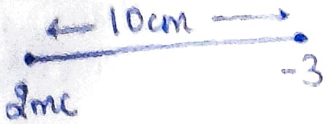
ε_r is also called dielectric constant

Also, $\frac{F_0}{F_m} = \frac{\varepsilon}{\varepsilon_0} = \varepsilon_r$

$F_0 \rightarrow$ Force in vacuum

$F_m \rightarrow$ Force in medium

Q Find electrostatic force b/w them.



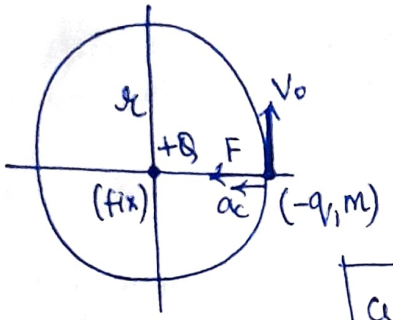
$$F = \frac{KQ_1Q_2}{r^2} = \frac{9\times10^9 \times 2\times10^{-3} \times 3\times10^{-6}}{(0.1)^2} = 5400\ N$$

Q Find min^m electrostatic force b/w two charge particles placed at a separation 5cm in air.

For min^m force $\Rightarrow$ Q must be min^m $\Rightarrow$ $\boxed{Q = 1.6\times10^{-19}C}$

$$F = \frac{KQ_1Q_2}{r^2} = \frac{9\times10^9 \times (1.6\times10^{-19})^2}{(0.05)^2} = \frac{9\times10^9 \times 2.56\times10^{-38}}{25\times10^{-4}} = \frac{9\times2.56}{25}\times10^{-25}N$$

Q Find V_0 for $-q$ to perform circular motion around $+Q$.

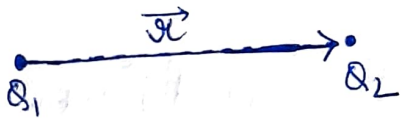


$$F = \frac{KQq}{r^2} \qquad \text{Also, } F_c = \frac{mv_0^2}{r}$$

$$\Rightarrow \frac{KQq}{r^2} = \frac{mv_0^2}{r} \Rightarrow \boxed{V_0 = \sqrt{\frac{KQq}{mr}}}$$

$$\boxed{\omega = \frac{V_0}{r} = \sqrt{\frac{KQq}{mr^3}}} \qquad \boxed{T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{mr^3}{KQq}}} \qquad \boxed{f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{KQq}{mr^3}}}$$

Coulomb's law in vector form:



$$\boxed{\vec{F} = \left(\frac{KQ_1Q_2}{r^2}\right)\hat{r}} \qquad \boxed{F = \frac{KQ_1Q_2}{r^3}\vec{r}} \qquad \left\{\hat{r} = \frac{\vec{r}}{r}\right\}$$

Here use Q_1, Q_2 with sign.

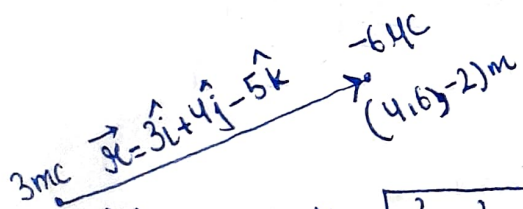
If Q_1, Q_2 are of same sign $\Rightarrow \vec{F} \parallel \vec{r}$

If Q_1, Q_2 are of opposite sign $\Rightarrow \vec{F} \uparrow\downarrow \vec{r}$

$\vec{r} \Rightarrow$ Tail: At the charge due to which $\vec{F}$ is to be found

Head: At the charge on which $\vec{F}$ is acting.

Q Find electrostatic force in vector form on $-6\mu C$.

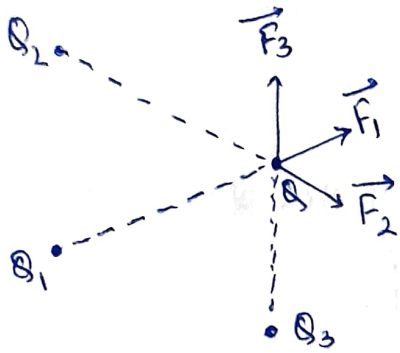


$$\vec{F} = \frac{9\times10^9 \times 3\times10^{-3} \times (-6\times10^{-6})(3\hat{i}+4\hat{j}-5\hat{k})}{(5\sqrt{2})^3}$$

$$|\vec{r}| = \sqrt{3^2+4^2+5^2} = \sqrt{50}$$

$$|\vec{r}| = 5\sqrt{2}\ m$$

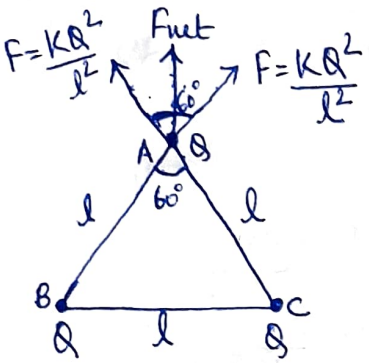
Principle of Superposition:



$$\vec{F_{net}} = \vec{F_1} + \vec{F_2} + \vec{F_3}$$

force acting on Q due to Q1 is independent of the presence or absence of Q2, Q3 etc.

Q

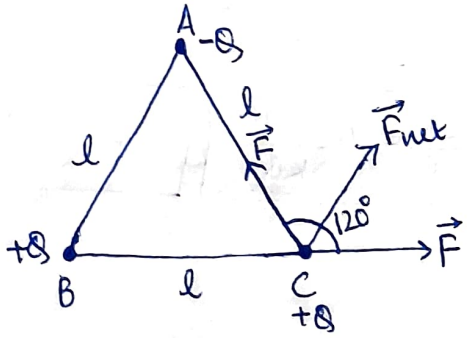


Find $\vec{F}_{elec}$ on A.

$$\vec{F_{net}} = \sqrt{F^2 + F^2 + 2F^2\cos 60°} = \sqrt{3F^2}$$

$$\vec{F_{net}} = F\sqrt{3} = \frac{\sqrt{3}KQ^2}{l^2}$$

Q

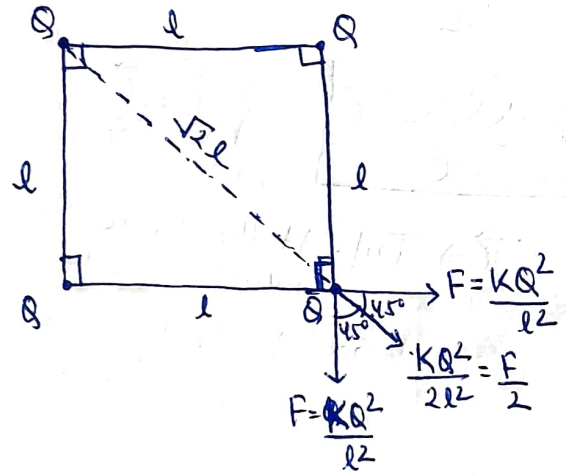


Find $\vec{F}_{elec}$ on C

$$\vec{F} = \frac{KQ^2}{l^2}$$

$$\vec{F}_{net} = \sqrt{F^2 + F^2 + 2F^2\cos 120°} = \sqrt{F^2} = F$$

Q



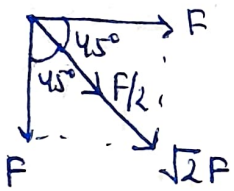
Find $\vec{F}_{electric}$ on any one of the charge.

$$\vec{F_{net}} = \sqrt{2}F + \frac{F}{2}$$

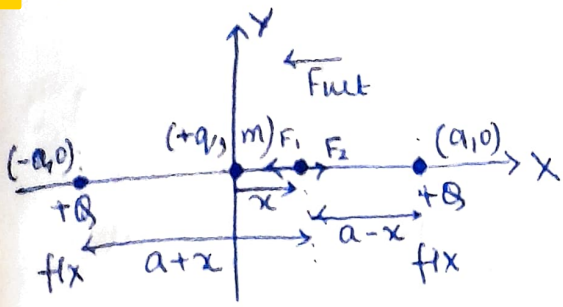
$$= F\left(\sqrt{2} + \frac{1}{2}\right)$$

$$= F\left(\frac{2\sqrt{2}+1}{2}\right)$$

$$\vec{F_{net}} = \frac{KQ^2}{l^2}\left(\frac{2\sqrt{2}+1}{2}\right)$$



Q Prove that if $+q$ is displaced slightly along x-axis, it will perform SHM. Also find Time period of SHM. $(x \ll a)$



$$F_{net} = F_1 - F_2 = \frac{KQq}{(a-x)^2} - \frac{KQq}{(a+x)^2}$$

$$= KQq\left\{\frac{1}{(a-x)^2} - \frac{1}{(a+x)^2}\right\} = \frac{KQq\,4ax}{(a^2-x^2)^2}$$

As $x \ll\ll a \Rightarrow (a^2-x^2)^2 \approx a^4$

$$\frac{KQq\,4ax}{a^4} = \frac{KQq\,4x}{a^3}$$

$$\therefore \boxed{F_{net} = \frac{4KQq}{a^3}x} \quad F \propto x \Rightarrow \text{SHM proved}$$

Now, $F = m\frac{dv}{dt}$

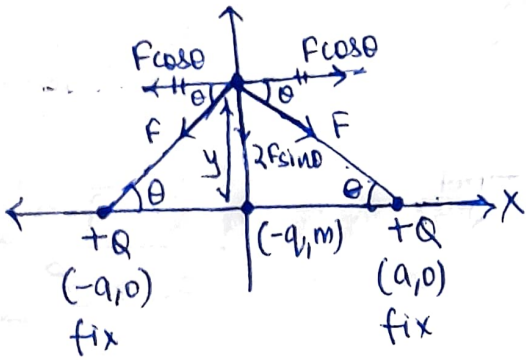
$$\frac{4KQqx}{a^3} = m\frac{dv}{dt} \Rightarrow \boxed{\frac{dv}{dt} = \frac{4KQqx}{ma^3}}$$

$$= \omega^2 x$$

$$\boxed{\omega = \sqrt{\frac{4KQq}{ma^3}}}$$

$$\boxed{T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{ma^3}{4KQq}}}$$

Q Prove that if $-q$ is displaced slightly along y-axis, it will perform SHM. Also find time period of SHM $(y \ll a)$



$$F_{net} = 2F\sin\theta = 2 \times \frac{KQq}{(\sqrt{a^2+y^2})^2} \times \frac{y}{\sqrt{a^2+y^2}}$$

$$= \frac{2KQqy}{(a^2+y^2)^{3/2}} = \frac{2KQqy}{a^3} \quad (y \ll a)$$

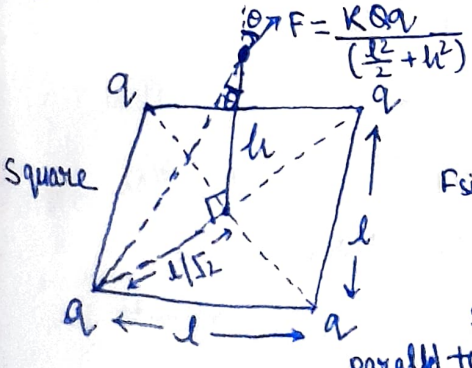
$$\therefore \boxed{F = \frac{2KQq}{a^3}y} \Rightarrow \text{SHM proved}$$

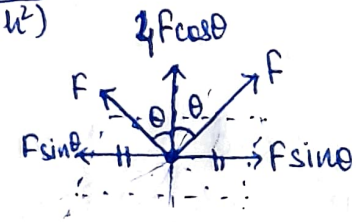
$$\boxed{\frac{dv}{dt} = \frac{2KQq}{ma^3}y}$$

$$\boxed{\omega = \sqrt{\frac{2KQq}{ma^3}}}$$

$$\boxed{T = 2\pi\sqrt{\frac{ma^3}{2KQq}}}$$

Q Find $\vec{F}$ electric on Q.





Similarly, all components parallel to plane will cancel out and ⊥ to plane will add up

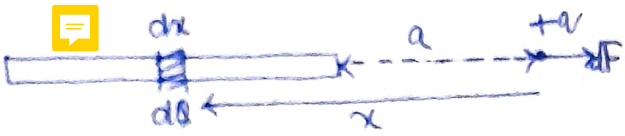
$= 4F\cos\theta$

$$F_{net} = 4F\cos\theta$$

$$= 4 \times \frac{KQq}{\left(\frac{l^2}{2}+h^2\right)} \times \frac{h}{\sqrt{\frac{l^2}{2}+h^2}}$$

$$\boxed{F_{net} = \frac{4KQqh}{\left(\frac{l^2}{2}+h^2\right)^{3/2}}}$$

Q Find electrostatic force between point charge $+q$ and uniform rod $(+Q, l)$



$$dF = \frac{K(dQ)q}{x^2}$$

$\lambda = \frac{Q}{l}$ $\quad dQ = \lambda dx$

$$\boxed{dQ = \frac{Q}{l}dx}$$

$$\Rightarrow dF = \frac{KQq\,dx}{l x^2}$$

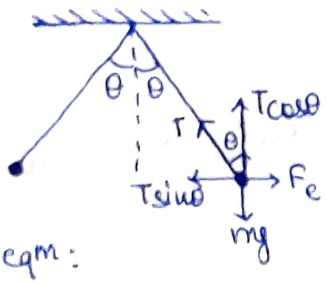
$$F = \int dF = \frac{KQq}{l}\int_a^{a+l}\frac{dx}{x^2} = \frac{KQq}{l}\left(\frac{-1}{x}\right)_a^{a+l}$$

$$= \frac{KQq}{l}\left(\frac{1}{a} - \frac{1}{a+l}\right)$$

$$\therefore \boxed{F = \frac{KQq}{a(a+l)}}$$

Q Given point charges of mass density $= \rho$. If this entire arrangement is submerged in a liquid of density σ. Equilibrium position remains unchanged. Find relative permittivity of liquid (ε_x).

Air



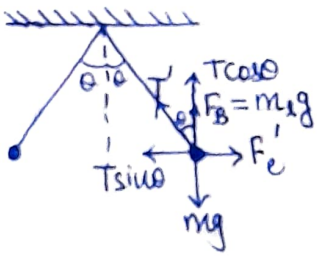
For eqm:

$T\cos\theta = mg$

$T\sin\theta = F_e$

$$\therefore \boxed{\tan\theta = F_e/mg} \quad -(1)$$

Liquid



$$F_B = m_l g = \sigma V g = \frac{\sigma mg}{\rho}$$

$$F_e' = \frac{F_e}{\varepsilon_x}$$

For eqm:

$T'\cos\theta = mg - mg\frac{\sigma}{\rho} = mg\left(1-\frac{\sigma}{\rho}\right)$

$T'\sin\theta = F_e/\varepsilon_x$

$$\therefore \boxed{\tan\theta = \frac{F_e}{\varepsilon_x mg\left(1-\frac{\sigma}{\rho}\right)}} \quad -(2)$$

Now as eqm position remains same in both mediums

$(1) = (2)$

$$\frac{\cancel{F_e}}{\cancel{mg}} = \frac{\cancel{F_e}}{\varepsilon_x mg\left(1-\frac{\sigma}{\rho}\right)}$$

$$\varepsilon_x\left(1-\frac{\sigma}{\rho}\right) = 1$$

$$\boxed{\varepsilon_x = \frac{\rho}{\rho-\sigma}}$$

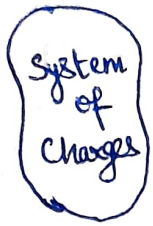
Electric Field:

→ It is a region or space around a charged particle or system, in which they have their electric effects.

Electric Field intensity: It is a physical quantity which tells about electric effects quantitatively.

It is popularly known as 'Electric Field'.

$q_0 \rightarrow$ test charge.



$$\vec{E} = \frac{\vec{F_e}}{q_0}$$

* Electric field at a point P due to system of charges is defined as electrostatic force experienced by test charge per unit test charge.

OR

Electric field at point P due to system of charges is defined as electrostatic force experienced by unit positive test charge (+1C).

→ Electric field is a vector quantity.

→ Unit: N/C = Volt/meter

→ Dimension: $\frac{MLT^{-2}}{IT} = [MLT^{-3}I^{-1}]$

→ If q_0 is placed at a point having $\vec{E} \Rightarrow q_0$ charge will experience a force $\boxed{\vec{F} = q_0\vec{E}}$

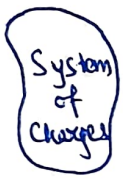
→ If $q_0 = +ve \Rightarrow \vec{F_e} \parallel \vec{E}$

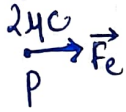
→ If $q_0 = -ve \Rightarrow \vec{F_e} \uparrow\downarrow \vec{E}$

→ Electric Field follows principle of superposition $\Rightarrow \boxed{\vec{E}_{net} = \vec{E_1} + \vec{E_2} + \vec{E_3} + \ldots + \vec{E_n}}$

→ We can change electric field by changing charge distribution and by changing medium.

Q① A 2μC charge placed at point P experiences a force $\vec{F_e} = (2\hat{i} - 4\hat{j} + 10\hat{k})$ mN. Find $\vec{E}$ at point P due to system of charges.



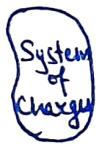


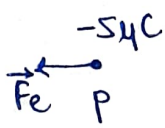
$$\vec{E} = \frac{\vec{F_e}}{q_0} = \frac{(2\hat{i} - 4\hat{j} + 10\hat{k}) \times 10^{-3}}{2 \times 10^{-6}} = (\hat{i} - 2\hat{j} + 5\hat{k}) \times 10^3$$

$$\boxed{\vec{E} = (\hat{i} - 2\hat{j} + 5\hat{k}) \times 10^3 \text{ N/C}}$$

* Now even if we remove 2μC, $\vec{E}$ at point P will remain same as charge distribution and medium is same.

② If at point P, −5μC is placed. Find $\vec{F_e}$ on it.

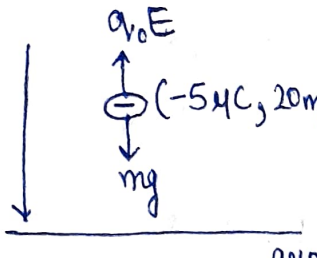




$$\vec{F_e} = q_0\vec{E} = -5 \times 10^{-6} \times (\hat{i} - 2\hat{j} + 5\hat{k}) \times 10^3$$

$$\boxed{\vec{F_e} = (-5\hat{i} + 10\hat{j} - 25\hat{k}) \times 10^{-3} \text{ N}}$$

Q Find direction and magnitude of electric field in the region.

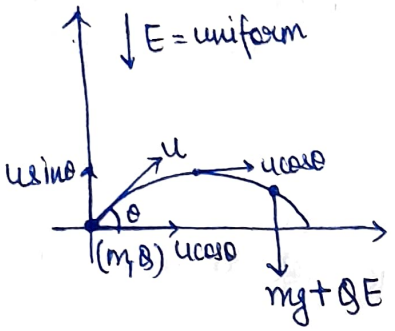


$q_0E = mg \Rightarrow E = \frac{mg}{q_0}$

$E = \frac{20 \times 10^{-6}\,kg \times 10\,m/s^2}{5 \times 10^{-6}\,C} = 40\ \frac{kg\,m/s^2}{C}$

$\boxed{E = 40\,N/C}$

Q



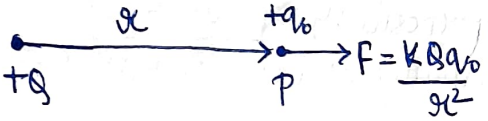
Find ① Time of flight ② Maxm Height ③ Speed at highest point ④ Range.

$mg + qE = mg_{eff} \Rightarrow \boxed{g_{eff} = g + \frac{qE}{m}}$

① $\boxed{T = \frac{2u\sin\theta}{g + \frac{qE}{m}}}$ ② $\boxed{H = \frac{u^2\sin^2\theta}{2\left(g + \frac{qE}{m}\right)}}$ ④ $\boxed{R = \frac{u^2\sin 2\theta}{g + \frac{qE}{m}}}$

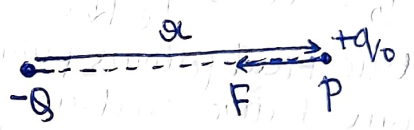
③ $\boxed{u\cos\theta}$

Electric Field due to point charge:



$\boxed{E = \frac{F}{q_0} = \frac{KQ}{r^2}}$

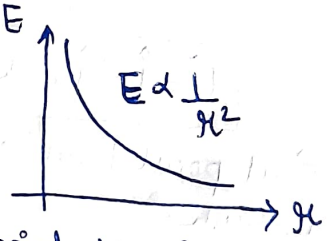
→ $\vec{E}$ due to +Q is radially outwards.



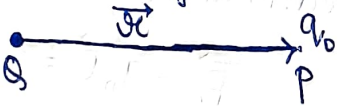
$\boxed{E = \frac{F}{q_0} = \frac{KQ}{r^2}}$

→ $\vec{E}$ due to -Q is radially inwards.

Graph of $\vec{E}$ v/s r:



Electric Field due to point charge in vector form:



$\boxed{\vec{F} = \frac{KQq_0}{r^3}\vec{r}}$

$\boxed{\vec{E} = \frac{\vec{F}}{q_0} = \frac{KQ}{r^3}\vec{r} = \frac{KQ}{r^2}\hat{r}}$

(Use Q with sign)

$\vec{r}$ ⇒ Tail: At the charge due to which $\vec{E}$ is to be found

Head: At the point in which $\vec{E}$ is to be found.

Q Find $\vec{E}$ at point P due to 2mC.

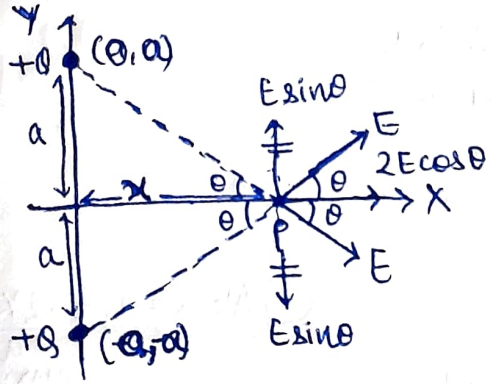
2mC $\vec{r} = 6\hat{i}+8\hat{j}-10\hat{k}$ → P

$(2,-1,4)$m $\qquad$ $(8,7,-6)$m

$|\vec{r}| = \sqrt{36+64+100} = \sqrt{200} = 10\sqrt{2}$ m

$\vec{E} = \frac{KQ\vec{r}}{r^3} = \frac{9\times10^9 \times 2\times10^{-3}}{(10\sqrt{2})^3}(6\hat{i}+8\hat{j}-10\hat{k})$

Q ① Find $\vec{E}$ at P.



$E = \frac{KQ}{a^2+x^2}$

$E_{net} = 2E\cos\theta = \frac{2KQ}{a^2+x^2}\left(\frac{x}{\sqrt{a^2+x^2}}\right)$

$$\boxed{E_{net} = \frac{2KQx}{(a^2+x^2)^{3/2}}}$$

② Draw $\vec{E}$ v/s x curve.

$$\boxed{E = \frac{2KQx}{(a^2+x^2)^{3/2}}}$$

(I) At $x=0$, $E=0$

(II) At $x\to\infty$, $E\to 0$ $\left(E_{far} = \frac{2KQx}{x^{3/2}} = \frac{2KQ}{x^2}\right)$ $x\to\infty$, $E\to 0$

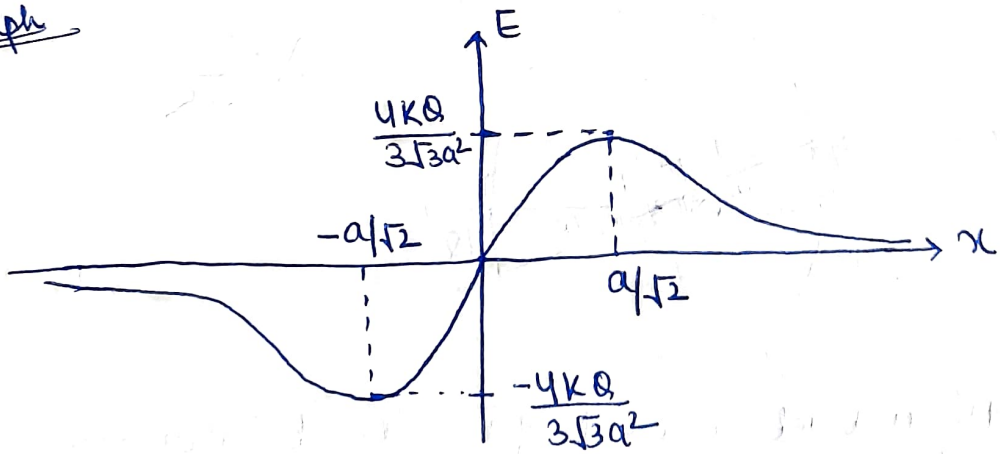
② (III) Maxima: $\frac{dE}{dx}=0 \Rightarrow 2KQ\left(\frac{(a^2+x^2)^{3/2} - x\left(\frac{3}{2}\right)(a^2+x^2)^{1/2}\cdot 2x}{(a^2+x^2)^3}\right) = 0$

$\Rightarrow (a^2+x^2)^{3/2} = 3x^2(a^2+x^2)^{1/2}$

$\Rightarrow a^2+x^2 = 3x^2 \Rightarrow a^2 = 2x^2 \Rightarrow$ $\boxed{x = \pm\frac{a}{\sqrt{2}}}$

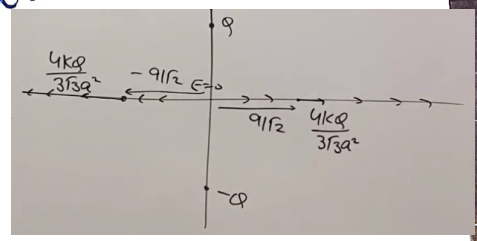
$E_{max} = \frac{2KQ\left(\frac{a}{\sqrt{2}}\right)}{\left(\frac{3a^2}{2}\right)^{3/2}} = \frac{\sqrt{2}KQa}{\frac{3\sqrt{3}}{2\sqrt{2}}a^3} = \frac{4KQ}{3\sqrt{3}a^2}$

Graph

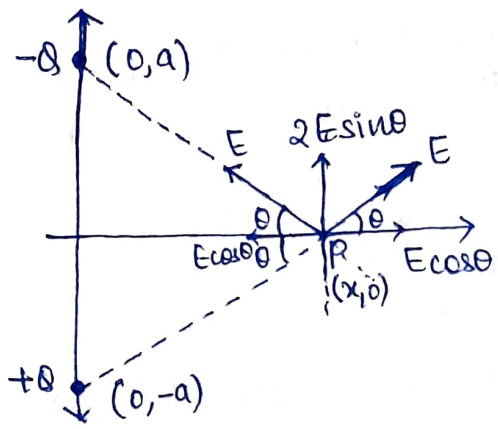


Graph is mirror image in I st and III rd quadrant.

As $x\to -x$, $f(-x) = -f(x) \Rightarrow$ odd fn



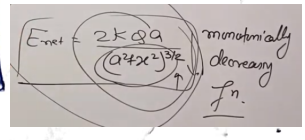
Q① Find electric field at point P.



$$E = \frac{KQ}{(a^2+x^2)}$$

$$E_{net} = 2E\sin\theta = \frac{2KQ}{(a^2+x^2)}\left(\frac{a}{\sqrt{a^2+x^2}}\right)$$

$$\boxed{E_{net} = \frac{2KQa}{(a^2+x^2)^{3/2}}}$$



② Draw $\vec{E}$ v/s x curve.

$$\boxed{E = \frac{2KQa}{(a^2+x^2)^{3/2}}}$$

① $x \to -x$

$f(-x) = f(x)$

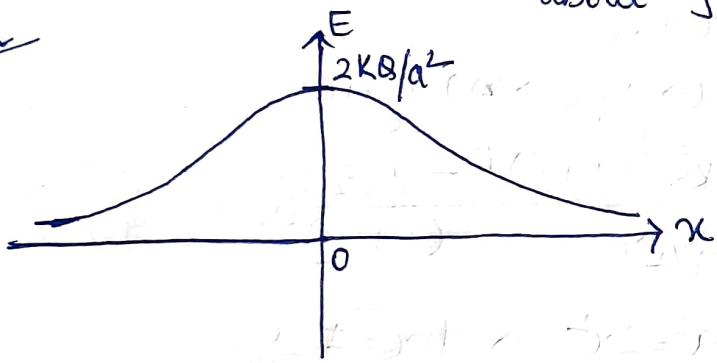
$\Rightarrow$ even fn $\Rightarrow$ Graph symmetric about y-axis

② At $x=0$, $E = \frac{2KQ}{a^2}$ (max value)

③ At $x \to \infty$, $E \to 0$

④ As $x \uparrow$, $E \downarrow$.

Graph

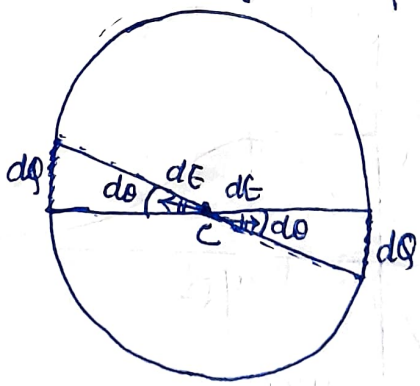


Electric Field due to uniform ring:

① At center:

Q charge is uniformly distributed.

$$\lambda = \frac{Q}{2\pi R}$$

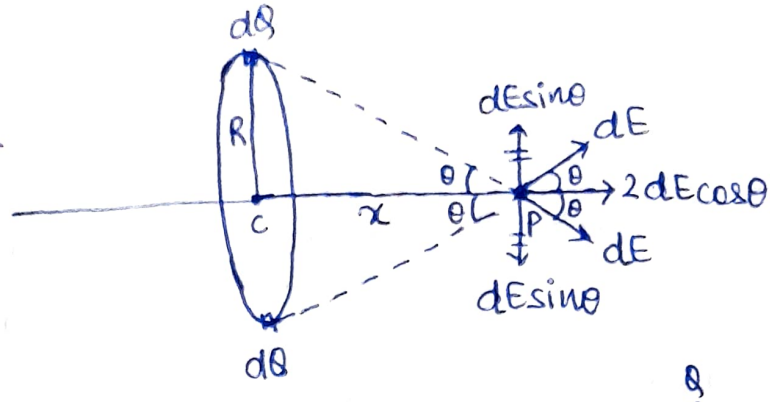


Similarly all $d\vec{E}$ will get cancelled by its diametrically opposite point and $\vec{E}_{net}$ at center will be zero.

* $\vec{E}$ at center of uniformly charged ring is zero.

② At axis of ring:

$\lambda = \frac{Q}{2\pi R}$



$dE = \frac{K dQ}{(\sqrt{R^2+x^2})^2}$

Here we can integrate by two ways, either $\int_0^Q dE\cos\theta$ or $\int_0^{Q/2} 2dE\cos\theta$

$$E_{net} = \int_0^Q dE\cos\theta = \int_0^Q \frac{K dQ}{(\sqrt{R^2+x^2})^2}\left(\frac{x}{\sqrt{R^2+x^2}}\right) = \frac{Kx}{(R^2+x^2)^{3/2}}\int_0^Q dQ$$

$$\star \boxed{E = \frac{KQx}{(R^2+x^2)^{3/2}}}$$

Now, ① At $x=0$, $E=0=E_{center}$

② E at very far $\Rightarrow x >>> R$

$E_{far} = \frac{KQ}{x^2}$ **similar to a point charge**

③ If $x \to \infty$, $E \to 0$

④ Maxima: $\frac{dE}{dx} = 0$

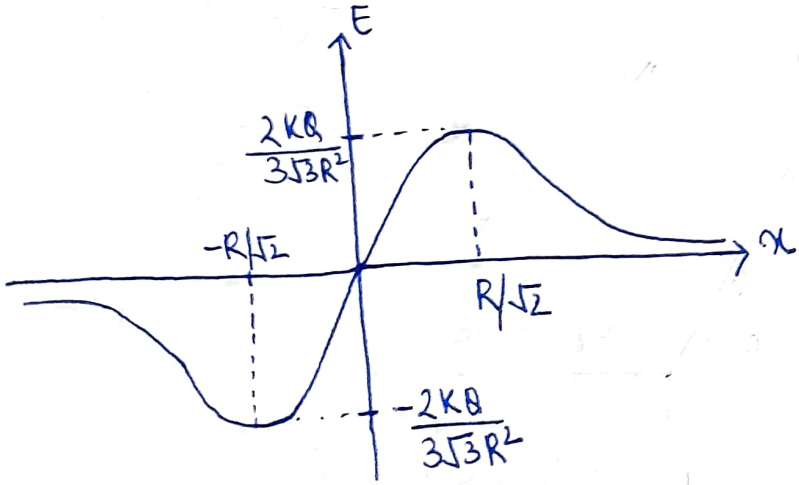
$$\Rightarrow \frac{(R^2+x^2)^{3/2} - x\left(\frac{3}{2}\right)(R^2+x^2)^{1/2}\cdot 2x}{(R^2+x^2)^3} = 0$$

$$\Rightarrow (R^2+x^2)^{3/2} = 3x^2(R^2+x^2)^{1/2}$$

$$\Rightarrow R^2+x^2 = 3x^2 \Rightarrow R^2 = 2x^2 \Rightarrow \boxed{x = \pm\frac{R}{\sqrt{2}}}$$

$$E_{max} = \frac{KQR}{\sqrt{2}\cdot 3\sqrt{3}R^3/2\sqrt{2}} = \frac{2KQ}{3\sqrt{3}R^2}$$

Graph:

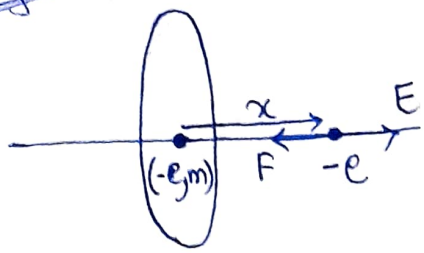


Q Prove that if $-e$ is displaced slightly along axis, it will perform SHM. Also find Time period of SHM.

No gravity

$F = q_0 E$

$\Rightarrow F = \frac{eKQx}{(R^2+x^2)^{3/2}}$

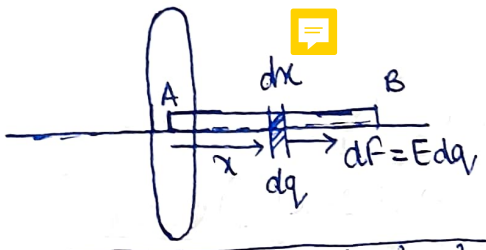


As $x \ll\ll R$, $\boxed{F = \frac{eKQ}{R^3}x}$ SHM proved

fixed uniform Ring

$\boxed{a = \frac{KQe}{mR^3}x}$ $\boxed{\omega = \sqrt{\frac{KQe}{mR^3}}}$ $\boxed{T = 2\pi\sqrt{\frac{mR^3}{KQe}}}$

Q Given Uniform Ring (Q,R) and uniform rod (q,R). Find electrostatic force b/w ring and rod.



$dF = E\,dq = \frac{KQx}{(R^2+x^2)^{3/2}}\left(\frac{q}{R}dx\right)$

$F = \int dF = \frac{KQq}{R}\int_0^R \frac{x\,dx}{(R^2+x^2)^{3/2}}$

$I = \int \frac{x\,dx}{(R^2+x^2)^{3/2}}$ let $R^2+x^2 = t^2$

$2x\,dx = 2t\,dt$

$x\,dx = t\,dt$

$I = \int \frac{t\,dt}{t^3} = \int \frac{dt}{t^2} = -\frac{1}{t} = \frac{-1}{\sqrt{R^2+x^2}}$

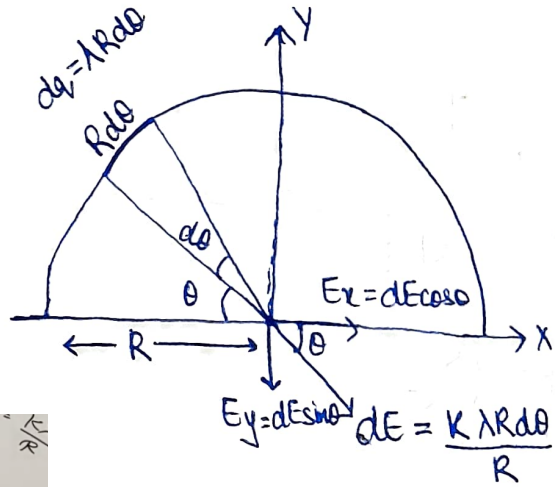
$F = \frac{KQq}{R}\left(\frac{-1}{\sqrt{R^2+x^2}}\right)_0^R$

$F = \frac{KQq}{R}\left(\frac{1}{R} - \frac{1}{\sqrt{2}R}\right) \Rightarrow \boxed{F = \frac{KQq}{R^2}\left(1-\frac{1}{\sqrt{2}}\right)}$

Electric Field due to Uniform Half Ring:

① At center

$\lambda = \frac{Q}{\pi R}$



$dE = \frac{K\lambda R\,d\theta}{R} = K\lambda\,d\theta$

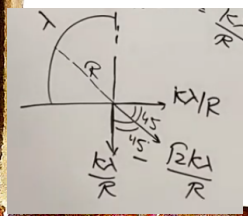
$E_x = \int dE\cos\theta = \int_0^{\pi} \frac{K\lambda}{R}d\theta\cos\theta = \frac{K\lambda}{R}(\sin\theta)_0^{\pi}$

$\boxed{E_x = 0}$

$E_y = \int dE\sin\theta = \int_0^{\pi} \frac{K\lambda}{R}d\theta\sin\theta = \frac{K\lambda}{R}(-\cos\theta)_0^{\pi}$

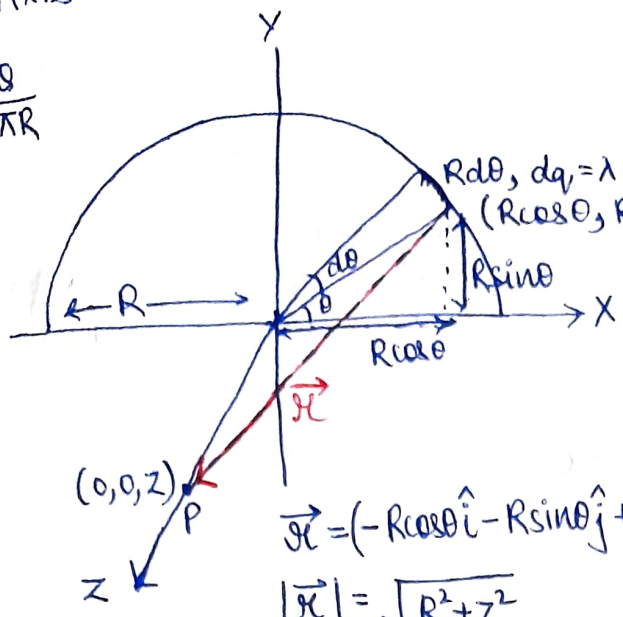
$\boxed{E_y = \frac{2K\lambda}{R}}$

$\therefore \boxed{E_{net} = \frac{2K\lambda}{R}}$



② At Axis

$\lambda = \frac{Q}{\pi R}$



$\vec{dE} = \frac{K dQ \vec{r}}{r^3}$

$\vec{dE} = \frac{K\lambda R d\theta(-R\cos\theta\hat{i} - R\sin\theta\hat{j} + z\hat{k})}{(R^2+z^2)^{3/2}}$

$\vec{E}_{net} = \int d\vec{E}$

$\vec{r} = (-R\cos\theta\hat{i} - R\sin\theta\hat{j} + z\hat{k})$

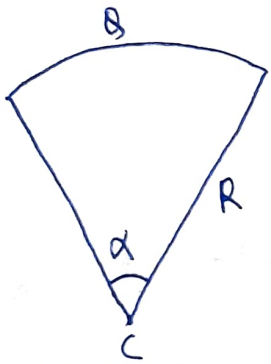
$|\vec{r}| = \sqrt{R^2+z^2}$

$= \frac{K\lambda R}{(R^2+z^2)^{3/2}}\left[-R\int_0^{\pi}\cos\theta d\theta\hat{i} - R\int_0^{\pi}\sin\theta d\theta\hat{j} + z\int_0^{\pi} d\theta\hat{k}\right]$

$= \frac{K\lambda R}{(R^2+z^2)^{3/2}}\left[-R\times 0\,\hat{i} - 2R\hat{j} + z\pi\hat{k}\right]$

$$\boxed{E = \frac{K\lambda R}{(R^2+z^2)^{3/2}}\left(-2R\hat{j} + \pi z\hat{k}\right)}$$

Electric Field due to a circular arc at the centre:



λ = Linear charge density

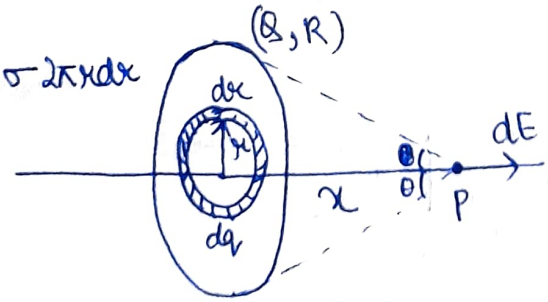
$$\boxed{\lambda = \frac{Q}{R\alpha}}$$

$$\boxed{E_{net\,at\,C} = \frac{2K\lambda}{R}\sin\left(\frac{\alpha}{2}\right)}$$

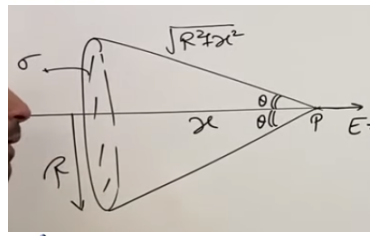
Electric Field due to uniform disc:

$$\boxed{\sigma = \frac{Q}{\pi R^2}}$$

$dq = \sigma 2\pi x dx$



We take an elemental ring and write dE due to it.



$dE = \frac{K dq x}{(x^2+x^2)^{3/2}} = \frac{\sigma 2\pi x dx x}{4\pi\varepsilon_0(x^2+x^2)^{3/2}} = \frac{\sigma x dx x}{2\varepsilon_0(x^2+x^2)^{3/2}}$

$E = \int dE = \frac{\sigma x}{2\varepsilon_0}\int_0^R \frac{x dx}{(x^2+x^2)^{3/2}} = \frac{\sigma x}{2\varepsilon_0}\left(\frac{-1}{\sqrt{x^2+x^2}}\right)_0^R = \frac{\sigma x}{2\varepsilon_0}\left(\frac{1}{x} - \frac{1}{\sqrt{R^2+x^2}}\right)$

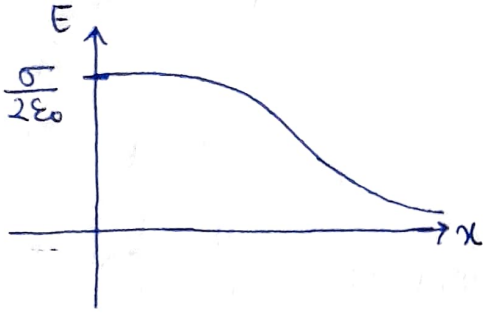
$$\boxed{E = \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{x}{\sqrt{R^2+x^2}}\right)} \text{ or } \boxed{E = \frac{\sigma}{2\varepsilon_0}(1-\cos\theta)} \quad \left(\cos\theta = \frac{x}{\sqrt{R^2+x^2}}\right)$$

Also, as $E = \frac{\sigma}{2\varepsilon_0}(1-\cos\theta) \Rightarrow x\uparrow, \theta\downarrow, \cos\theta\uparrow, E\downarrow$

At near point, $x \ll\ll R$

$$E = \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{x}{R}\right) = \frac{\sigma}{2\varepsilon_0}$$

Graph:



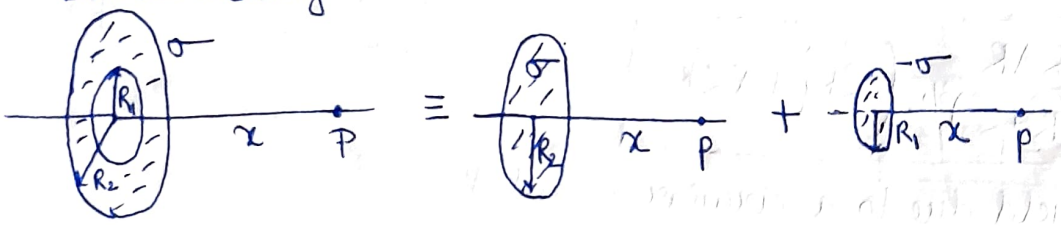
Special case:

If $R \to \infty$, then we say disc is an infinitely large sheet

$$E = \frac{\sigma}{2\varepsilon_0}$$ (uniform)

Q Find $\vec{E}$ at point P.

Disc with cavity



For Cavity we can write $\sigma - \sigma = 0$

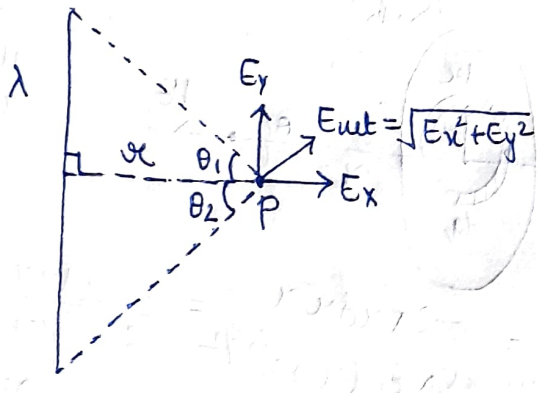
$$E = E_\sigma + E_{-\sigma}$$

$$E = \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{x}{\sqrt{R_2^2+x^2}}\right) + \frac{-\sigma}{2\varepsilon_0}\left(1 - \frac{x}{\sqrt{R_1^2+x^2}}\right)$$

Another method is to use integration

$$E = \frac{\sigma x}{2\varepsilon_0}\left(\frac{1}{\sqrt{R_1^2+x^2}} - \frac{1}{\sqrt{R_2^2+x^2}}\right)$$

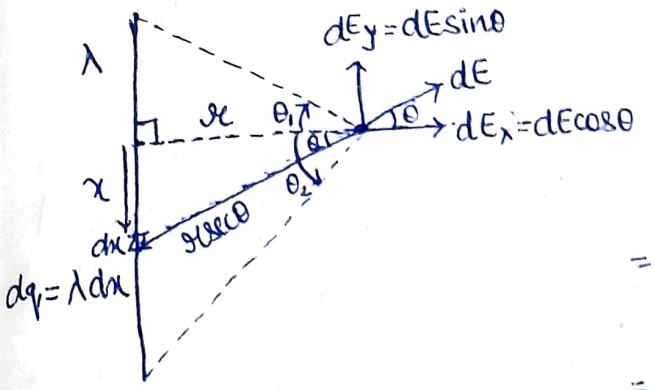
Electric Field due to Uniform wire:



$$E_\perp = E_x = \frac{K\lambda}{r}(\sin\theta_1 + \sin\theta_2)$$

$$E_{\parallel} = E_y = \frac{K\lambda}{r}\left|\cos\theta_1 - \cos\theta_2\right|$$

Derivation:



$$dE = \frac{K\lambda dx}{(r\sec\theta)^2}$$

$$E_y = \int dE_y = \int dE\sin\theta$$

$$= \int \frac{K\lambda dx}{r^2\sec^2\theta}\sin\theta$$

$$= \frac{K\lambda}{r}\int_{-\theta_2}^{\theta_1}\sin\theta = \frac{K\lambda}{r}(+\cos\theta)_{+\theta_1}^{-\theta_2}$$

$$\boxed{E_y = \frac{K\lambda}{r}\left|\cos\theta_2 - \cos\theta_1\right|}$$

$$E_x = \int dE_x = \int dE\cos\theta$$

$$= \int \frac{K\lambda dx}{r^2\sec^2\theta}\cos\theta$$

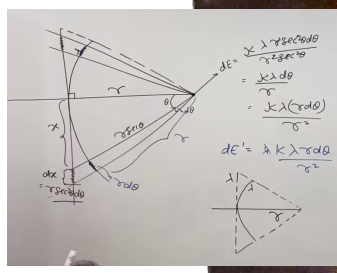
$$= \int \frac{K\lambda r\sec^2\theta\, d\theta\cos\theta}{r^2\sec^2\theta}$$

$$= \frac{K\lambda}{r}\int_{-\theta_2}^{\theta_1}\cos\theta\, d\theta = \frac{K\lambda}{r}(\sin\theta)_{-\theta_2}^{\theta_1}$$

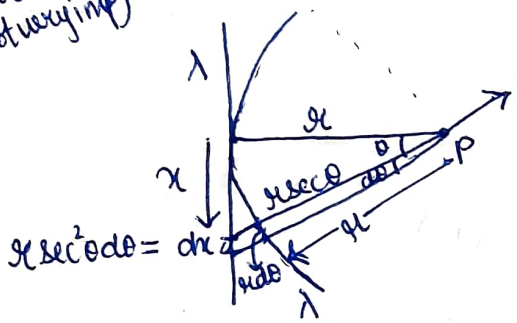
$$= \frac{K\lambda}{r}(\sin\theta_1 - \sin(-\theta_2))$$

$$\boxed{E_x = \frac{K\lambda}{r}(\sin\theta_1 + \sin\theta_2)}$$

$x = r\tan\theta$
$dx = r\sec^2\theta\, d\theta$
Substitute in place of dx in integration.



Note: (not very imp)

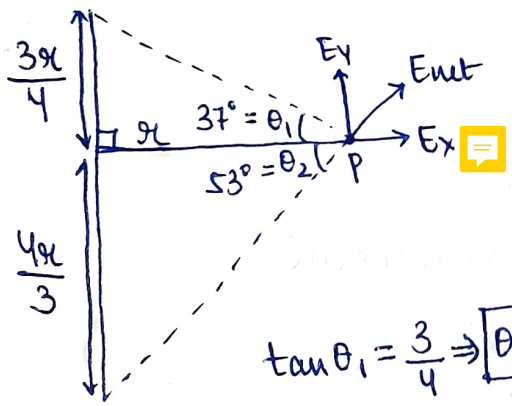


$$dE = \frac{K\lambda r\sec^2\theta\, d\theta}{r^2\sec^2\theta}$$

$$= \frac{K\lambda d\theta}{r} = \frac{K\lambda(r\,d\theta)}{r^2}$$

Here we can say that the E due to a circular arc with P as center and 1st distance from straight wire as radius will be same in magnitude and direction as that by straight wire.

Q Find $\vec{E}_{net}$ at point P.



$$\tan\theta_1 = \frac{3}{4} \Rightarrow \boxed{\theta_1 = 37^\circ}$$

$$\tan\theta_2 = \frac{4}{3} \Rightarrow \boxed{\theta_2 = 53^\circ}$$

$$E_x = \frac{K\lambda}{r}(\sin\theta_1 + \sin\theta_2) = \frac{K\lambda}{r}\left(\frac{3}{5} + \frac{4}{5}\right)$$

$$\boxed{E_x = \frac{7K\lambda}{5r}}$$

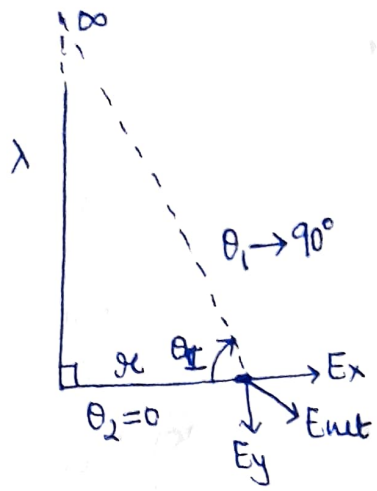
$$E_y = \frac{K\lambda}{r}\left|\cos\theta_1 - \cos\theta_2\right| = \frac{K\lambda}{r}\left(\frac{4}{5} - \frac{3}{5}\right)$$

$$\boxed{E_y = \frac{K\lambda}{5r}}$$

$$E_{net} = \sqrt{(E_x)^2 + (E_y)^2} = \frac{K\lambda}{5r}\sqrt{7^2 + 1^2}$$

$$\boxed{E_{net} = \frac{\sqrt{2}K\lambda}{r}}$$

Electric Field due to Semi-infinite wire:



$$E_x = \frac{K\lambda}{x}(\sin 90 + \sin 0)$$

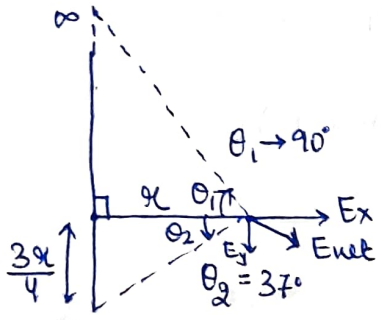
$$\boxed{E_x = \frac{K\lambda}{x}}$$

$$E_y = \frac{K\lambda}{x}|\cos 90 - \cos 0|$$

$$\boxed{E_y = \frac{K\lambda}{x}}$$

$$\boxed{E_{net} = \sqrt{2}\,\frac{K\lambda}{x}}$$

Q Find $\vec{E}_{net}$ at point P.

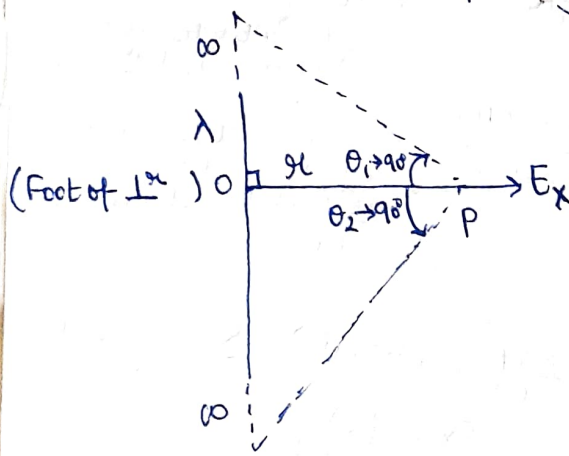


$$E_x = \frac{K\lambda}{x}(\sin 90 + \sin 37) \Rightarrow \boxed{E_x = \frac{8K\lambda}{5x}}$$

$$E_y = \frac{K\lambda}{x}|\cos 90 - \cos 37| \Rightarrow \boxed{E_y = \frac{4K\lambda}{5x}}$$

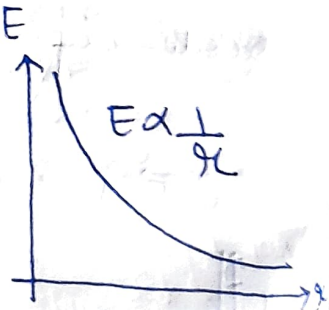
$$E_{net} = \frac{4K\lambda}{5x}\sqrt{1^2+2^2} \Rightarrow \boxed{E_{net} = \frac{4K\lambda}{\sqrt{5}x}}$$

Electric Field due to Infinitely long wire:



$$E_x = E_{net} = \frac{K\lambda}{x}(2\sin 90^\circ)$$

$$\therefore \boxed{E_{net} = \frac{2K\lambda}{x}}$$



Vector Form:

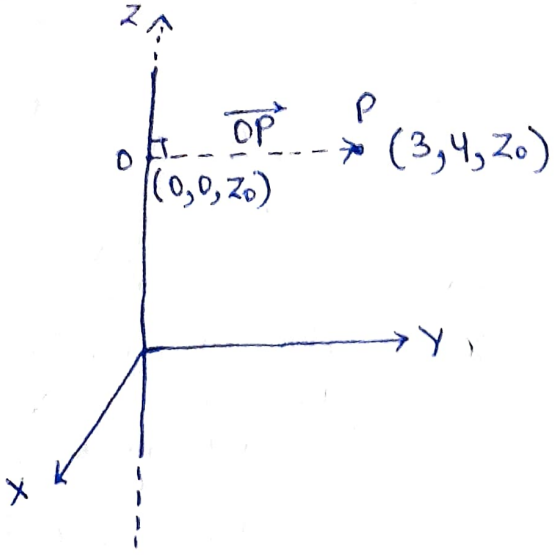
$$\boxed{\vec{E} = \frac{2K\lambda}{x}(\hat{OP}) = \frac{2K\lambda}{x^2}\vec{OP}}$$

(Use λ with sign)

If $\lambda \to +ve \Rightarrow \vec{E} \parallel \vec{OP}$

If $\lambda \to -ve \Rightarrow \vec{E} \uparrow\downarrow \vec{OP}$

Q



An infinitely long wire (λ) is kept on z axis. Find $\vec{E}$ at P in vector form

$\overrightarrow{OP} = 3\hat{i}+4\hat{j}$ $\quad |\overrightarrow{OP}| = 5$

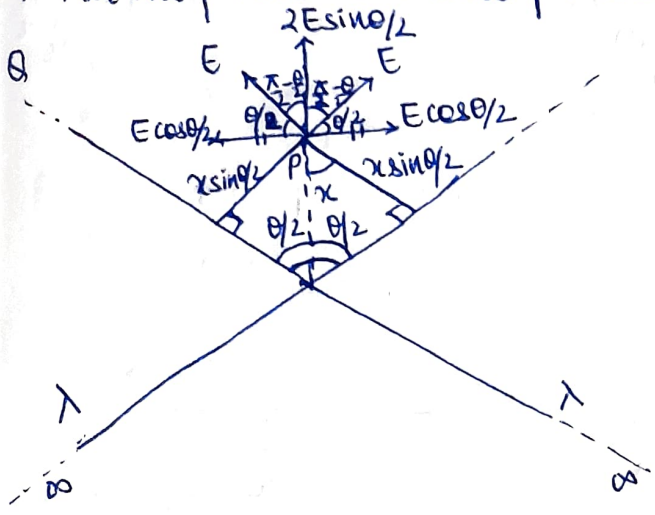
$$\vec{E} = \frac{2K\lambda}{x}(\hat{OP}) = \frac{2K\lambda}{5}\left(\frac{\overrightarrow{OP}}{|\overrightarrow{OP}|}\right)$$

$$\boxed{\vec{E} = \frac{2K\lambda}{25}(3\hat{i}+4\hat{j})}$$

→ $\vec{E}$ is independent of z_0.

★ All the points on a line parallel to infinite wire will have equal E.

Q



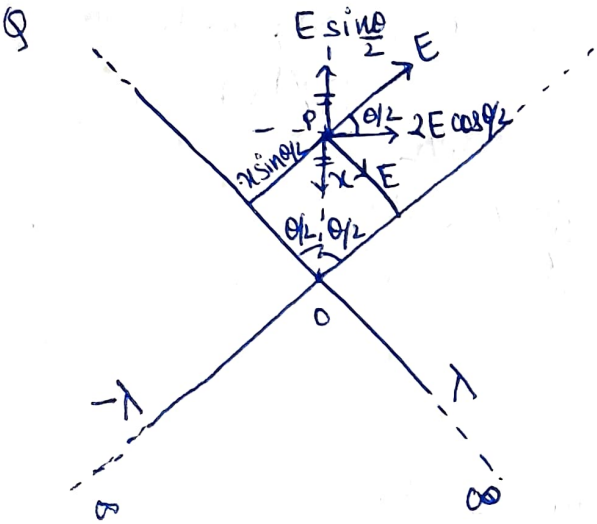
Find $\vec{E}$ at point P.

$$E_{net} = 2E\sin\frac{\theta}{2}$$

$$= 2\left(\frac{2K\lambda}{x\sin\frac{\theta}{2}}\right)\left(\sin\frac{\theta}{2}\right)$$

$$\boxed{E_{net} = \frac{4K\lambda}{x}}$$

Q



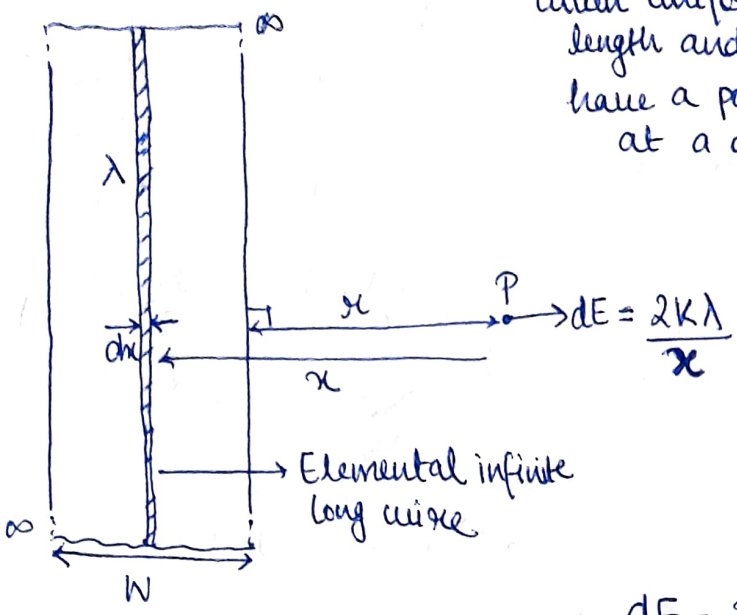
Find $\vec{E}$ at point P.

$$E_{net} = 2E\cos\frac{\theta}{2}$$

$$= 2\left(\frac{2K\lambda}{x\sin\frac{\theta}{2}}\right)\cos\frac{\theta}{2}$$

$$\boxed{E_{net} = \frac{4K\lambda}{x}\cot\frac{\theta}{2}}$$

Q



Given Uniform Rectangular sheet of infinite length and width W with σ. Also we have a point P (in the same plane of sheet) at a distance x from sheet.
Find $\vec{E}$ at P.

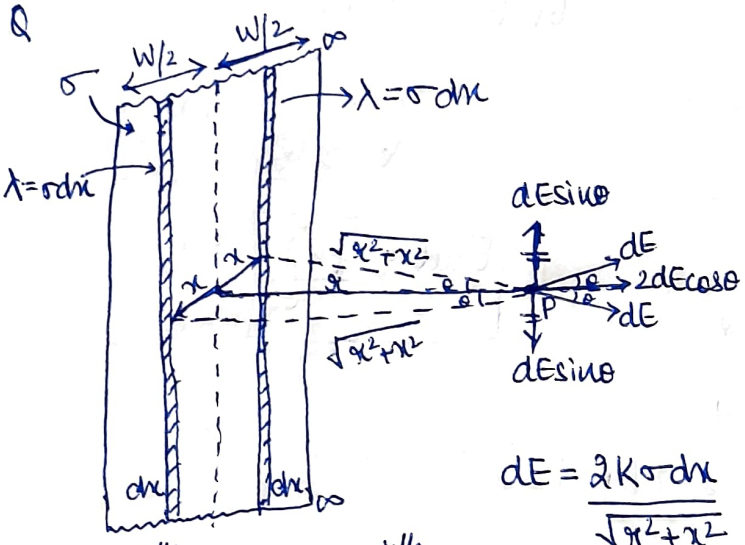
$$dE = \frac{2K\lambda}{x} \quad \text{—(1)}$$

Now, let length of wire be l
Charge on wire = λl
Also charge can be given as = σ l dx
So, $\lambda l = \sigma l\, dx \Rightarrow \boxed{\lambda = \sigma\, dx}$

$$dE = \frac{2K\sigma\, dx}{x} \Rightarrow E = \int dE = 2K\sigma \int_{x}^{x+W} \frac{dx}{x}$$

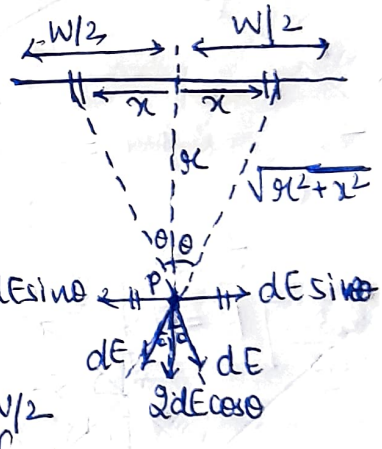
$$\boxed{E = 2K\sigma \ln\left(\frac{x+W}{x}\right)}$$

Q



Find $\vec{E}$ at point P.

Top View



$$dE = \frac{2K\sigma\, dx}{\sqrt{x^2+x^2}}$$

$$E = \int_0^{W/2} 2\,dE\cos\theta = 2\int_0^{W/2} \frac{2K\sigma\, dx}{\sqrt{x^2+x^2}}\left(\frac{x}{\sqrt{x^2+x^2}}\right) = \frac{4\sigma x}{4\pi\varepsilon_0}\int_0^{W/2} \frac{dx}{x^2+x^2} = \frac{\sigma x}{\pi\varepsilon_0}\left(\frac{1}{x}\tan^{-1}\left(\frac{x}{x}\right)\right)_0^{W/2}$$

$$\boxed{E = \frac{\sigma}{\pi\varepsilon_0}\tan^{-1}\left(\frac{W}{2x}\right)}$$

Special Case:
If $W \to \infty \Rightarrow$ This sheets converts to infinite sheet.

$$E = \frac{\sigma}{\pi\varepsilon_0}\tan^{-1}(\infty) = \frac{\sigma}{\pi\varepsilon_0}\left(\frac{\pi}{2}\right) = \frac{\sigma}{2\varepsilon_0}$$

$$\boxed{E = \frac{\sigma}{2\varepsilon_0}} \rightarrow \vec{E} \text{ due to infinite sheet}$$
It is uniform as independent of x.

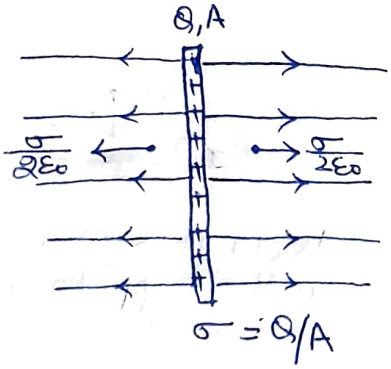
Electric Field due to infinite sheet (Non-Conducting):

$$\sigma = \frac{Q}{A}$$

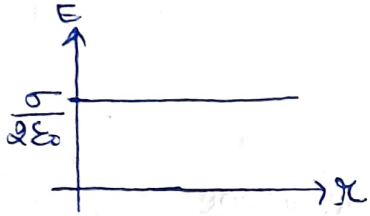
$$E = \frac{\sigma}{2\varepsilon_0} = \frac{Q}{2A\varepsilon_0}$$

(Uniform)

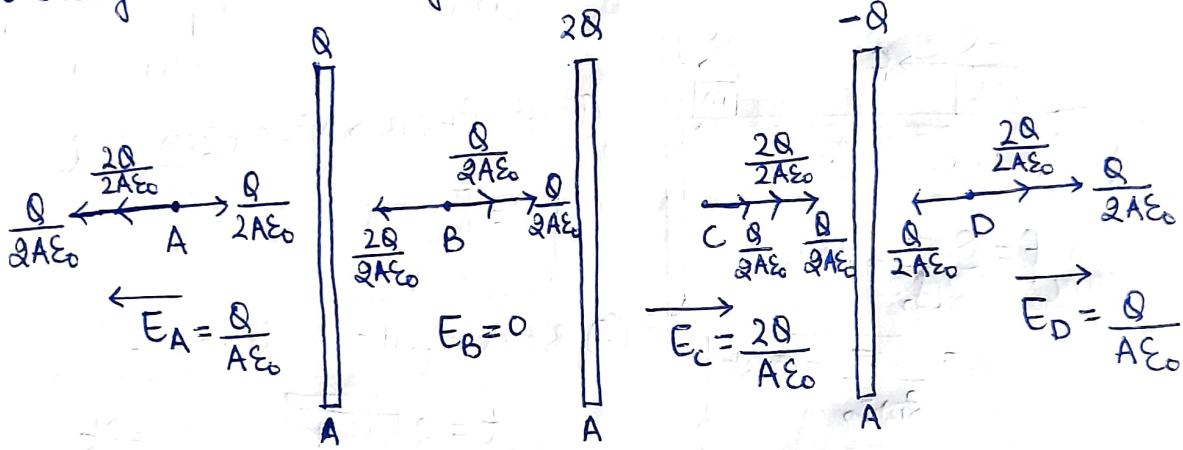
Side View



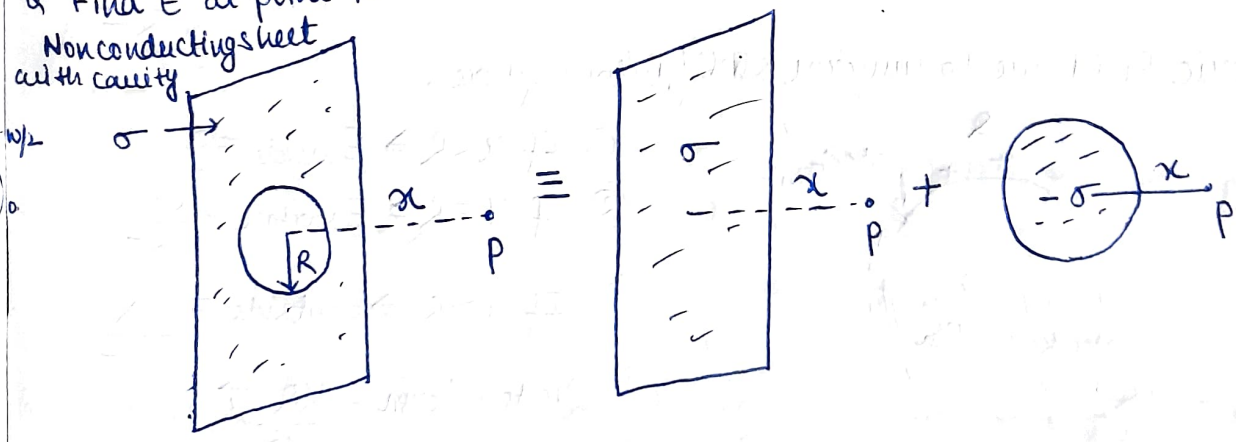
Graph of E V/s x :



Q Given 3 large non-conducting sheets. Find $\vec{E}$ at points A, B, C, D



Q Find $\vec{E}$ at point P.

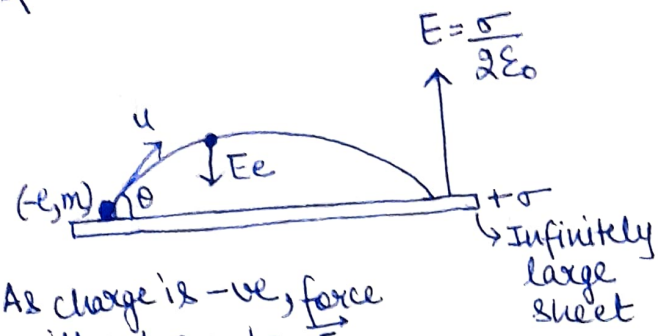


$$E_P = E_\sigma + E_{-\sigma} = \frac{\sigma}{2\varepsilon_0} + \frac{-\sigma}{2\varepsilon_0}\left(1 - \frac{x}{\sqrt{x^2+R^2}}\right)$$

$$\boxed{E_P = \frac{\sigma x}{2\varepsilon_0\left(\sqrt{R^2+x^2}\right)}}$$

Q Gravity free space

Find ① T ② Hmax ③ R



As charge is -ve, force will act opp. to $\vec{E}$.

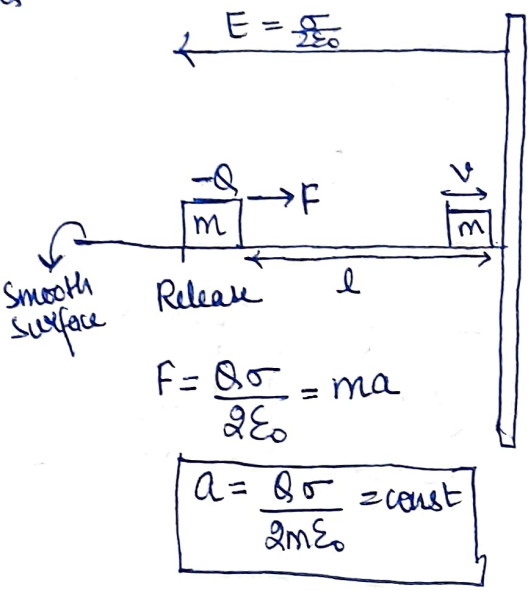
Force on e = $\frac{\sigma e}{2\varepsilon_0}$

So, $\frac{\sigma e}{2\varepsilon_0} = ma \Rightarrow \boxed{a = \frac{\sigma e}{2m\varepsilon_0}}$

① $\boxed{T = \frac{2u\sin\theta}{\frac{\sigma e}{2m\varepsilon_0}}}$ ② $\boxed{H = \frac{u^2\sin^2\theta}{2\left(\frac{\sigma e}{2m\varepsilon_0}\right)}}$

③ $\boxed{R = \frac{u^2\sin 2\theta}{\frac{\sigma e}{2m\varepsilon_0}}}$

Q



Find ① Speed of block when it is about to strike the sheet

② Find time period of oscillatory motion of block.

$F = \frac{Q\sigma}{2\varepsilon_0} = ma$

$\boxed{a = \frac{Q\sigma}{2m\varepsilon_0} = \text{const}}$

① $v^2 = u^2 + 2as \Rightarrow v^2 = 0 + 2\left(\frac{Q\sigma}{2m\varepsilon_0}\right)l$

$\boxed{v = \sqrt{\frac{Q\sigma l}{m\varepsilon_0}}}$

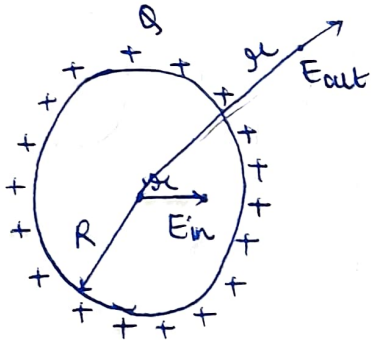
② $s = ut + \frac{1}{2}at^2 \Rightarrow l = \frac{1}{2}\left(\frac{Q\sigma}{2m\varepsilon_0}\right)t^2$

$\boxed{t = 2\sqrt{\frac{m\varepsilon_0 l}{Q\sigma}}}$

$T = 2t$

$\boxed{T = 4\sqrt{\frac{m\varepsilon_0 l}{Q\sigma}}}$

Electric Field due to uniform shell/Hollow sphere:



① If $r < R \Rightarrow E_{inside} = 0$

② If $r = R \Rightarrow E_{surface} = \frac{KQ}{R^2}$

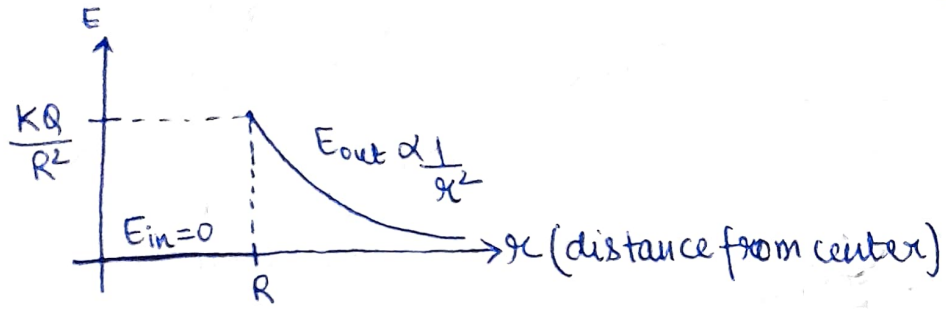
③ If $r > R \Rightarrow E_{outside} = \frac{KQ}{r^2}$

Vector form = $\frac{KQ}{r^3}\vec{r}$ (Q with sign)

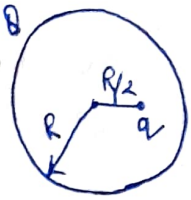
→ Inside the shell, $\vec{E}$ due to uniform spherical shell will be zero

→ For outside points uniform spherical shell will behave like point charge placed at center.

Graph E v/s x:



Q Find $\vec{F_e}$ b/w Q and q.

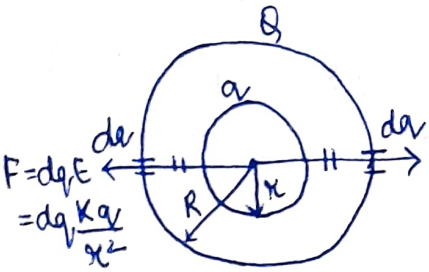


$F = qE$

$F = qE_{inside} = q \times 0$ $\therefore$ $\boxed{F = 0}$

By Newtons 3rd law, q also applies F=0 on Q.

Q Given two concentric uniform shells. Find $\vec{F_e}$ b/w Q and q.



Method 1: We find $\vec{F_e}$ on Q due to q.

We observe that all forces gets cancelled by diametric opposite point.

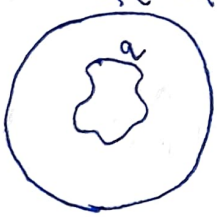
$\therefore$ $\boxed{F_{net} = 0}$

Method 2: We find $\vec{F_e}$ on q due to Q

As q is inside Q $\Rightarrow$ $E_{inside} = 0$

$F = qE_{in} \Rightarrow$ $\boxed{F_{net} = 0}$

Q



q is non-uniformly distributed

Find $\vec{F_e}$ b/w Q and q.

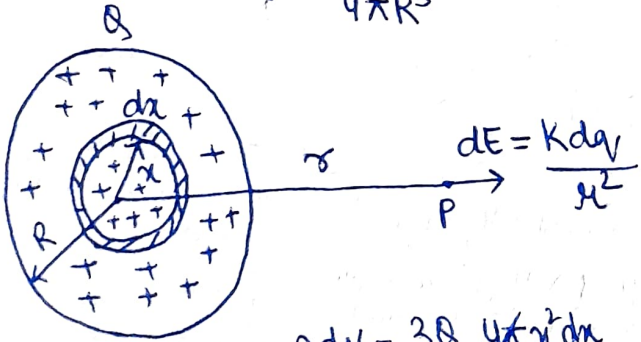
Here we cannot write $\vec{F_e}$ by q on Q as we do not know the geometry of body and charge is non-uniform. But we can write F due to Q on q. By Newtons 3rd law both will be equal.

$F = qE_{in}$ $\therefore$ $\boxed{F = 0}$

Electric Field due to uniform Solid Sphere:

(I) Outside

$\rho = \frac{3Q}{4\pi R^3}$



Elemental shell $\Rightarrow dq = \rho dV = \frac{3Q}{4\pi R^3} 4\pi x^2 dx$

$dq = \frac{3Qx^2 dx}{R^3}$

$dE = \frac{K dq}{x^2} = \frac{K\, 3Q x^2 dx}{R^3 x^2}$

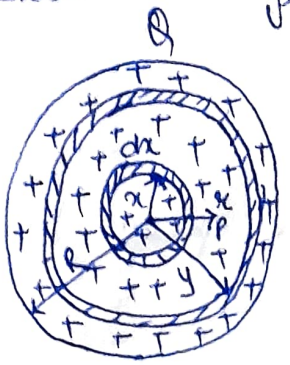
$E = \int dE = \frac{3KQ}{R^3 x^2} \int_0^R x^2 dx$

$E = \frac{3KQ}{R^3 x^2} \left(\frac{R^3}{3}\right)$

$\boxed{E_{out} = \frac{KQ}{x^2}}$

thus, for outside points, this sphere behaves like a point charge placed at center

(II) Inside



$\rho = \frac{3Q}{4\pi R^3}$

→ For shell of radius x, P behaves as outside point where we need to find $\vec{E}$.

→ For shell of radius y, P behaves as inside point ($y > x$), so $\vec{E}$ inside these shells must be zero.

$$E = \int_0^R dE = \int_0^x dE + \cancel{\int_x^R dE}^{\,0}$$

$dq = \rho\, dV$

$$(\text{For shell } x) = \frac{3Qx^2dx}{R^3}$$

$$E = \int_0^x dE = \int_0^x \frac{K\,dQ}{x^2} = \int_0^x \frac{K\,3Qx^2dx}{R^3x^2}$$

$$E = \frac{3KQ}{R^3x^2}\int_0^x x^2dx = \frac{\cancel{3}KQ}{R^3\cancel{x^2}}\left(\frac{x^3}{\cancel{3}}\right)$$

$$\boxed{E_{inside} = \frac{KQx}{R^3}}$$

Now, $\frac{KQx}{R^3} = \frac{1}{4\pi\varepsilon_0}\frac{\rho V x}{R^3} = \frac{1}{4\pi\varepsilon_0}\frac{\rho \cancel{4}\pi\cancel{R^3}\,x}{\cancel{3}\,R^{\cancel{3}}}$

$$\boxed{E_{inside} = \frac{\rho x}{3\varepsilon_0}}$$

x → distance from center
R → Radius of Solid Sphere

Another Method:

$$E_{in} = \frac{KQ_{in}}{x^2}$$

$$E_{in} = \frac{KQx^3}{R^3x^2}$$

$$\boxed{E_{in} = \frac{KQx}{R^3}}$$

$\frac{4}{3}\pi R^3 \longrightarrow Q$

$\frac{4}{3}\pi x^3 \longrightarrow \frac{Q}{\frac{4}{3}\pi R^3}\,\frac{4}{3}\pi x^3$

$$\boxed{Q_{in} = \frac{Qx^3}{R^3}}$$

Inside ($x \le R$):

$E_{in} \propto x$

$$\boxed{E_{in} = \frac{KQx}{R^3} = \frac{\rho x}{3\varepsilon_0}}$$

Vector Form ⇒

$$\boxed{\vec{E}_{in} = \frac{KQ\vec{x}}{R^3}}$$

$$\boxed{\vec{E}_{in} = \frac{\rho x\hat{x}}{3\varepsilon_0} = \frac{\rho\vec{x}}{3\varepsilon_0}}$$

Outside ($x \ge R$):

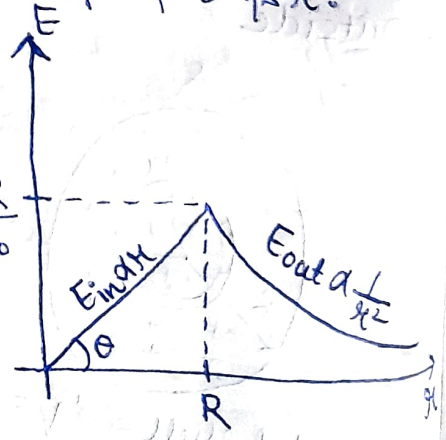
$$\boxed{E_{out} = \frac{KQ}{x^2}}$$

($\vec{x}$: Tail ⇒ At centre of sphere; Head ⇒ At pt. on which $\vec{E}$ is to be found)

$$\boxed{\vec{E}_{out} = \frac{KQ\hat{x}}{x^2} = \frac{KQ\vec{x}}{x^3}}$$

$$\boxed{E_{out} \propto \frac{1}{x^2}}$$

Graph of E v/s x:



$$\tan\theta = \frac{\rho}{3\varepsilon_0}$$

Q Uniform Solid sphere is centered at (0,0,0). Find $\vec{E}$ at ① $A\left(\frac{R}{2},\frac{R}{2},0\right)$ ② $B\left(\frac{R}{2},\frac{R}{2},\frac{R}{2}\right)$ ③ $C(R,R,R)$

(ρ, R)

① $A\left(\frac{R}{2},\frac{R}{2},0\right)$

Distance of A from centre = $\sqrt{\left(\frac{R}{2}\right)^2+\left(\frac{R}{2}\right)^2} = \frac{R}{\sqrt{2}}$ (OA)

$OA = \frac{R}{\sqrt{2}} < R \Rightarrow$ A lies inside sphere

$$\boxed{\vec{E} = \frac{\rho}{3\varepsilon_0}\left(\frac{R}{2}\hat{i}+\frac{R}{2}\hat{j}\right)}$$

② $B\left(\frac{R}{2},\frac{R}{2},\frac{R}{2}\right)$

$OB = \sqrt{\left(\frac{R}{2}\right)^2+\left(\frac{R}{2}\right)^2+\left(\frac{R}{2}\right)^2} = \frac{\sqrt{3}R}{2}$

$OB = \frac{\sqrt{3}R}{2} < R \Rightarrow$ B lies inside

$$\boxed{\vec{E} = \frac{\rho}{3\varepsilon_0}\left(\frac{R}{2}\hat{i}+\frac{R}{2}\hat{j}+\frac{R}{2}\hat{k}\right)}$$

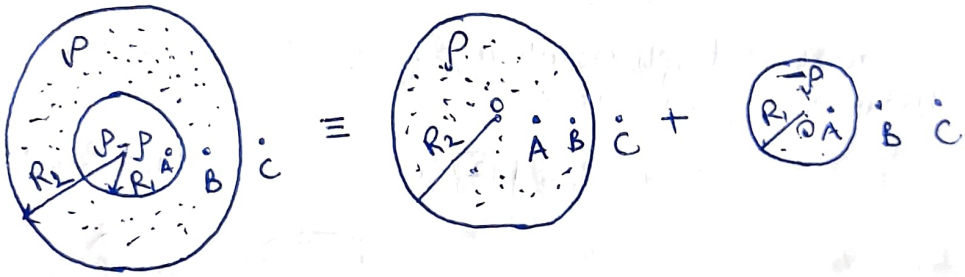
③ $C(R,R,R)$

$OC = \sqrt{R^2+R^2+R^2} = \sqrt{3}R.$

$OC = \sqrt{3}R > R \Rightarrow$ C lies Outside

$$\boxed{\vec{E}_C = \frac{KQ\vec{r}}{r^3} = \frac{K\rho\frac{4}{3}\pi R^3}{(\sqrt{3}R)^3}\left(R\hat{i}+R\hat{j}+R\hat{k}\right)}$$

Q Given Solid sphere with cavity. Find $\vec{E}$ at point A, B, and C.

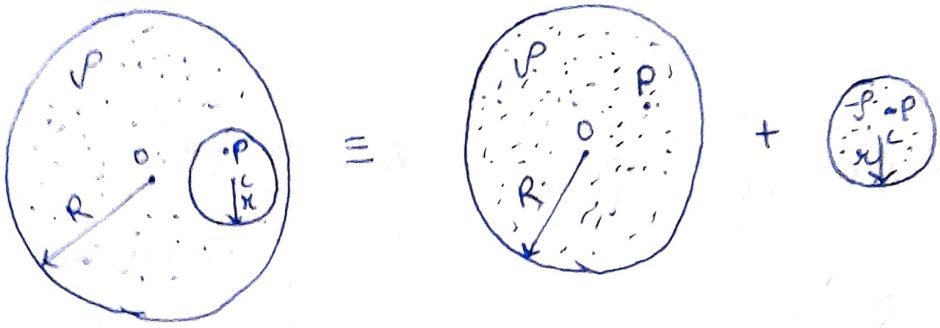


① $\vec{E}_A = \frac{\rho(\vec{OA})}{3\varepsilon_0} + \frac{(-\rho)(\vec{OA})}{3\varepsilon_0} = 0$

② $\vec{E}_B = \frac{\rho(\vec{OB})}{3\varepsilon_0} + \frac{\left(-\rho\frac{4}{3}\pi R_1^3\right)(\vec{OB})}{4\pi\varepsilon_0 (OB)^3}$

③ $\vec{E}_C = \frac{\rho\left(\frac{4}{3}\pi R_2^3\right)(\vec{OC})}{4\pi\varepsilon_0 (OC)^3} + \frac{-\rho\left(\frac{4}{3}\pi R_1^3\right)(\vec{OC})}{4\pi\varepsilon_0 (OC)^3}$

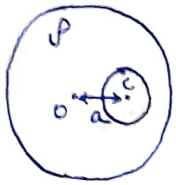
Q Find $\vec{E}$ at point P. Given Solid sphere with cavity.
(ρ, R)



$$\vec{E}_P = \vec{E}_p + \vec{E}_{-p} = \frac{\rho(\vec{OP})}{3\varepsilon_0} + \frac{(-\rho)(\vec{CP})}{3\varepsilon_0} = \frac{\rho}{3\varepsilon_0}(\vec{OP} - \vec{CP}) = \frac{\rho}{3\varepsilon_0}(\vec{OP} + \vec{PC})$$

$$\boxed{\vec{E}_P = \frac{\rho}{3\varepsilon_0}(\vec{OC})} \quad \text{(independent of P)}$$

Note:

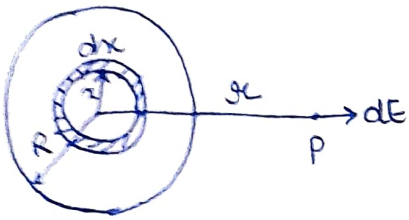


$\vec{E}$ inside cavity will be uniform

$$\boxed{E = \frac{\rho a}{3\varepsilon_0}}$$

Q Given a non-uniform solid sphere where $\rho = \rho_0 x$ where x is distance from center. Find $\vec{E}$ at distance x from center

① $x \geq R$



$$dq = \rho_0 x \, 4\pi x^2 dx$$

$$dE = \frac{K dq}{x^2} = \frac{\rho_0 x 4\pi x^2 dx}{4\pi\varepsilon_0 x^2} = \frac{\rho_0 x^3 dx}{\varepsilon_0 x^2}$$

$$E = \int dE = \frac{\rho_0}{\varepsilon_0 x^2}\int_0^R x^3 dx$$

$$\boxed{E = \frac{\rho_0 R^4}{4\varepsilon_0 x^2}} \qquad \boxed{E \propto \frac{1}{x^2}}$$

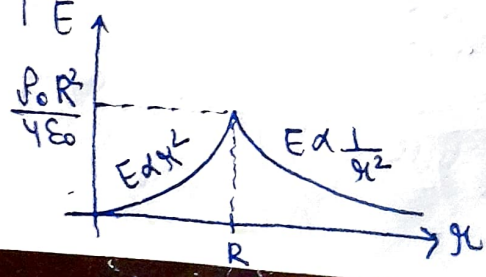
② $x \leq R$ $\quad dq = \rho_0 x \, 4\pi x^2 dx$

$$E = \int_0^R dE = \int_0^x dE + \cancel{\int_x^R dE}$$

$$E = \int_0^x \frac{K dq}{x^2} = \int_0^x \frac{\rho_0 x 4\pi x^2 dx}{4\pi\varepsilon_0 x^2} = \int_0^x \frac{\rho_0 x^3 dx}{\varepsilon_0 x^2}$$

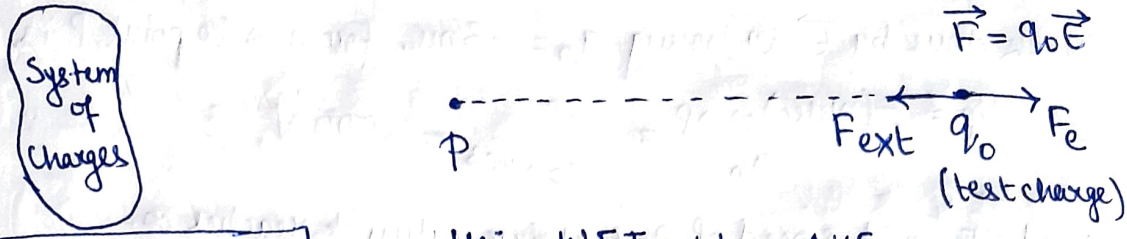
$$E = \frac{\rho_0}{\varepsilon_0 x^2}\int_0^x x^3 dx = \frac{\rho_0}{\varepsilon_0 \cancel{x^2}} \frac{x \cancel{y^2}}{4}$$

$$\boxed{E = \frac{\rho_0 x^2}{4\varepsilon_0}} \qquad \boxed{E \propto x^2}$$



Electric Potential (V):

→ Electric Potential at point P due to system of charges is defined as work done by external agent to bring unit positive charge from ∞ to point P slowly, against electrostatic forces due to system of charges.



$$V_P = \left[\frac{(W_{ext})_{\infty \to P}}{q_0}\right]_{\Delta K = 0}$$

Using WET ⇒ $W_{all} = \Delta KE$

$W_{ext} + W_{electric} = 0$

$W_{ext} = -W_{electric}$

$$V_P = \left[\frac{-(W_{elec})_{\infty \to P}}{q_0}\right] = \left[\frac{(W_{elec})_{P \to \infty}}{q_0}\right]$$

→ Electric Potential at point P due to system of charges is defined as -ve of work done by electric field due to system of charges to bring unit +ve charge from ∞ to point P. or it is equal to the work done per unit charge by the EF on moving it from point P to infinity

Now, $\vec{F} = q_0\vec{E}$ ($\vec{E}$ is non-uniform due to system of charges)

$$V_P = \frac{-(W_{elec})_{\infty \to P}}{q_0} = \frac{-\int_{\infty}^{P} q_0\vec{E}\cdot d\vec{x}}{q_0} = -\int_{\infty}^{P}\vec{E}\cdot d\vec{x}$$

$$V_P = \frac{(W_{elec})_{P \to \infty}}{q_0} = \frac{\int_{P}^{\infty} q_0\vec{E}\cdot d\vec{x}}{q_0} = \int_{P}^{\infty}\vec{E}\cdot d\vec{x}$$

$$\therefore \boxed{V_P = -\int_{\infty}^{P}\vec{E}\cdot d\vec{x} = \int_{P}^{\infty}\vec{E}\cdot d\vec{x}}$$

* $\int\vec{E}\cdot d\vec{x}$ is called line integral of Electric Field

→ Unit of V = Joule/Coulomb = J/C = Volt

→ Dimension = $\frac{ML^2T^{-2}}{IT} = [ML^2T^{-3}I^{-1}]$

→ Scalar Quantity

→ Can be +ve, -ve or zero

→ We take ∞ (very far) as Reference point and we assum $V_\infty = 0$

→ V due to multiple charges can be added scalarly.

$$V = V_1 + V_2 + V_3 + \dots$$

For continuous body = $\int dV$

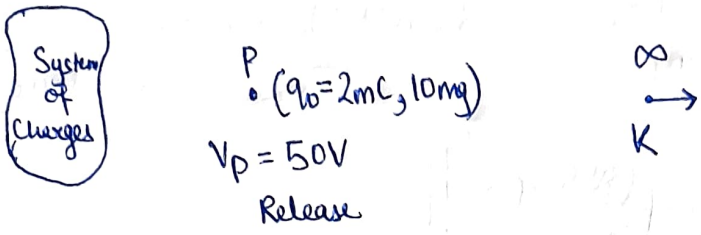
Q To bring a charge $q_0 = 5mC$ from ∞ to point P slowly, $W_{ext} = 0.5J$. Find V_P.

$$V_P = \left[\frac{(W_{ext})_{\infty \to P}}{q_0}\right]_{\Delta K = 0} = \frac{0.5}{5\times10^{-3}} = 100\,V$$

Q Work done by $\vec{E}$ to bring $q_0 = -3mC$ from ∞ to point P is 6J. Find V_p

$$V_P = \frac{-(W_{elec})_{\infty \to P}}{q_0} = \frac{-6}{-3\times10^{-3}} = 2000\,V$$

Q Find KE and speed of 2mC charge when it reaches ∞.



Using WET $\Rightarrow W_{all} = \Delta KE$

$W_{elec} = K_f - K_i^{\,0}$

$W_{elec} = K$

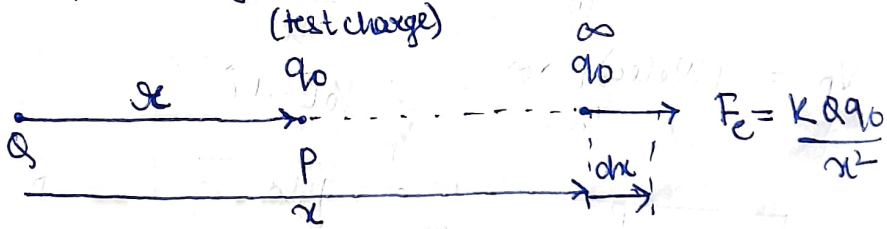
$\boxed{K = 0.1J}$

$$V_P = \frac{(W_{elec})_{P\to\infty}}{q_0} \Rightarrow (W_{elec})_{P\to\infty} = V_P q_0$$

$$\boxed{(W_{elec})_{P\to\infty} = 50\times2\times10^{-3} = 0.1J}$$

Now, $K = \frac{1}{2}mv^2 \Rightarrow 0.1 = \frac{1}{2}(10\times10^{-6})v^2 \Rightarrow \boxed{v = 100\sqrt{2}}$ (speed)

Electric Potential due to point charge:



$\vec{dW} = \vec{F_e}\cdot\vec{dx}$

$dW = \frac{KQq_0\,dx\cos0^\circ}{x^2}$

$$W_{elec} = \int dW = \int_x^\infty \frac{KQq_0\,dx}{x^2} = KQq_0\int_x^\infty \frac{dx}{x^2}$$

$$= KQq_0\left(\frac{-1}{x}\right)_x^\infty = KQq_0\left(\frac{-1}{\infty} + \frac{1}{x}\right)$$

$$\boxed{W_{elec} = \frac{KQq_0}{x}}$$

$$V_P = \frac{(W_{elec})_{P\to\infty}}{q_0} = \frac{\frac{KQq_0}{x}}{q_0}$$

Ex

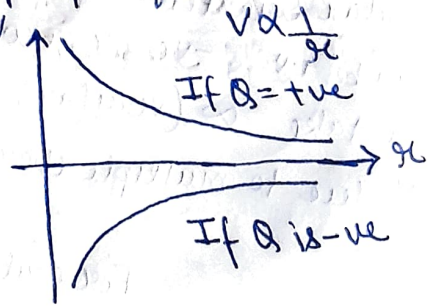
$$\star\ \boxed{V_P = \frac{KQ}{x}}$$

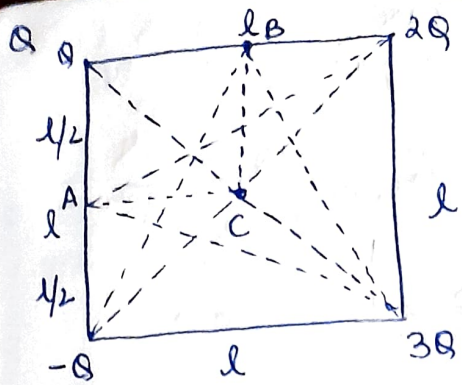
If Q = +ve, V = +ve

If Q = -ve, V = -ve

If $x \to \infty$, $V \to 0$

Graph of V v/s x:





Distance of A from 2Q or 3Q

$= \sqrt{\left(\frac{\ell}{2}\right)^2 + \ell^2} = \frac{\sqrt{5}\ell}{2}$

Find ① V_A ② V_B ③ V_C

① $V_A = \frac{KQ}{\ell/2} + \frac{K(-Q)}{\ell/2} + \frac{K(2Q)}{\sqrt{5}\ell/2} + \frac{K(3Q)}{\sqrt{5}\ell/2}$

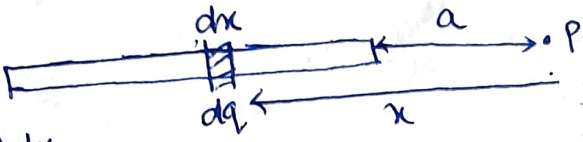
$$\boxed{V_A = \frac{K(5Q)}{\sqrt{5}\ell/2}}$$

② $V_B = \frac{KQ}{\ell/2} + \frac{K(2Q)}{\ell/2} + \frac{K(-Q)}{\sqrt{5}\ell/2} + \frac{K(3Q)}{\sqrt{5}\ell/2}$

$$\boxed{V_B = \frac{K(3Q)}{\ell/2} + \frac{K(2Q)}{\sqrt{5}\ell/2}}$$

③ $V_C = \frac{KQ}{\ell/\sqrt{2}} + \frac{K(2Q)}{\ell/\sqrt{2}} + \frac{K(3Q)}{\ell/\sqrt{2}} + \frac{K(-Q)}{\ell/\sqrt{2}} = \frac{K(5Q)}{\ell/\sqrt{2}}$

Q Find V at point P due to uniform rod (Q, ℓ)



$\lambda = \frac{Q}{\ell}$

$dq = \lambda dx$

$dq = \frac{Q}{\ell} dx$

$dV = \frac{K dq}{x} = \frac{KQ\,dx}{\ell x}$

$V = \int dV = \frac{KQ}{\ell} \int_a^{a+\ell} \frac{dx}{x} = \frac{KQ}{\ell} \ln\left(\frac{a+\ell}{a}\right)$

$$\boxed{V = \frac{KQ}{\ell} \ln\left(\frac{a+\ell}{a}\right)}$$

Electric Potential due to Ring:

① At center

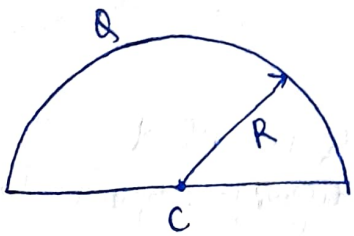
$V_C = \frac{K(dQ_1)}{R} + \frac{K(dQ_2)}{R} + \frac{K(dQ_3)}{R} + \cdots$

$V_C = \frac{K}{R}(dQ_1 + dQ_2 + dQ_3 + \cdots)$

$$\boxed{V_C = \frac{KQ}{R}}$$

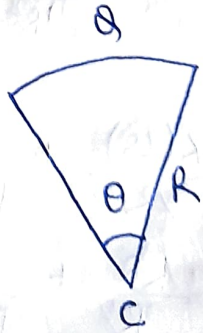
it doesn't matter if the charge distribution is uniform or not since the sum of all charges dQ1 + dQ2 + ... = Q

Ex Semicircular ring



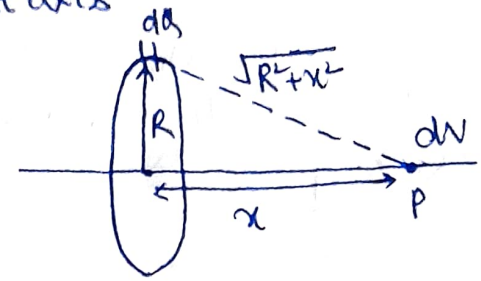
$$\boxed{V_C = \frac{KQ}{R}}$$

Ex Circular arc



$$\boxed{V_C = \frac{KQ}{R}}$$

② At axis

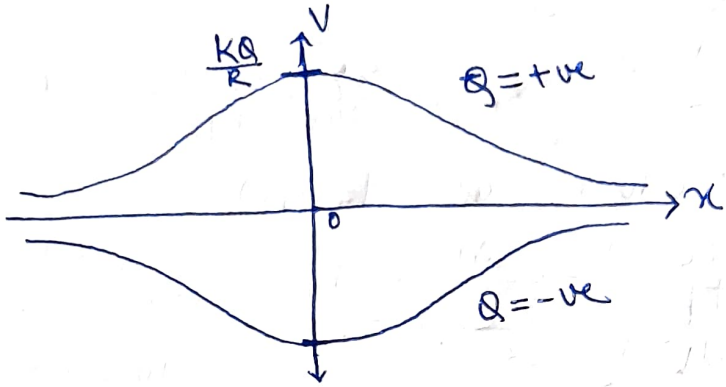


$$dV = \frac{K\,dQ}{\sqrt{R^2+x^2}}$$

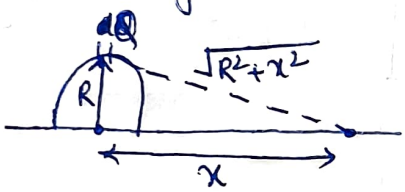
$$V = \int dV = \frac{K}{\sqrt{R^2+x^2}} \int_0^Q dQ = \frac{KQ}{\sqrt{R^2+x^2}}$$

$$\boxed{V = \frac{KQ}{\sqrt{R^2+x^2}}}$$

Graph of V v/s x :

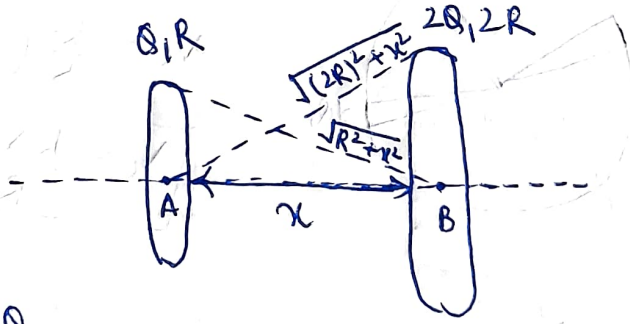


Ex Semi-Circular Ring



$$\boxed{V = \frac{KQ}{\sqrt{R^2+x^2}}}$$

Q Given two coaxial charged rings. Find ① V_A ② V_B ③ $V_A - V_B$



① $V_A = \frac{KQ}{R} + \frac{2KQ}{\sqrt{4R^2+x^2}}$

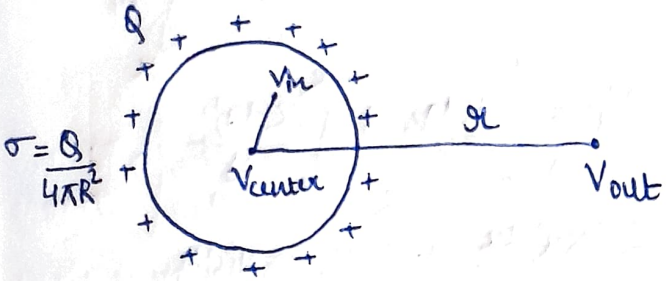
② $V_B = \frac{K(2Q)}{2R} + \frac{KQ}{\sqrt{R^2+x^2}}$

$= \frac{KQ}{R} + \frac{KQ}{\sqrt{R^2+x^2}}$

③ $V_A - V_B = \frac{2KQ}{\sqrt{4R^2+x^2}} - \frac{KQ}{\sqrt{R^2+x^2}}$

$V_A - V_B$ is called potential difference b/w A and B.

Electric Potential due to uniform spherical shell: (Hollow sphere)



① If $r > R \Rightarrow V_{outside} = \frac{KQ}{r}$

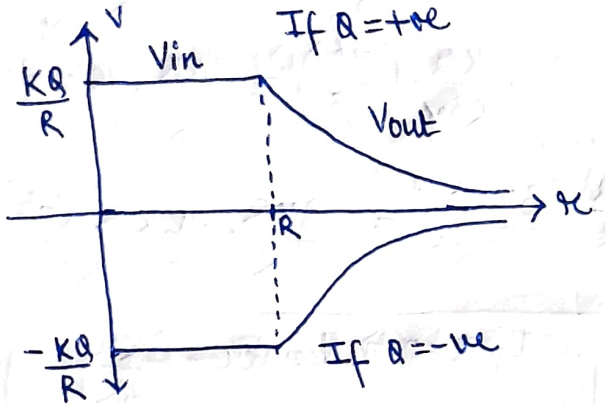
② If $r < R \Rightarrow V_{inside} = \frac{KQ}{R}$

③ If $r = R \Rightarrow V_{surface} = \frac{KQ}{R}$

④ $V_{center} = \frac{KQ}{R}$ (even if charge is non-uniform) placed

→ For outside points shell will behave like point charge at center.

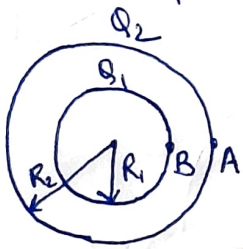
Graph of V v/s r:



$r \le R \Rightarrow V_{in} = \frac{KQ}{R}$

$r \ge R \Rightarrow V_{out} = \frac{KQ}{r}$

Q Given two uniform concentric shells. Find ① V_A ② V_B ③ Potential difference b/w the shells.



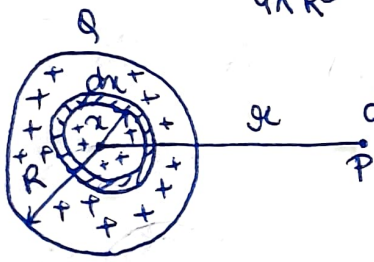
① $V_A = \frac{KQ_2}{R_2} + \frac{KQ_1}{R_2}$

② $V_B = \frac{KQ_2}{R_2} + \frac{KQ_1}{R_1}$

③ $V_A - V_B = KQ_1\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

Electric Potential due to uniform Solid sphere:

Ⓘ Outside $\rho = \frac{3Q}{4\pi R^3}$



$dq = \rho dV$

$= \frac{3Q}{4\pi R^3} \times 4\pi x^2 dx$

$dq = \frac{3Q x^2 dx}{R^3}$

$dV = \frac{Kdq}{r} = \frac{K\,3Q x^2 dx}{R^3 r}$

$V = \int dV = \frac{3KQ}{R^3 r}\int_0^R x^2 dx$

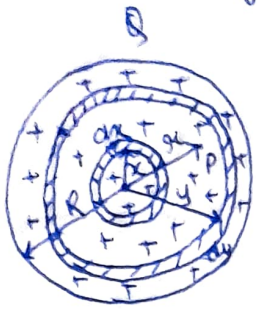
$V = \frac{3KQ}{R^3 r}\left(\frac{R^3}{3}\right)$

$\boxed{V_{out} = \frac{KQ}{r}}$

we could say directly it will behave like a point charge placed at the center

(II) Inside

$\rho = \frac{3Q}{4\pi R^3}$



$$V_p = \int_0^R dV = \int_0^x dV + \int_x^R dV$$

$$V_p = \int_0^x \frac{K dq}{x} + \int_x^R \frac{K dq'}{y}$$

$$V_p = \int_0^x \frac{K\,3Qx^2dx}{R^3 x} + \int_x^R \frac{K\,3Q y^2 dy}{R^3 y}$$

$$dq = \frac{3Qx^2dx}{R^3}$$

$$dq' = \frac{3Qy^2dy}{R^3}$$

$$V_p = \frac{3KQ}{R^3 x}\int_0^x x^2dx + \frac{3KQ}{R^3}\int_x^R y\,dy$$

$$V_p = \frac{3KQ}{R^3 x}\left(\frac{x^3}{3}\right) + \frac{3KQ}{R^3}\left(\frac{R^2-x^2}{2}\right)$$

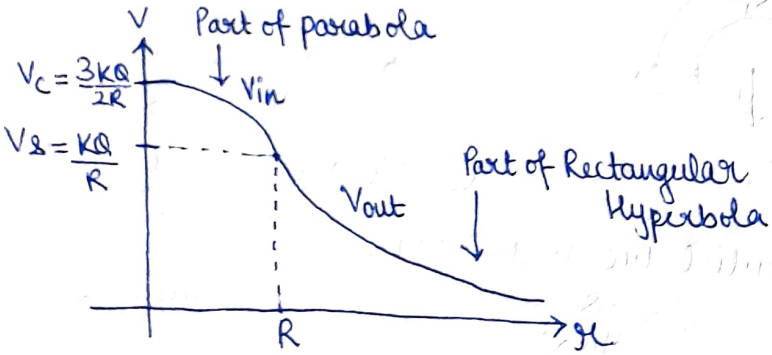
$$V_p = \frac{KQx^2}{R^3} + \frac{3KQ}{2R} - \frac{3KQx^2}{2R^3} = \frac{3KQ}{2R} - \frac{1}{2}\frac{KQx^2}{R^3}$$

$$\boxed{V_{inside} = \frac{KQ}{2R^3}(3R^2 - x^2)}$$

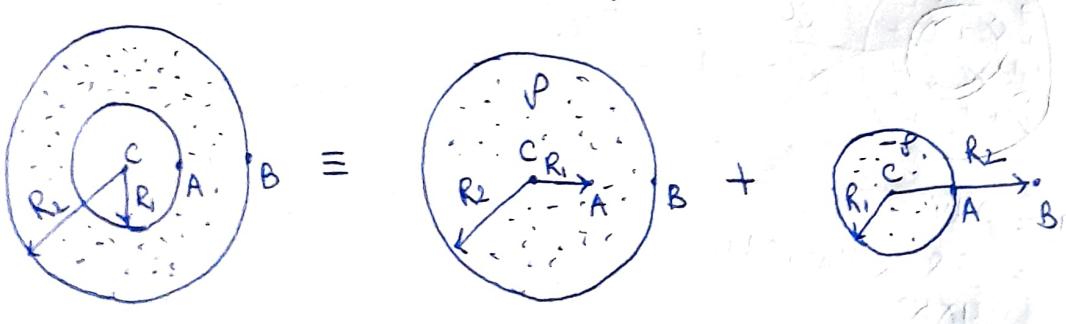
① If $x=0 \Rightarrow V_{center} = \frac{3KQ}{2R}$

② If $x=R \Rightarrow V_{surface} = \frac{KQ}{R}$

Graph of V v/s x:



Q Given solid sphere with concentric cavity. Find ① V_A ② V_B ③ V_C



① $V_A = V_p + V_{-p}$

$$= \frac{K\left(\rho \frac{4}{3}\pi R_2^3\right)}{2R_2^3}\left(3R_2^2 - R_1^2\right) + \frac{K\left(-\rho \frac{4}{3}\pi R_1^3\right)}{R_1}$$

② $V_B = V_p + V_{-p}$

$$= \frac{K\left(\rho \frac{4}{3}\pi R_2^3\right)}{R_2} + \frac{K\left(-\rho \frac{4}{3}\pi R_1^3\right)}{R_2}$$

③ $$V_C = \frac{3}{2}\frac{K\left(\rho \frac{4}{3}\pi R_2^3\right)}{R_2} + \frac{3}{2}\frac{K\left(-\rho \frac{4}{3}\pi R_1^3\right)}{R_1}$$

Electric Potential Difference:

→ Difference b/w electric potential on 2 points

→ Unit: J/C = Volts

→ Dimension = $\frac{MLT^{-2}}{IT} = [MLT^{-3}I^{-1}]$

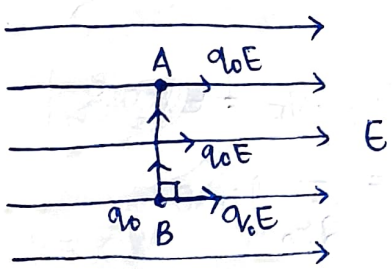
$$V_{BA} = V_B - V_A = \left[\frac{W_{ext}}{q_0}\right]_{\substack{A\to B \\ \Delta K = 0}} = \frac{-(W_{elec})_{A\to B}}{q_0}$$

$$= \frac{(W_{elec})_{B\to A}}{q_0}$$

$$V_B - V_A = -\frac{\int_{\vec{r}_A}^{\vec{r}_B} q_0 \vec{E}\cdot d\vec{r}}{q_0} = -\int_{\vec{r}_A}^{\vec{r}_B} \vec{E}\cdot d\vec{r} = \int_{\vec{r}_B}^{\vec{r}_A} \vec{E}\cdot d\vec{r}$$

Electric Potential difference due to uniform electric field:

Case I:

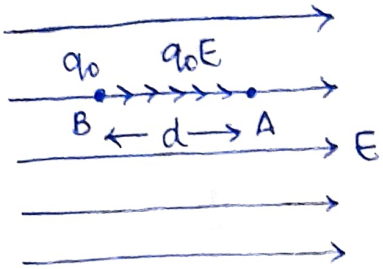


$$V_B - V_A = \frac{(W_{elec})_{B\to A}}{q_0}$$

$$dW = q_0 E\, ds \cos 90^\circ = 0$$

$$\therefore \boxed{V_B - V_A = 0}$$

Case 2:



$$V_{BA} = V_B - V_A = \frac{(W_{elec})_{B\to A}}{q_0}$$

$$= \frac{q_0 E d\cos 0^\circ}{q_0} = Ed$$

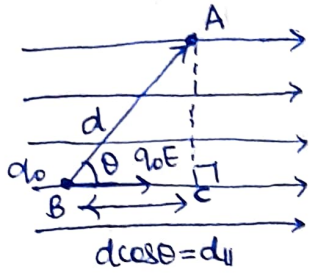
$$\therefore \boxed{V_B - V_A = Ed}$$

$$\boxed{E = \frac{\Delta V}{d}} \rightarrow \text{Potential Gradient}$$

$d \Rightarrow$ displacement along E.F

* Electric field is always from Higher to lower V.
* V will ↑ as we move opposite to E.F.

Case 3:



$$V_{BA} = V_B - V_A = \frac{(W_{elec})_{B\to A}}{q_0} = \frac{q_0 E d\cos\theta}{q_0}$$

$$\therefore \boxed{V_B - V_A = E(d\cos\theta)}$$

We know A and C has same potential (By case ①):

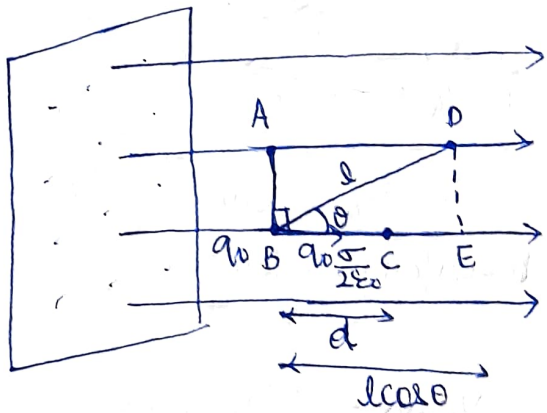
$$V_B - V_A = V_B - V_C$$
$$= E(d\cos\theta)$$

$$\boxed{V_{BA} = E d_{\parallel}}$$

$$\boxed{E = \frac{\Delta V}{d_{\parallel}}}$$

$$\boxed{\begin{aligned} V_B - V_A &= E d\cos\theta \\ &= \vec{E}\cdot\vec{d} \\ &= \vec{E}\cdot\vec{BA} \\ &= -\vec{E}\cdot\vec{AB} \end{aligned}}$$

Electric Potential difference due to uniform large sheet:



$$E = \frac{\sigma}{2\varepsilon_0} = \frac{Q}{2A\varepsilon_0}$$

① $V_A = V_B$

$V_A - V_B = 0$

② $V_B - V_C = Ed$

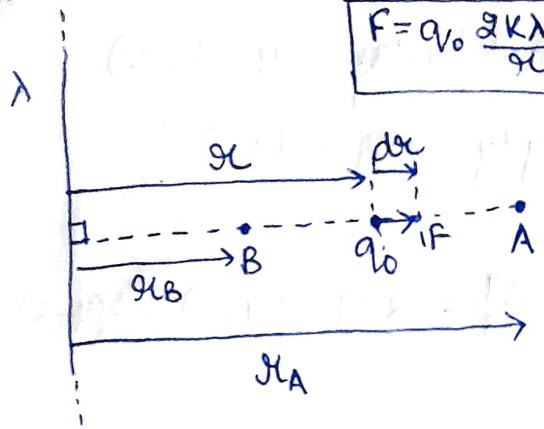
$= \frac{\sigma}{2\varepsilon_0} d$

③ $V_B - V_D = V_B - V_E = El\cos\theta$

$V_B - V_D = \frac{\sigma}{2\varepsilon_0} l\cos\theta$

Electric Potential difference due to uniform large wire:

$$F = q_0 \frac{2K\lambda}{r}$$



$$V_B - V_A = \frac{(W_{elec})_{B\to A}}{q_0}$$

$$W_{BA} = \int dW = \int_{r_A} \frac{2K\lambda}{r} q_0 \, dr \cos 0^\circ$$

$$= 2K\lambda q_0 \int_{r_B} \frac{dr}{r} = 2K\lambda q_0 \ln \frac{r_A}{r_B}$$

$$V_{BA} = \frac{W_{BA}}{q_0} = \frac{2K\lambda q_0 \ln \frac{r_A}{r_B}}{q_0}$$

$$\boxed{V_B - V_A = 2K\lambda \ln\left(\frac{r_A}{r_B}\right)}$$

Relation between electric field and electric potential:

$$dV = \frac{-dW_{elec}}{q_0} = -\frac{q_0 \vec{E}\cdot d\vec{r}}{q_0} = -\vec{E}\cdot d\vec{r}$$

$$dV = -\vec{E}\cdot d\vec{r} = -(E_x\hat{i} + E_y\hat{j} + E_z\hat{k})\cdot(dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$= -E_x dx - E_y dy - E_z dz$$

Now, if $y, z = \text{const}$
$\Rightarrow dy = dz = 0$

$$E_x = -\left(\frac{dV}{dx}\right)_{Y,Z=\text{const.}} = -\frac{\partial V}{\partial x}$$

If $x, z = \text{const}$
$\Rightarrow dx = dz = 0$

$$E_y = -\left(\frac{dV}{dy}\right)_{X,Z=\text{const.}} = -\frac{\partial V}{\partial y}$$

If $x, y = \text{const}$
$\Rightarrow dx = dy = 0$

$$E_z = -\left(\frac{dV}{dz}\right)_{X,Y=\text{const.}} = -\frac{\partial V}{\partial z}$$

$$\boxed{\vec{E} = -\frac{\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j} - \frac{\partial V}{\partial z}\hat{k}}$$

$$\vec{E} = -\left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)V$$

($\nabla \rightarrow$ del operator)

$$\boxed{\vec{E} = -(\text{gradient})V = -\nabla V}$$

Q $V = x^2 - yz$. Find $\vec{E}$ at $(1,2,3)$.

$$E_x = -\frac{\partial V}{\partial x} = -(2x - 0) = -2x$$

$$E_y = -\frac{\partial V}{\partial y} = -(0 - z) = z$$

$$E_z = -\frac{\partial V}{\partial z} = -(0 - y) = y$$

$$\vec{E} = -2x\hat{i} + z\hat{j} + y\hat{k}$$

$$\boxed{\vec{E}_{(1,2,3)} = -2\hat{i} + 3\hat{j} + 2\hat{k}}$$

Q $\vec{E} = -3x^2\hat{i} + 2y\hat{j} + 3\hat{k}$. If V at (1,2,3) is 10V. Then find V at (2,3,4)

$$dV = -\vec{E}\cdot d\vec{x} = -(-3x^2\hat{i} + 2y\hat{j} + 3\hat{k})\cdot(dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$\int_{10}^{V} dV = \int_1^2 3x^2 dx - \int_2^3 2y\,dy - \int_3^4 3\,dz$$

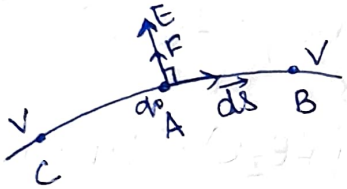
$$\Rightarrow \quad V - 10 = (x^3)_1^2 - (y^2)_2^3 - 3(z)_3^4 = (8-1) - (9-4) - 3(4-3)$$

$$\Rightarrow \quad V - 10 = 7 - 5 - 3 = -1$$

$$\boxed{V = 9 \text{ Volts}}$$

Equipotential Surface:

→ A surface (real or imaginary) on which electric potential at every point is same is known as equipotential surface.



$$\boxed{V_A = V_B = V_C = V}$$

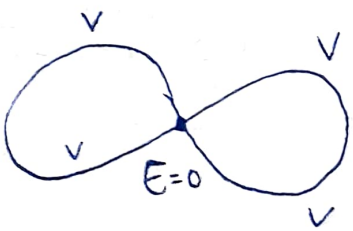
→ Work done by electric field to move a charge on equipotential surface is zero.

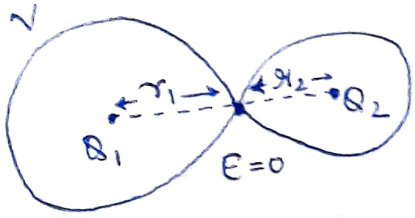
→ Work done by external agent against electric forces to move a charge on equipotential surface is zero.

→ For work done to be zero, either electric field must be zero at equipotential surface or electric field is ⊥ to equipotential surface.

→ Two different equipotential surface will never intersect each other.

→ An equipotential surface can intersect itself, but electric field must be zero at intersection point.





Given an equipotential surface
Find separation b/w Q_1 and Q_2.

$r = r_1 + r_2$ — ①

At intersection point, $E=0$ ⇒ E due to both charges are equal and opp.

$$\Rightarrow \frac{KQ_1}{r_1^2} = \frac{KQ_2}{r_2^2} \Rightarrow \boxed{\frac{r_2}{r_1} = \sqrt{\frac{Q_2}{Q_1}}} - ②$$

Also potential is same at each point.

$$V = \frac{KQ_1}{r_1} + \frac{KQ_2}{r_2} = \frac{KQ_1}{r_1} + \frac{KQ_2}{r_1\sqrt{\frac{Q_2}{Q_1}}} = \frac{KQ_1}{r_1} + \frac{K\sqrt{Q_1Q_2}}{r_1} = \frac{K(Q_1+\sqrt{Q_1Q_2})}{r_1}$$

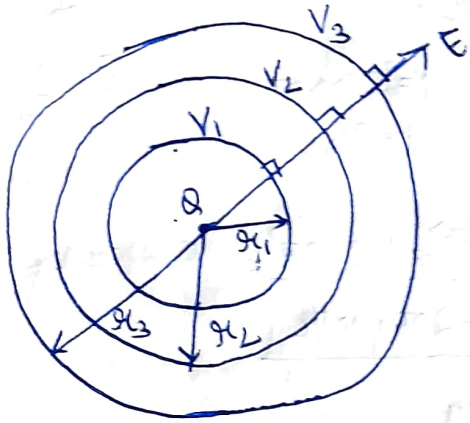
$$\boxed{r_1 = \frac{K(Q_1+\sqrt{Q_1Q_2})}{V}}$$ Similarly $$\boxed{r_2 = \frac{K(Q_2+\sqrt{Q_1Q_2})}{V}}$$

Sep b/w $Q_1Q_2 = r = r_1 + r_2 = \frac{K}{V}\left(Q_1+Q_2+2\sqrt{Q_1Q_2}\right) \Rightarrow \boxed{r = \frac{K}{V^2}\left(\sqrt{Q_1}+\sqrt{Q_2}\right)^2}$

Some important equipotential surfaces:

① Point Charge:

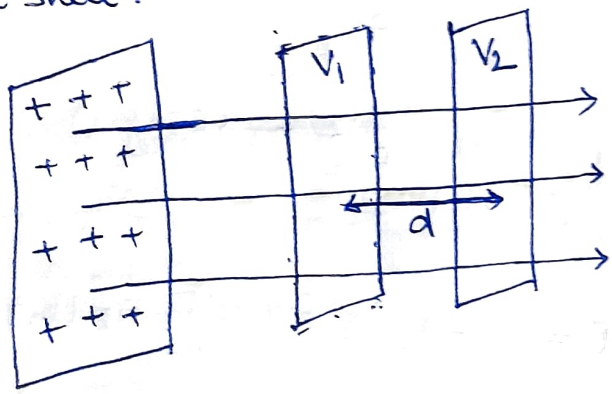
$V = \frac{KQ}{r} = \text{const}$



★ Equipotential surface is imaginary spherical surfaces centred at Q.

★ $V_1 > V_2 > V_3$

② Large Sheet:

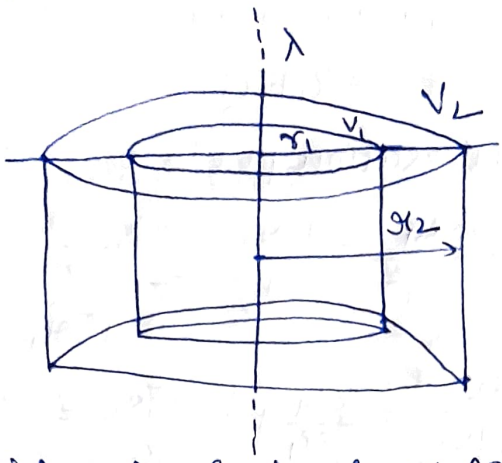


$$E = \frac{\sigma}{2\varepsilon_0}$$

$$\boxed{V_1 - V_2 = Ed = \frac{\sigma}{2\varepsilon_0}d}$$

→ Equipotential surface is parallel planes

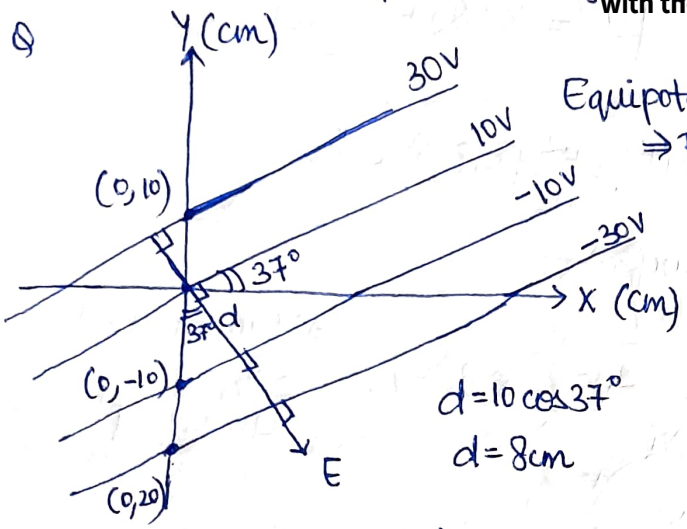
③ Infinite Wire:



$$V_1 - V_2 = 2k\lambda \ln\frac{r_2}{r_1} = \frac{\lambda}{2\pi\varepsilon_0}\ln\frac{r_2}{r_1}$$

→ Equipotential surface is imaginary coaxial cylinders with the infinite wire as the axis of the cylinder.

Q



Equipotential surfaces ⇒ parallel planes.

Find $\vec{E}$ in the region

First we draw ELOF which must be ⊥ to surface.

$d = 10\cos 37°$

$d = 8\,cm$

$$E = \frac{\Delta V}{d} = \frac{10-(-10)\,V}{0.08\,m} = \frac{20\,V}{0.08\,m}$$

$$\boxed{E = 250\ V/m}$$

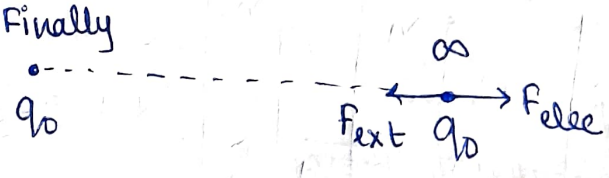
$$\vec{E} = E\sin 37°\hat{i} + E\cos 37°(-\hat{j}) = 250\left(\frac{3}{5}\right)\hat{i} - 250\left(\frac{4}{5}\right)\hat{j}$$

$$\boxed{\vec{E} = (150\hat{i} - 200\hat{j})\ N/C}$$

Electric Potential Energy:

System of Charge

Finally



Slowly

$$(W_{ext})_{\infty \to Here} = \Delta K + \Delta U = U_f - U_i \quad (\text{for reference } U_\infty = 0)$$

Note: Next can be stored as KE and PE but we bring charge from ∞ slowly so all the Next is stored in the form of potential energy

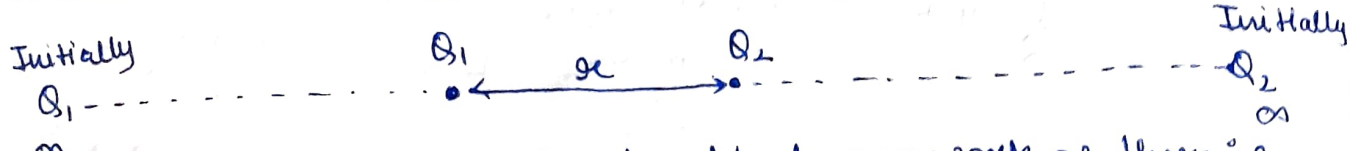
$$U_f = [(W_{ext})_{\infty \to Here}]_{\Delta K \to 0} = (-W_{elec})_{\infty \to Here} = (W_{elec})_{Here \to \infty}$$

$$\therefore \boxed{U_f - U_i = [(W_{ext})_{\infty \to Here}]_{\Delta K \to 0}}$$

→ Unit : Joule
→ $1eV = 1.6 \times 10^{-19} J$
→ Dimension = $[ML^2T^{-2}]$

→ Scalar
→ can be +ve, -ve or zero
→ We assume $U_\infty = 0$

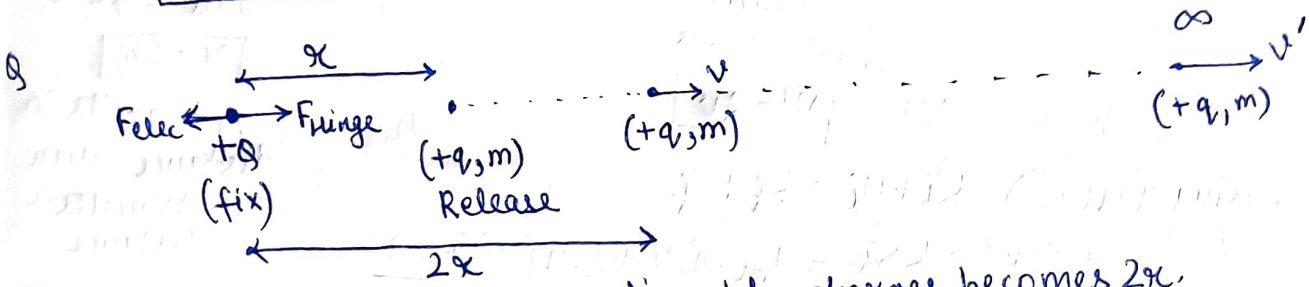
Electric Potential Energy of two point charges:



For bringing Q_1 from ∞ we do not need to do any work as there is no other charge at the final position. But when we bring Q_2 we need to work against electrostatic forces of Q_1.

$$U_{system} = 0 + V_1 Q_2 = 0 + \left(\frac{KQ_1}{r}\right) Q_2$$

* $\boxed{U = \frac{KQ_1Q_2}{r}}$ (Put Q_1, Q_2 with sign)



① Find speed of +q when separation b/w charges becomes $2r$.
We cannot use M/C as $F_{singe} = F_{ext}$ but as $W_{singe} = 0$ we can use TMEC

Using TMEC ⇒ $K_i + U_i = K_f + U_f$

$$0 + 0 + \frac{KQq}{r} = 0 + \frac{1}{2}mv^2 + \frac{KQq}{2r}$$

$$= \frac{KQq}{2r} = \frac{1}{2}mv^2$$

$$\boxed{v = \sqrt{\frac{KQq}{mr}}}$$

② Find speed of +q when separation b/w charges becomes very large.

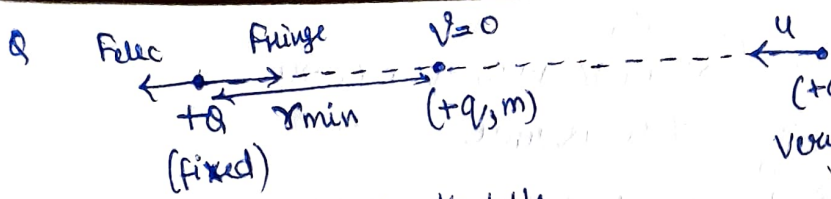
Using TMEC ⇒ $K_i + U_i = K_f + U_f$

$$= 0 + 0 + \frac{KQq}{r} = 0 + \frac{1}{2}mv'^2 + 0$$

$$= \frac{KQq}{r} = \frac{1}{2}mv'^2$$

$$= \boxed{v' = \sqrt{\frac{2KQq}{mr}}}$$

Q



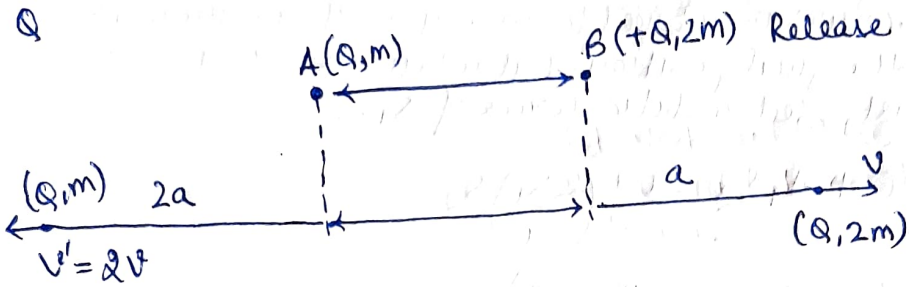
Find min^m separation b/w charges.

Using TMEC ⇒ $K_i + U_i = K_f + U_f$

$$\frac{1}{2}mu^2 + 0 = 0 + \frac{KQq}{r_{min}}$$

$$\boxed{r_{min} = \frac{2KQq}{mu^2}}$$

Q



Find speed of particle B when it travelled distance a.

As $F_{ext} = 0$

$a_{com} = 0 \Rightarrow x_{com} = 0$

$\Rightarrow m_1x_1 = m_2x_2$

$2m(a) = m(x)$

$\boxed{x = 2a}$

∴ A travells 2a distance when B travells a distance.

M/C ⇒ $P_i = P_f$

$0 = 2mv - mv'$

$mv' = 2mv$

$\boxed{v' = 2v}$

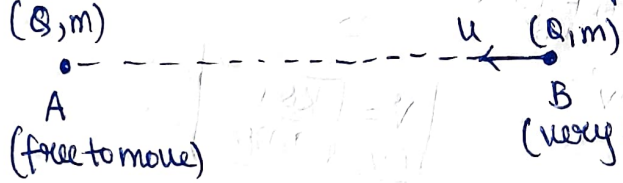
Now, using TMEC ⇒ $K_i + U_i = K_f + U_f$

$$= 0 + 0 + \frac{KQ^2}{a} = \frac{1}{2}(2m)v^2 + \frac{1}{2}m(2v)^2 + \frac{KQ^2}{4a}$$

$$= \frac{3KQ^2}{4a} = 3mv^2$$

$$\boxed{v = \sqrt{\frac{KQ^2}{4ma}}}$$

Q



Find min^m separation b/w charges

r_{min} will be when both will have same speed (say v)

using M/C ⇒ $0 + mu = mv + mv \Rightarrow \boxed{v = \frac{u}{2}}$ — (1)

Using TMEC ⇒ $K_i + U_i = K_f + U_f$

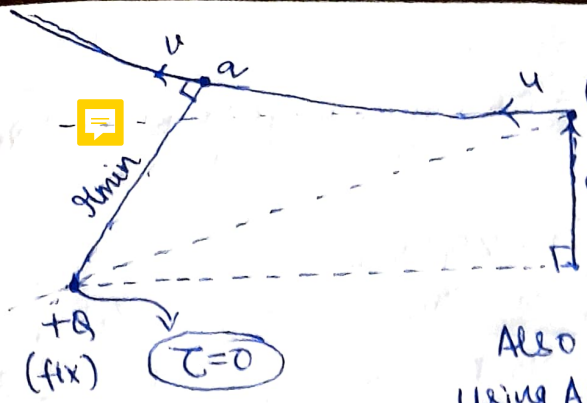
$$= 0 + \frac{1}{2}mu^2 + 0 = 2 \times \frac{1}{2}mv^2 + \frac{KQ^2}{r_{min}}$$

$$= \frac{1}{2}mu^2 = \frac{mu^2}{4} + \frac{KQ^2}{r_{min}}$$

$$= \frac{1}{4}mu^2 = \frac{KQ^2}{r_{min}}$$

$$\boxed{r_{min} = \frac{4KQ^2}{mu^2}}$$

Q



Find min^m separation b/w charges.

Given $KQq = mu^2d$

Using TMEC = $K_i + U_i = K_f + U_f$

$= \frac{1}{2}mu^2 + 0 = \frac{1}{2}mv^2 + \frac{KQq}{r_{min}}$ — ①

Also $\tau = 0$ about pt +Q.

Using AMCT $\Rightarrow L_i = L_f$

$mud = mv\, r_{min}$

$\boxed{v = \frac{ud}{r_{min}}}$ — ②

Put ② in ①

$= \frac{1}{2}mu^2 = \frac{1}{2}m\frac{u^2d^2}{r_{min}^2} + \frac{KQq}{r_{min}}$

$= mu^2 = \frac{mu^2d^2}{r_{min}^2} + \frac{2KQq}{r_{min}} \Rightarrow mu^2r_{min}^2 = mu^2d^2 + 2KQq\, r_{min}$

$= mu^2r_{min}^2 - 2KQq\, r_{min} - mu^2d^2 = 0$

$= mu^2r_{min}^2 - 2mu^2d\, r_{min} - mu^2d^2 = 0 \quad \{KQq = mu^2d\}$

$= r_{min}^2 - 2d\, r_{min} - d^2 = 0$

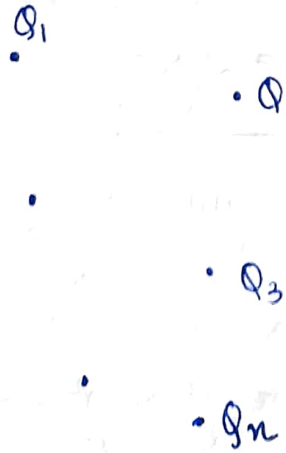
$r_{min} = \frac{2d \pm \sqrt{4d^2+4d^2}}{2} = \frac{2d \pm 2\sqrt{2}d}{2} = \frac{d \pm \sqrt{2}d}{2}$

r_{min} cannot be -ve $\Rightarrow d - \sqrt{2}d$ is rejected

$\boxed{r_{min} = d + \sqrt{2}d}$

Electric Potential energy of a system of charges:

n-charged particles

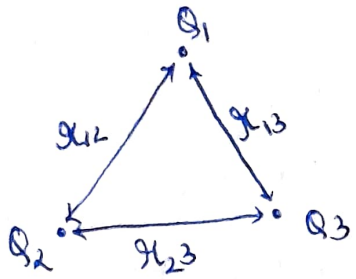


$U_{system} = (W_{ext})_{\infty \to Here}$

$= (U_{12} + U_{13} + \cdots + U_{1n}) + (U_{23} + U_{24} + \cdots + U_{2n}) + (U_{34} + U_{35} + \cdots + U_{3n}) + \cdots + (U_{n-1,n})$

$\boxed{\text{Total number of terms} = {}^nC_2 = \frac{n(n-1)}{2}}$

3 point Charges:



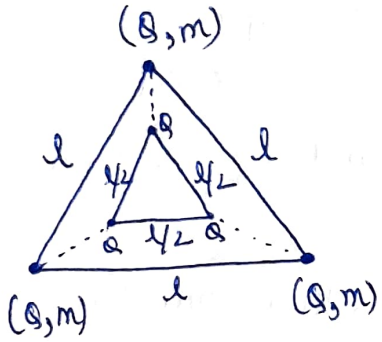
$U = (W_{ext})_{\infty \to Here}$

$= 0 + \frac{KQ_1Q_2}{r_{12}} + \left\{\frac{KQ_1}{r_{13}} + \frac{KQ_2}{r_{23}}\right\} Q_3$

$$\boxed{U = \frac{KQ_1Q_2}{r_{12}} + \frac{KQ_1Q_3}{r_{13}} + \frac{KQ_2Q_3}{r_{23}}}$$

No of terms $= {}^3C_2 = \frac{3!}{2!(3-2)!} = 3$

Q



Find Wext and Welec to bring these charges on an equilateral triangle of side $l/2$ slowly.

$W_{ext} = \Delta TME = \Delta U + \Delta K^{\nearrow 0}$

$\therefore W_{ext} = U_f - U_i$

$= \left(\frac{3KQ^2}{l/2}\right) - \left(\frac{3KQ^2}{l}\right)$

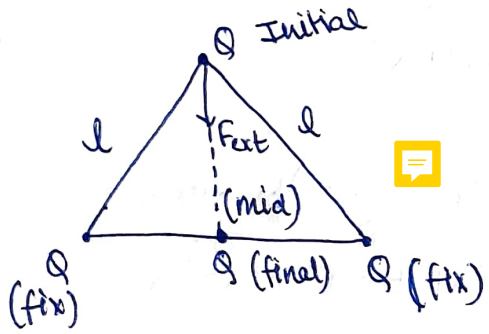
$$\boxed{W_{ext} = \frac{3KQ^2}{l}}$$

$W_{elec} = -W_{ext} = -\Delta U$

$= -(U_f - U_i)$

$$\boxed{W_{elec} = -\frac{3KQ^2}{l}}$$

Q



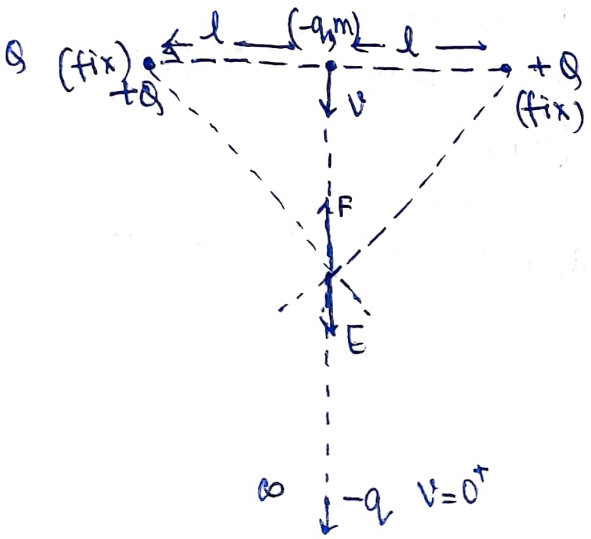
Find Wext and Welec to shift the charge Q from initial to final position.

$W_{ext} = U_f - U_i = \left\{2\frac{KQ^2}{l/2} + \frac{KQ^2}{l}\right\} - \left\{3\frac{KQ^2}{l}\right\}$

$$\boxed{W_{ext} = \frac{2KQ^2}{l}}$$

$$\boxed{W_{elec} = -\frac{2KQ^2}{l}}$$

Q



Find min^m value of v so that -q gets out of electric field due to +Q and never comes back.

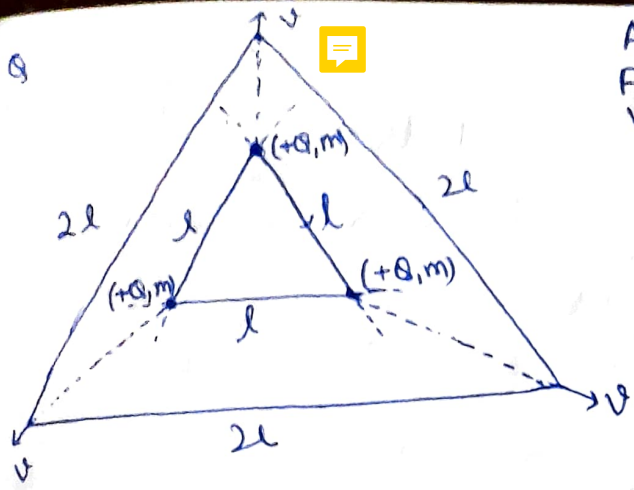
Using TMEC $\Rightarrow$ $K_i + U_i = K_f + U_f$

$= \frac{1}{2}mv^2 + 2\times -\frac{KQq}{l} + \frac{KQ^2}{2l} = 0 + \frac{KQ^2}{2l}$

$= \frac{1}{2}mv^2 = \frac{2KQq}{l}$

$$\boxed{v = \sqrt{\frac{4KQq}{ml}}}$$

escape velocity



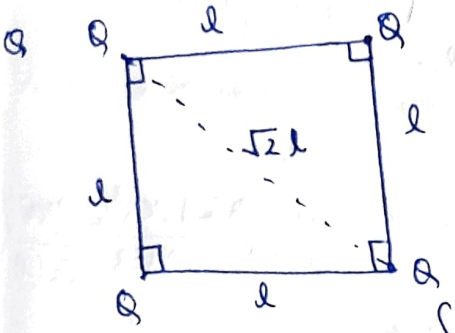
All charges are released initially. Find their speeds when they are at vertices of a Δ whose side is 2l.

Using TMEC → $K_i + U_i = K_f + U_f$

$$\Rightarrow 0 + \frac{3KQ^2}{l} = 3 \times \frac{1}{2}mv^2 + 3\frac{KQ^2}{2l}$$

$$= \frac{3KQ^2}{2l} = \frac{3}{2}mv^2$$

$$\boxed{V = \sqrt{\frac{KQ^2}{ml}}}$$



Find $U_{electric}$ of this system.

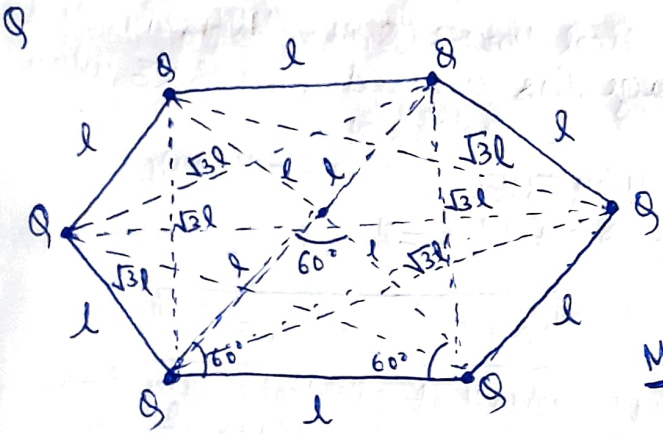
No. of terms = ${}^4C_2 = \frac{4!}{2!(4-2)!} = 6$

$$U_{sys} = \frac{4KQ^2}{l} + \frac{2KQ^2}{\sqrt{2}l}$$

Trick for Symmetrical System: Take any charge and write U due to it and then multiply it by no. of charges/2

By trick:

$$U = Q\left(\frac{2KQ}{l} + \frac{KQ}{\sqrt{2}l}\right) \times \frac{4}{2} = \frac{4KQ^2}{l} + \frac{2KQ^2}{\sqrt{2}l}$$



Find U_{system}

No of terms = ${}^6C_2 = 15$

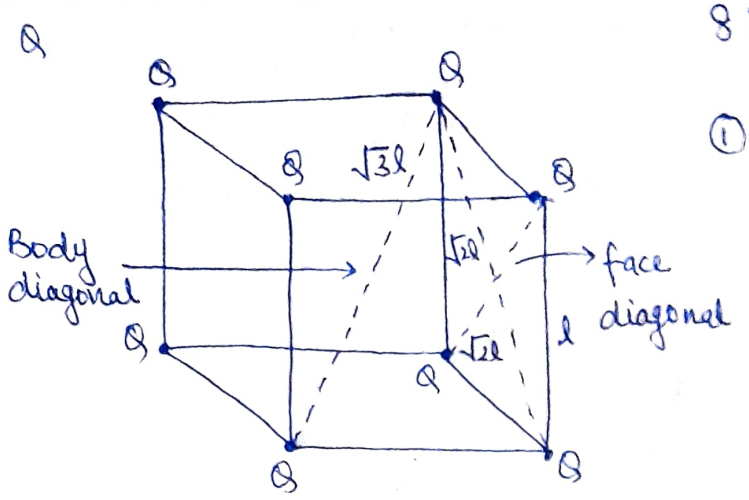
$$U = 6\left(\frac{KQ^2}{l}\right) + 6\left(\frac{KQ^2}{\sqrt{3}l}\right) + 3\left(\frac{KQ^2}{2l}\right)$$

Method 2: Using trick

$$U = Q\left(\frac{2KQ}{l} + \frac{2KQ}{\sqrt{3}l} + \frac{KQ}{2l}\right) \times \frac{6}{2}$$

$$U = \left(\frac{2KQ^2}{l} + \frac{2KQ^2}{\sqrt{3}l} + \frac{KQ^2}{2l}\right) \times 3$$

$$\boxed{U = \frac{6KQ^2}{l} + \frac{6KQ^2}{\sqrt{3}l} + \frac{3KQ^2}{2l}}$$



8 identical charges (Q, m)
Cube : l No of terms = 12

① Find U_{system}.

$$U = 12\left(\frac{KQ^2}{l}\right) + 12\left(\frac{KQ^2}{\sqrt{2}l}\right) + 4\left(\frac{KQ^2}{\sqrt{3}l}\right)$$

② All charges are released simultaneously. Find their speeds when they are at vertices of a cube of side $2l$.

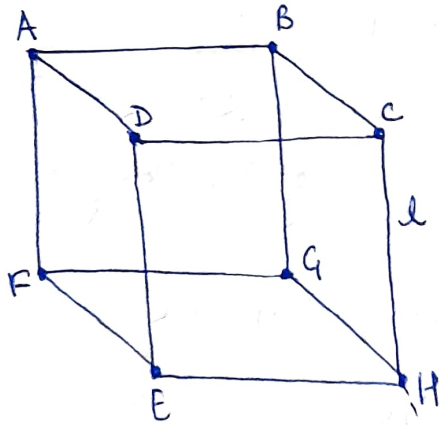
Using TMEC $\Rightarrow K_i + U_i = K_f + U_f$

$$= 0 + 12\frac{KQ^2}{l} + 12\frac{KQ^2}{\sqrt{2}l} + 4\frac{KQ^2}{\sqrt{3}l} = 8\times\frac{1}{2}mv^2 + \left\{\frac{12KQ^2}{2l} + \frac{12KQ^2}{2\sqrt{2}l} + \frac{4KQ^2}{\sqrt{3}(2l)}\right.$$

$$\Rightarrow \frac{1}{2}\left\{\frac{12KQ^2}{l} + \frac{12KQ^2}{\sqrt{2}l} + \frac{4KQ^2}{\sqrt{3}l}\right\} = 4mv^2$$

$$\boxed{v = \sqrt{\frac{1}{8m}\left\{\frac{12KQ^2}{l} + \frac{12KQ^2}{\sqrt{2}l} + \frac{4KQ^2}{\sqrt{3}l}\right\}}}$$

③



8 identical charges (Q, m). All fixed initial charge H is released. Find its KE when it reaches ∞

Using TMEC $\Rightarrow K_i + U_i = K_f + U_f$

$\rightarrow 0 + U_{28} = K_H + U_{21}$

$$\boxed{K_H = U_{28} - U_{21} = U_7}$$

We know that potential energy of the system is converted to KE. When H is released K_H at ∞ will only be because of 7 terms of the 28 in which charge H was involved

$$\boxed{U = \frac{3KQ^2}{l} + \frac{3KQ^2}{\sqrt{2}l} + \frac{KQ^2}{\sqrt{3}l} = K_H}$$

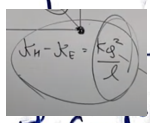
④ Now charge E is released. Find its KE when it reaches ∞.

Using TMEC ⇒ $K_i + U_i = K_f + U_f$

$= 0 + U_{21} = K_E + U_{15}$

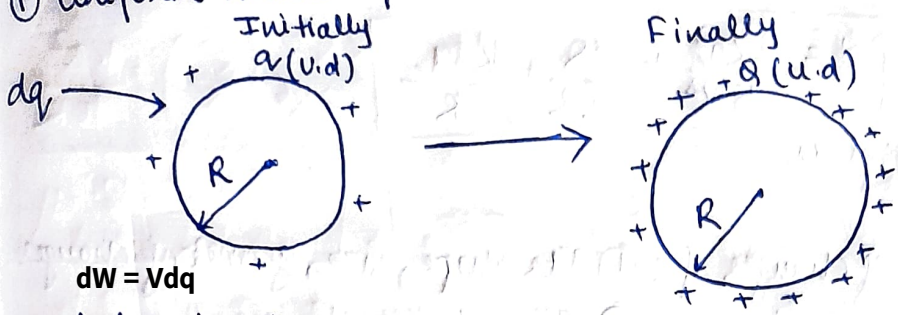
$$K_E = U_{21} - U_{15} = U_6$$

$$K_E = \frac{2KQ^2}{\ell} + \frac{3KQ^2}{\sqrt{2}\ell} + \frac{KQ^2}{\sqrt{3}\ell}$$



Self energy (Electric Potential Energy of Continuous Bodies):

① Uniform Hollow Sphere:



$U = (W_{ext})_{\infty \to Here}$

dW = Vdq

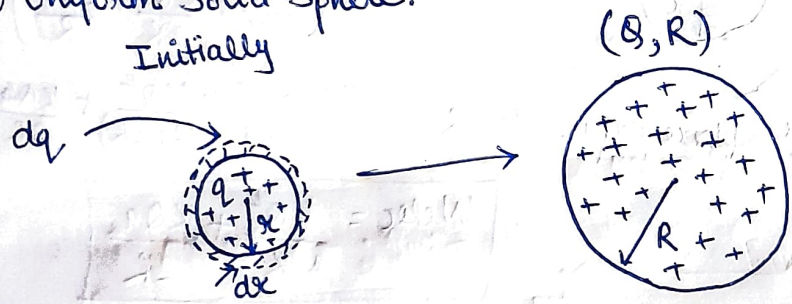
$$dW = \frac{Kq\,dq}{R}$$

$$W_{ext} = \int dW = \frac{K}{R}\int_0^Q q\,dq = \frac{K}{R}\left(\frac{q^2}{2}\right)_0^Q = \frac{KQ^2}{2R}$$

$$U_{self} = \frac{KQ^2}{2R}$$

If Q = +ve or Q = −ve ⇒ U_{self} = +ve

② Uniform Solid Sphere:



$U = (W_{ext})_{\infty \to Here}$

$$dW = Vdq = \frac{Kq}{x}dq$$

$$\rho = \frac{3Q}{4\pi R^3}$$

$$q = \rho\,\frac{4}{3}\pi x^3 = \frac{Q}{R^3}x^3$$

$$dq = \frac{Q}{R^3}3x^2\,dx$$

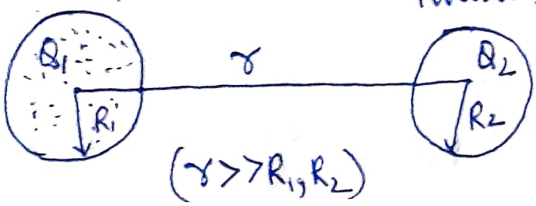
$$W_{ext} = \int dW = K\int \frac{q}{x}dq = K\int \frac{\left(\frac{Q}{R^3}x^3\right)\left(\frac{Q}{R^3}3x^2dx\right)}{x} = \frac{3KQ^2}{R^6}\int_0^R x^4\,dx$$

$$= \frac{3KQ^2}{R^6}\left(\frac{x^5}{5}\right)_0^R = \frac{3KQ^2}{R^6}\left(\frac{R^5}{5}\right) = \frac{3}{5}\frac{KQ^2}{R}$$

$$U_{self} = \frac{3KQ^2}{5R}$$

Also, $(U_{self})_{SS} > (U_{self})_{HS}$

Q Solid Sphere Hollow Sphere Find U_{system}.

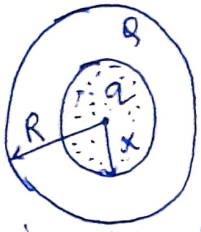


$(r >> R_1, R_2)$

$U = (U_{self})_{SS} + (U_{self})_{HS} + U_{inter.}$

$$U = \frac{3KQ_1^2}{5R_1} + \frac{KQ_2^2}{2R_2} + \frac{KQ_1Q_2}{r}$$

Q



Find U_{system}

$U = (U_{self})_{SS} + (U_{self})_{HS} + U_{interaction}$

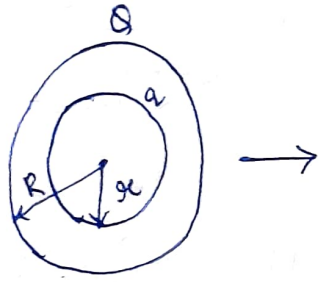
applying $U=qV$ for interaction energy

$$U = \frac{3Kq^2}{5x} + \frac{KQ}{2R} + \frac{KQq}{R}$$

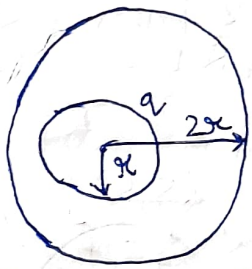
2 concentric solid and Hollow sphere

Note: Self Energy of a body changes if: ① Its shape, size, geometry changes
② Its charge changes.

Q



2 concentric Hollow spheres



Larger sphere expands to 2R

Find W_{elec}.

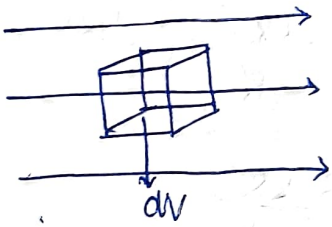
$W_{elec} = U_i - U_f$

$= \left(\frac{Kq^2}{2x} + \frac{KQ^2}{2R} + \frac{KQq}{R}\right)$

$- \left(\frac{Kq^2}{2r} + \frac{KQ^2}{2(2R)} + \frac{KQq}{2R}\right)$

$$W_{elec} = \frac{KQ^2}{4R} + \frac{KQq}{2R}$$

Energy Density: Energy stored due to electric field.



dU = Electric Energy Stored

dV = Volume

$$\text{Energy Density} = \frac{dU}{dV} = \frac{1}{2}\varepsilon E^2 \text{ (in medium)}$$

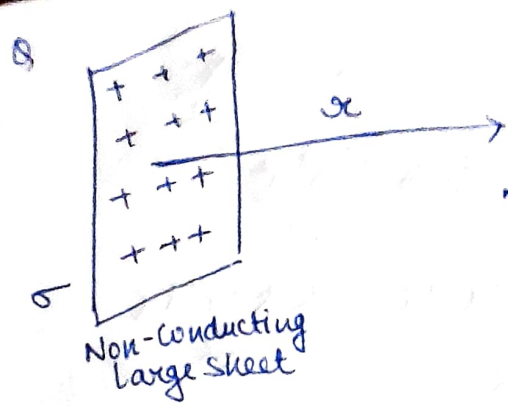
$$\frac{dU}{dV} = \frac{1}{2}\varepsilon_0 E^2 \text{ (in air/vacuum)}$$

→ Unit = Joule/m^3

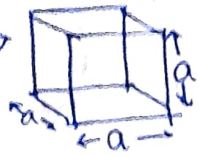
→ Dimension = $\frac{ML^2T^{-2}}{L^3}$

$= [ML^{-1}T^{-2}]$

→ Scalar



$E = \frac{\sigma}{2\varepsilon_0}$

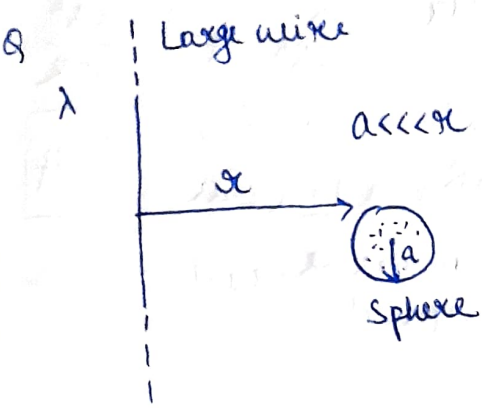


Find EPE stored in this cube due to E.F.

Energy density $= \frac{dU}{dV} = \frac{1}{2}\varepsilon_0\left(\frac{\sigma^2}{4\varepsilon_0^2}\right)$

$$\frac{dU}{dV} = \frac{\sigma^2}{8\varepsilon_0}$$

$$\boxed{U = \left(\frac{\sigma^2}{8\varepsilon_0}\right)a^3} \quad \{U = (E\cdot d)V\}$$



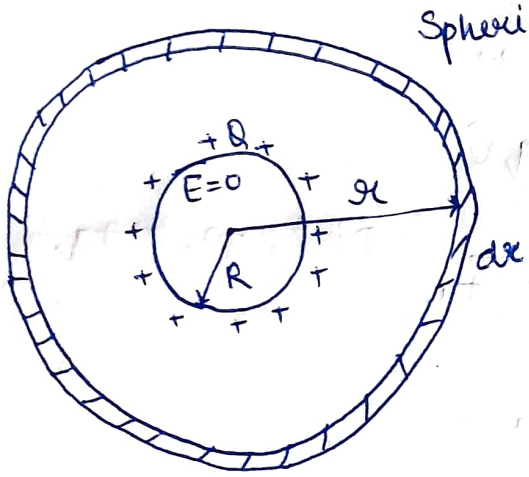
Find EPE stored in this sphere due to E.F.

$$E = \frac{2K\lambda}{x} = \frac{\lambda}{2\pi\varepsilon_0 x}$$

$$\frac{dU}{dV} = \frac{1}{2}\varepsilon_0\frac{\lambda^2}{4\pi^2\varepsilon_0^2 x^2} = \frac{\lambda^2}{8\pi^2\varepsilon_0 x^2}$$

$$\boxed{U = \frac{\lambda^2}{8\pi^2\varepsilon_0 x^2}\left(\frac{4}{3}\pi a^3\right)}$$

Energy due to Electric field of Hollow sphere:



Spherical Element

$$E = \frac{KQ}{x^2}$$

$$\frac{dU}{dV} = \frac{1}{2}\varepsilon_0\left(\frac{K^2Q^2}{x^4}\right)$$

$$dU = \frac{\varepsilon_0 K^2Q^2}{2x^4}\left(4\pi x^2 dx\right)$$

$$dU = \frac{KQ^2dx}{2x^2}$$

$$U = \int dU = \frac{KQ^2}{2}\int_R^\infty \frac{dx}{x^2} = \frac{KQ^2}{2}\left(\frac{-1}{x}\right)_R^\infty$$

$$U = \frac{KQ^2}{2}\left(-\frac{1}{\infty} - \left(\frac{-1}{R}\right)\right) = \frac{KQ^2}{2R}$$

$$\boxed{U = \frac{KQ^2}{2R} = U_{self}}$$

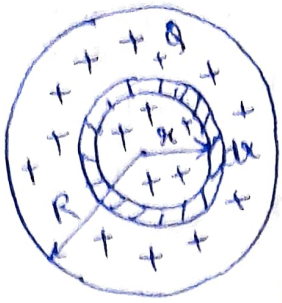
This is also the energy due to electric field of solid sphere at outside of the sphere.

thus, the self energy of a charged body is stored in its electric field

Energy due to electric field of solid sphere:

Inside:

$$U = U_{0\to R} + U_{R\to\infty} = U_{0\to R} + \frac{KQ^2}{2R}$$



$$E = \frac{KQx}{R^3} \Rightarrow \frac{dU}{dV} = \frac{1}{2}\varepsilon_0 \frac{K^2Q^2x^2}{R^6}$$

$$dU = \frac{\varepsilon_0 K^2Q^2x^2}{2R^6} \times 4\pi x^2 dx = \frac{KQ^2x^4dx}{2R^6}$$

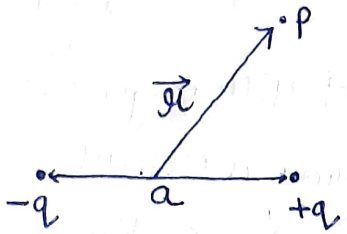
$$U_{0\to R} = \int dU = \frac{KQ^2}{2R^6}\int_0^R x^4dx = \frac{KQ^2}{2R^6}\left(\frac{R^5}{5}\right) = \frac{KQ^2}{10R}$$

$$U = \frac{KQ^2}{10R} + \frac{KQ^2}{2R} = \frac{KQ^2}{10R}(1+5) = \frac{6KQ^2}{10R} = \frac{3KQ^2}{5R}$$

$$\boxed{U = \frac{3KQ^2}{5R} = U_{self}}$$

Electric Dipole:

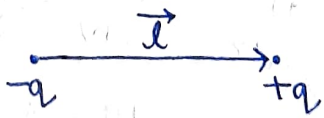
→ It is a neutral system which has two point charges of equal magnitude and opposite signs, placed at very small separation.



a → small

★ $\boxed{a << r}$

Electric dipole moment: $\vec{P} = +q\vec{l}$



Dirⁿ: -q to +q

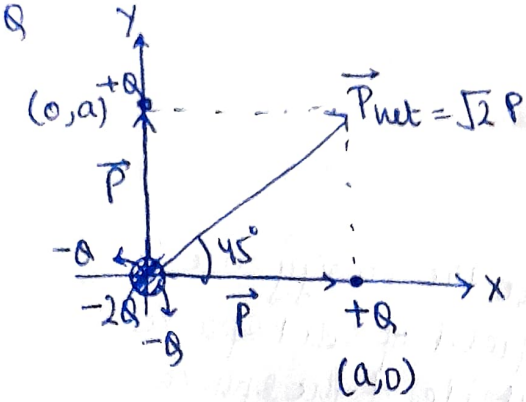
→ Unit = C-m

→ Dimension = [ITL]

→ CGS unit: 1 Debye = 3.3×10^{-30} C-m

→ Vector Quantity

Q



We can break -2Q into -Q and -Q each and create a dipole with each +Q.

Find Electric dipole moment of system

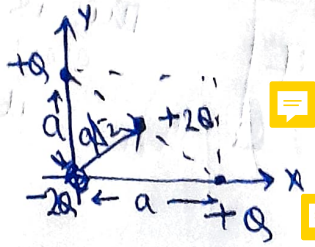
$P = Qa$

$P_{net} = \sqrt{2}P$

$\boxed{P_{net} = \sqrt{2}Qa}$

Method 2:

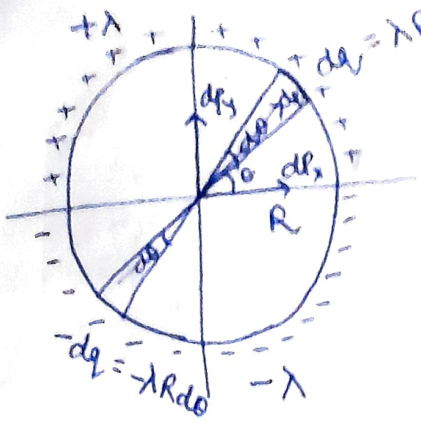
We can also use tricks similar to COM



$P = 2Q \times \frac{a}{\sqrt{2}}$

$\boxed{P = \sqrt{2}Qa}$

Q



Find Electric Dipole moment of this system

$dP = (\lambda R d\theta)\, 2R = 2\lambda R^2 d\theta$

$P_x = \int dP_x = \int dP\cos\theta$

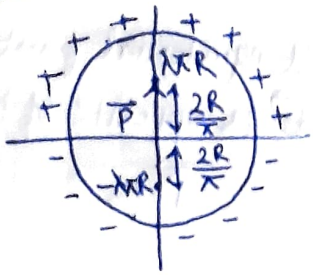
$= 2\lambda R^2 \int_0^{\pi} \cos\theta\, d\theta = 0$

$P_y = \int dP_y = \int dP\sin\theta$

$= 2\lambda R^2 \int_0^{\pi} \sin\theta\, d\theta = 4\pi R^2$

$\therefore \boxed{\vec{P}_{net} = 4\pi R^2 \hat{j}}$

Method 2:



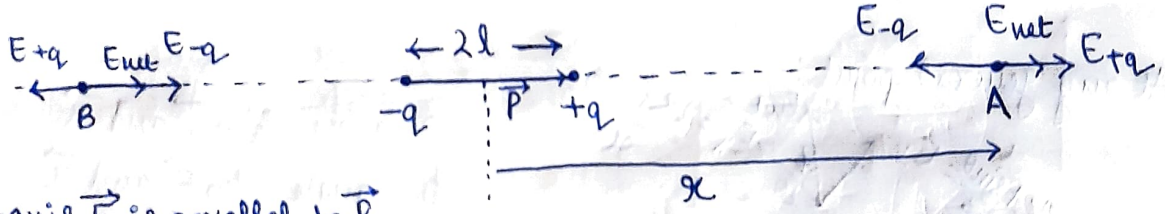
$P = (\lambda\pi R)\left(\frac{4R}{\lambda}\right) = 4\pi R^2$

$\boxed{\vec{P} = 4\pi R^2 \hat{j}}$

Electric Field due to electric dipole :

① At Axis :

$\boxed{P = q(2l)}$

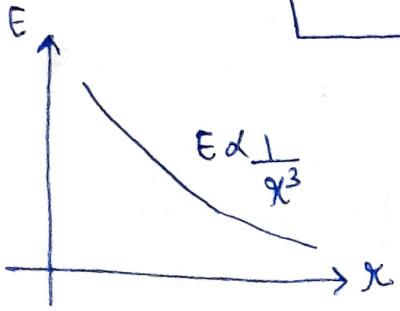


* At axis $\vec{E}$ is parallel to $\vec{P}$

$E_{net} = E_{+q} - E_{-q} = \frac{Kq}{(x-l)^2} - \frac{Kq}{(x+l)^2} = Kq\left\{\frac{1}{(x-l)^2} - \frac{1}{(x+l)^2}\right\}$

$= Kq\left\{\frac{(x+l)^2 - (x-l)^2}{(x^2-l^2)^2}\right\} = Kq\left\{\frac{4xl}{(x^2-l^2)}\right\} = \frac{2kPx}{(x^2-l^2)^2}$ $(P = q(2l))$

as $x >> 2l \Rightarrow$ $\boxed{E_{axis} = \frac{2kP}{x^3}}$ $\boxed{E_{axis} \propto \frac{1}{x^3}}$

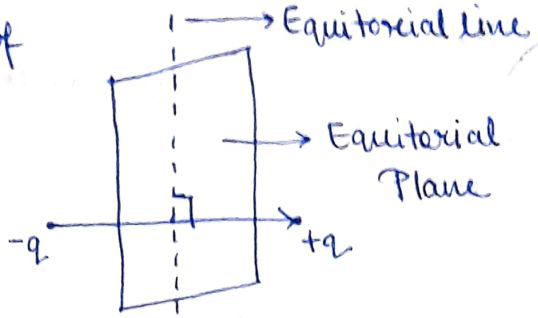


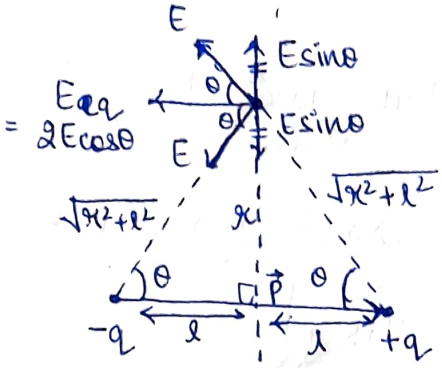
$\boxed{\vec{E}_{axis} = \frac{2kP}{x^3}\hat{P} = \frac{2k\vec{P}}{x^3}}$

② At equitorial point (at Perpendicular Bisector):

There can be ∞ number of ⊥ bisectors, all will lie on equitorial plane.

however, there can only be one axis





$E_{eq} = 2E\cos\theta$

$E = \frac{Kq}{r^2+l^2}$

$P = q(2l)$

★ At equitorial point $\vec{E}$ is opposite to $\vec{P}$.

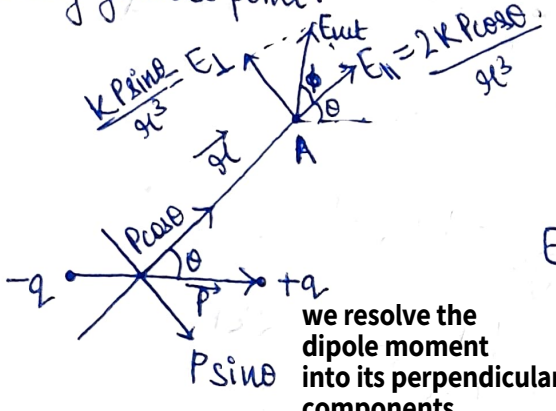
$$E_{eq} = 2E\cos\theta = 2\left(\frac{Kq}{r^2+l^2}\right)\left(\frac{l}{\sqrt{r^2+l^2}}\right) = \frac{KP}{(r^2+l^2)^{3/2}}$$

As $r >>> l \Rightarrow$

$$\boxed{E_{eq} = \frac{KP}{r^3}}$$

$$\boxed{\vec{E}_{eq} = \frac{KP}{r^3}(-\hat{P}) = -\frac{K\vec{P}}{r^3}}$$

③ At any general point:



we resolve the dipole moment into its perpendicular components

θ: angle b/w $\vec{P}$ and $\vec{r}$

ϕ: angle b/w $\vec{E}$ and $\vec{r}$

$$E_{net} = \sqrt{E_\parallel^2 + E_\perp^2}$$

$$= \frac{KP}{r^3}\sqrt{4\cos^2\theta + \sin^2\theta}$$

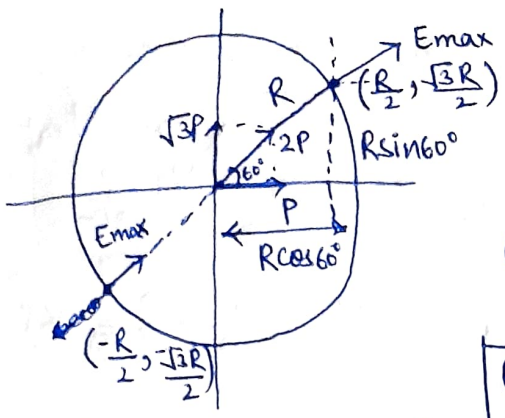
$$\boxed{E_{net} = \frac{KP}{r^3}\sqrt{1+3\cos^2\theta}}$$

$$\tan\phi = \frac{E_\perp}{E_\parallel} = \frac{\sin\theta}{2\cos\theta} = \frac{\tan\theta}{2}$$

$$\boxed{\phi = \tan^{-1}\left(\frac{\tan\theta}{2}\right)}$$

$$\boxed{\text{Angle b/w } \vec{E} \text{ and } \vec{P} = \theta + \phi = \theta + \tan^{-1}\left(\frac{\tan\theta}{2}\right)}$$

Q



Find point on circle at which electric field is maxm. Also find this maxm electric field.

$$P_{net} = \sqrt{P^2 + (\sqrt{3}P)^2} = 2P$$

$$E = \frac{KP_{net}}{R^3}\sqrt{1+3\cos^2\theta}$$ For $E_{max} \Rightarrow \cos\theta = 1$, $\theta = 0°$

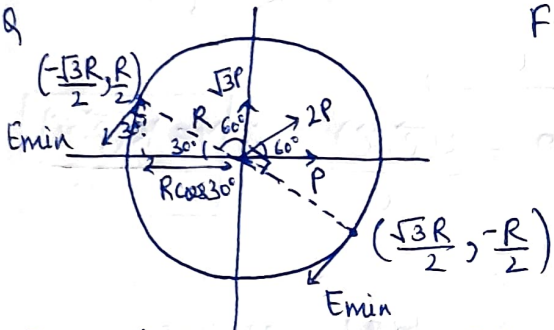
$$E_{max} = \frac{2KP_{net}}{R^3} \equiv \text{Axial Point}$$

$$E_{max} = \frac{2k(2P)}{R^3} = \frac{4KP}{R^3}$$

Pts are $\left(\frac{R}{2}, \frac{\sqrt{3}R}{2}\right)$ and $\left(\frac{-R}{2}, \frac{-\sqrt{3}R}{2}\right)$

$$\vec{E_{max}} = E_{max}\cos 60°\,\hat{i} + E_{max}\sin 60°\,\hat{j}$$
$$= \frac{2KP}{R^3}\hat{i} + \frac{2\sqrt{3}KP}{R^3}\hat{j}$$

Q



Find point on circle at which electric field is minm. Also find this minm electric field.

$$P_{net} = \sqrt{P^2 + (\sqrt{3}P)^2} = 2P$$

$$E = \frac{KP_{net}}{R^3}\sqrt{1+3\cos^2\theta}$$ For Emin $\Rightarrow \cos\theta = 0$, $\theta = \frac{\pi}{2}$

$$E_{min} = \frac{KP_{net}}{R^3} \equiv \text{Equitorial Point}$$

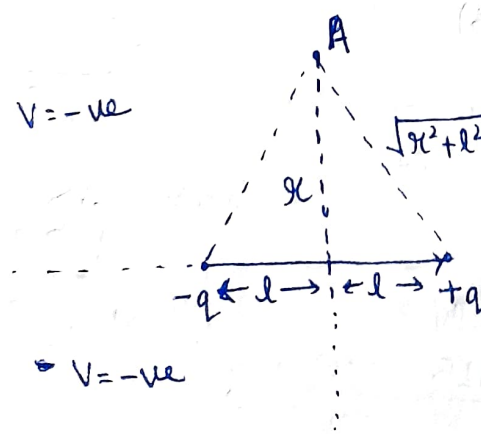
$$E_{min} = \frac{K(2P)}{R^3} = \frac{2KP}{R^3}$$

Pts are $\left(\frac{-\sqrt{3}R}{2}, \frac{R}{2}\right)$ and $\left(\frac{\sqrt{3}R}{2}, \frac{-R}{2}\right)$

$$\vec{E_{min}} = -E\sin 30°\,\hat{i} - E\cos 30°\,\hat{j}$$
$$= -\frac{KP}{R^3}\hat{i} - \frac{\sqrt{3}KP}{R^3}\hat{j}$$

Electric Potential due to dipole:

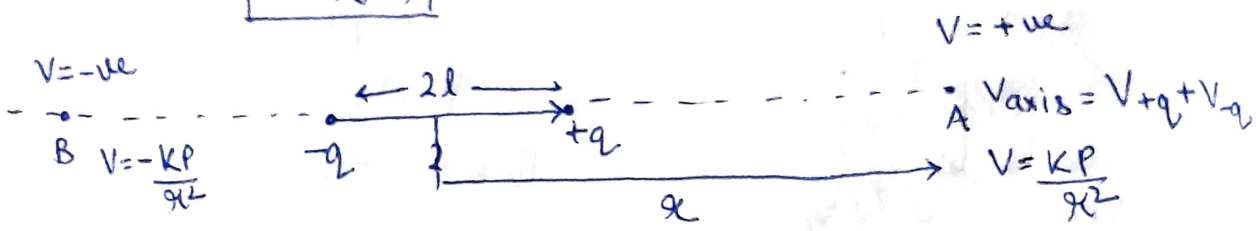
① Equitorial Point:



$$V_{eq} = V_{+q} + V_{-q}$$
$$= \frac{Kq}{\sqrt{x^2+l^2}} - \frac{Kq}{\sqrt{x^2+l^2}}$$
$$= 0$$

$$\therefore V_{eq} = 0$$

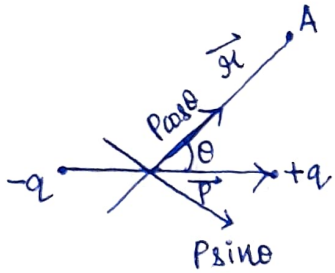
② At axis: $\boxed{P = q(2l)}$



$$V_{axis} = \frac{Kq}{(x-l)} + \frac{K(-q)}{x+l} = Kq\left\{\frac{1}{x-l} - \frac{1}{x+l}\right\} = Kq\left\{\frac{2l}{x^2-l^2}\right\} = \frac{KP}{x^2-l^2}$$

As $x >> l \Rightarrow$ $\boxed{V_{axis} = \frac{KP}{x^2}}$

③ At any general point:

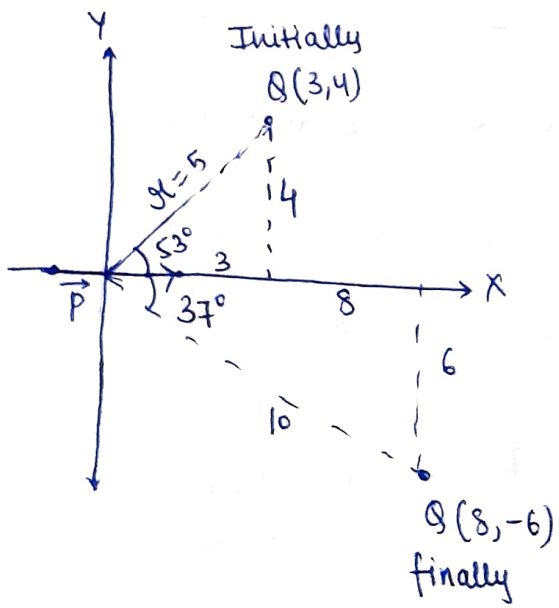


$$V = \frac{KP\cos\theta}{x^2} + 0$$

$$\boxed{V = \frac{KP\cos\theta}{x^2}}$$ θ: angle b/w $\vec{P}$ and $\vec{x}$

$$V = \frac{KPx\cos\theta}{x^3} = \frac{k(\vec{P}\cdot\vec{x})}{x^3} \quad \therefore \boxed{V = \frac{k(\vec{P}\cdot\vec{x})}{x^3}}$$

Q



Find work done by electric field to shift Q charge from initial to final position.

$$V_1 = \frac{KP\cos 53°}{5^2} = \frac{KP \times 3}{125}$$

$$V_2 = \frac{KP\cos 37°}{10^2} = \frac{KP \times 4}{5 \times 100} = \frac{KP}{125}$$

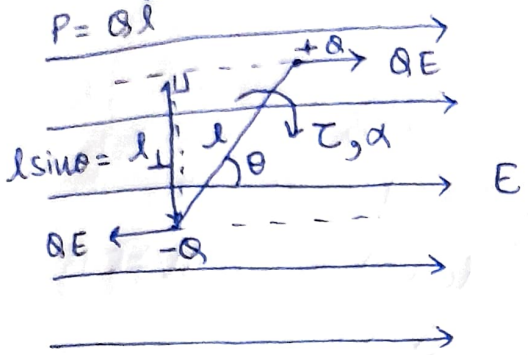
$$W_{elec} = U_i - U_f$$
$$= QV_1 - QV_2$$
$$= Q(V_1 - V_2)$$
$$= Q\left\{\frac{3KP}{125} - \frac{KP}{125}\right\}$$

$$\boxed{W_{elec} = \frac{2KPQ}{125}}$$

$$\boxed{W_{ext} = -\frac{2KPQ}{125}}$$

Electric Dipole in Uniform Electric field:

θ : Angle b/w $\vec{P}$ and $\vec{E}$



$\vec{F} = 0 \Rightarrow \vec{a}_{com} = 0$

But still we can have τ

Both forces on charge make a force couple.

$\boxed{\tau = F\, l_\perp}$ F = Any one of the force

$l_\perp$ = ⊥ distance b/w LOAF

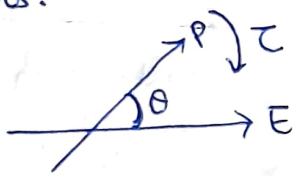
$\tau = QE(l\sin\theta)$

$\boxed{\tau = PE\sin\theta}$ $\boxed{\vec{\tau} = \vec{P} \times \vec{E}}$

This torque is independent of axis.

If Hinged	If not Hinged
$PE\sin\theta = I_{Hinge}\,\alpha$	$PE\sin\theta = I_{com}\,\alpha$

Equilibrium Positions:



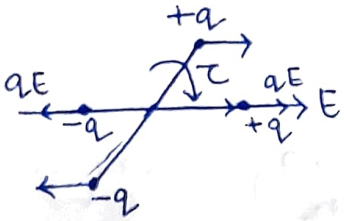
$\tau = PE\sin\theta$

For equilibrium $\tau = 0$

$\Rightarrow \sin\theta = 0$

$\therefore \boxed{\theta = 0 \text{ or } \theta = \pi}$

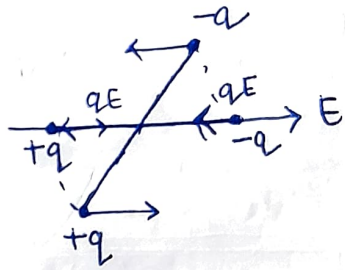
If $\theta = 0°$



F = 0 and $\tau = 0$

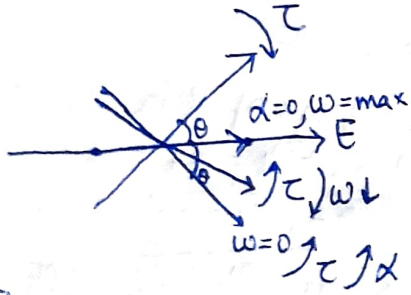
$\therefore \theta = 0°$ is stable Equilibrium position

If $\theta = \pi$



$\therefore \theta = \pi°$ is unstable equilibrium position

$\theta = 0°$ Stable Equilibrium position:



It will perform oscillatory motion.

$\tau = PE\sin\theta$

$\theta \approx$ small $\Rightarrow \sin\theta \approx \theta$

$\boxed{\tau_{res} = PE\theta}$ Angular SHM

$\boxed{\alpha = \left(\frac{PE}{I}\right)\theta} \Rightarrow \boxed{\omega = \sqrt{\frac{PE}{I}}}$

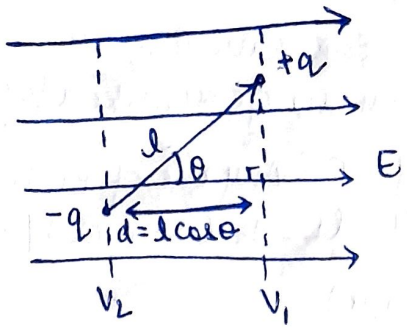
$\boxed{T = 2\pi\sqrt{\frac{I}{PE}}}$

If Hinged $\Rightarrow T = 2\pi\sqrt{\frac{I_{Hinge}}{PE}}$

If not Hinged $\Rightarrow T = 2\pi\sqrt{\frac{I_{com}}{PE}}$

Potential Energy of dipole placed in uniform electric field :

θ : Angle b/w $\vec{P}$ and $\vec{E}$

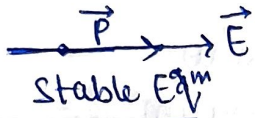


$$\boxed{P = ql}$$

$$U = +qV_1 + (-q)V_2$$
$$= -q(V_2 - V_1)$$
$$= -q(El\cos\theta) \quad \{\Delta V = Ed\}$$

$$\boxed{U = -PE\cos\theta = -\vec{P}\cdot\vec{E}}$$

$U_{min} \Rightarrow \cos\theta = 1$
$\theta = 0$



Stable Eqm

$$\boxed{U_{min} = -PE}$$

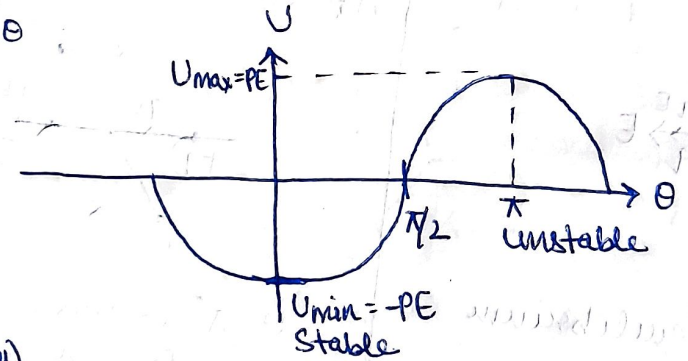
$U_{max} \Rightarrow \cos\theta = -1$
$\theta = \pi$

Unstable Eqm

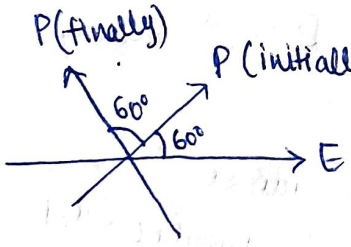
$$\boxed{U_{max} = PE}$$

Graph: U v/s θ

$U = -PE\cos\theta$



Q



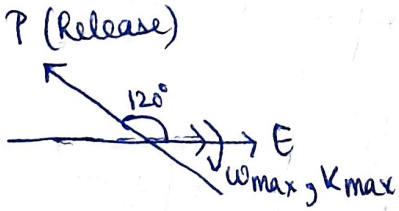
Find ① Wext ② Welec

① $W_{ext} = U_f - U_i = -PE\cos 120° - \{-PE\cos$ Method 2

$$\boxed{W_{ext} = PE}$$

② $W_{elec} = U_i - U_f = -PE$

Q



Find KE_{max} and ω_{max} of this dipole.

Using TMEC $\Rightarrow K_i + U_i = K_f + U_f$

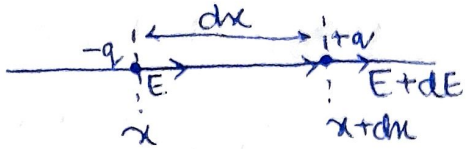
$= 0 - PE\cos 120° = K_{max} - PE\cos$

$$\boxed{KE_{max} = PE + \frac{PE}{2} = \frac{3PE}{2}}$$

Now $\Rightarrow KE = \frac{1}{2} I\omega_{max}^2$

$$\boxed{\omega_{max} = \sqrt{\frac{3PE}{I}}}$$

Force on dipole placed along Non-uniform electric field:



$P = q(dx)$

$F = q(E+dE) - qE = qdE$

$F = (qdx)\frac{dE}{dx} = P\frac{dE}{dx}$

$\therefore \boxed{F = P\frac{dE}{dx}}$ $dx \rightarrow$ sep b/w $-q$ and $+q$

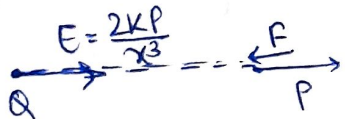
Find Electric force on dipole

Method 1: $F = P\frac{dE}{dx} = P\frac{d}{dx}\left(\frac{KQ}{x^2}\right)$

$= KQP\frac{d}{dx}\left(\frac{1}{x^2}\right)$

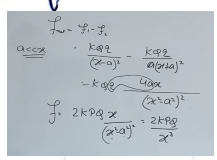
$\boxed{F = -\frac{2KQP}{x^3}}$ (-ve means attractive)

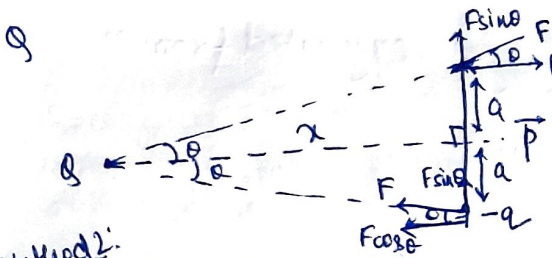
Method 2:



We write force on Q due to P
By NL 3rd law force on dipole is equal and opposite.

$\boxed{F = \frac{2KPQ}{x^3}}$





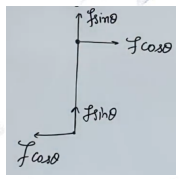
Find electric force b/w Q and P

$P = q(2a)$

Method 1:

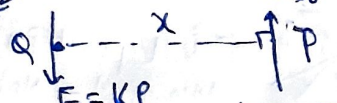
$F_{net} = 2F\sin\theta$

$= 2\left(\frac{KQq}{a^2+x^2}\right)\left(\frac{a}{\sqrt{a^2+x^2}}\right)$

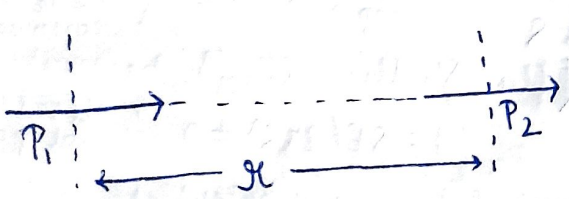


$\boxed{F_{net} = \frac{KPQ}{(a^2+x^2)^{3/2}} = \frac{KPQ}{x^3}}$

Method 2:



$\boxed{F = EQ = \frac{KPQ}{x^3}}$



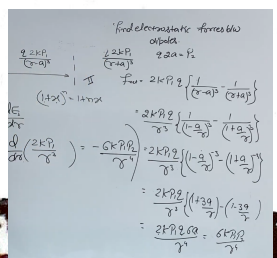
Find electrostatic forces b/w dipoles.

$F = P_2\frac{dE_1}{dx} = P_2\frac{d}{dx}\left(\frac{2KP_1}{x^3}\right)$

$F = P_2\, 2KP_1\left(\frac{-3}{x^4}\right)$

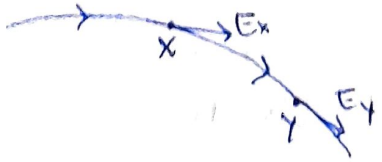
$\boxed{F = -\frac{6KP_1P_2}{x^4}}$

can also be solved by basic method



Electric lines of force (ELOF):

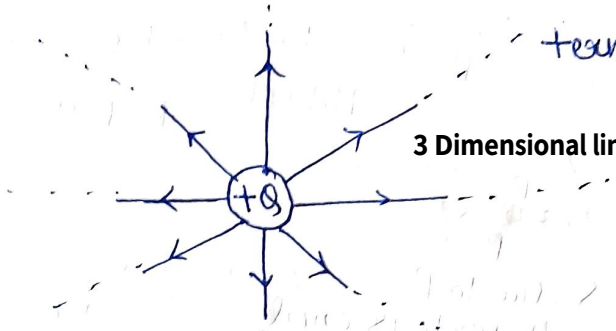
→ They are imaginary lines or curve, tangent to which at point will provide direction of electric field at that point.



→ ELOF are lines or curve which give diagramatic representation of electric field in the region.

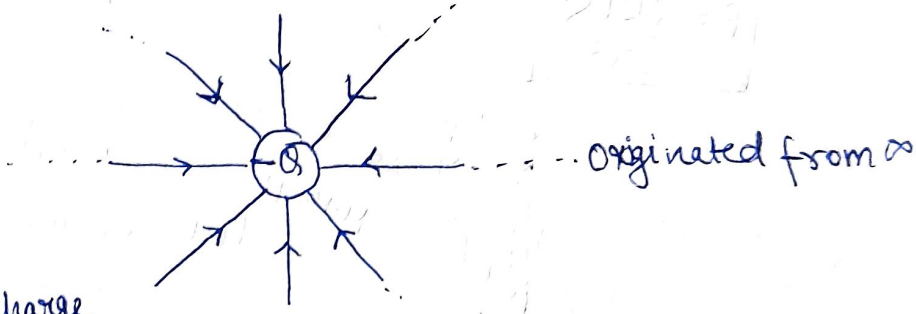
Properties:

1. ELOF originates from either positive charge or ∞.
2. ELOF terminates to either negative charge or ∞.
3. ELOF due to positive isolated point charge will be radially outwards

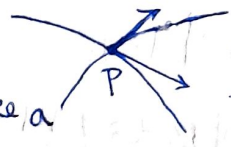


3 Dimensional lines

4. ELOF due to isolated negative charge is radially inwards.

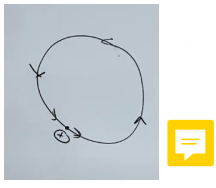


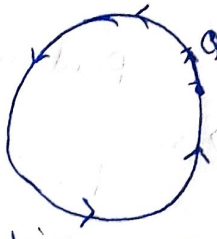
5. No. of ELOF ∝ Charge
6. Two ELOF will never intersect



$\vec{E}$ at point P is in two dirⁿ ⇒ Not possible

7. ELOF due to Static charge will never make a closed loop.





If it would have made a closed loop, $W = 0$ {As $\vec{F_e}$ is a conservative force}

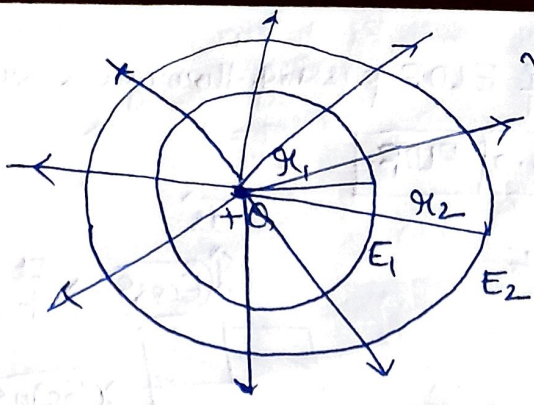
$W = QE(dl_1) + QE(dl_2) + \cdots$

$W = QE(2\pi R) \neq 0$ ∴ closed loop not possible

8. ELOF is always from higher potential to lower potential.
9. Line density ≈ No. of ELOF per unit cross sectional area

* Line Density ∝ Magnitude of EF

Ex



n lines

$$\text{line density}_1 = \frac{n}{4\pi r_1^2}$$

$$\text{line density}_2 = \frac{n}{4\pi r_2^2}$$

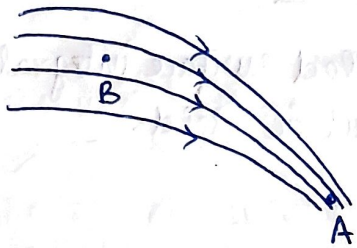
$$\frac{\text{line density}_1}{\text{line density}_2} = \left(\frac{r_2}{r_1}\right)^2 \quad -(1)$$

$$\frac{E_1}{E_2} = \frac{KQ/r_1^2}{KQ/r_2^2} = \left(\frac{r_2}{r_1}\right)^2 \quad -(2)$$

From (1) and (2)

$$\boxed{\frac{\text{line density}_1}{\text{line density}_2} = \frac{E_1}{E_2}}$$

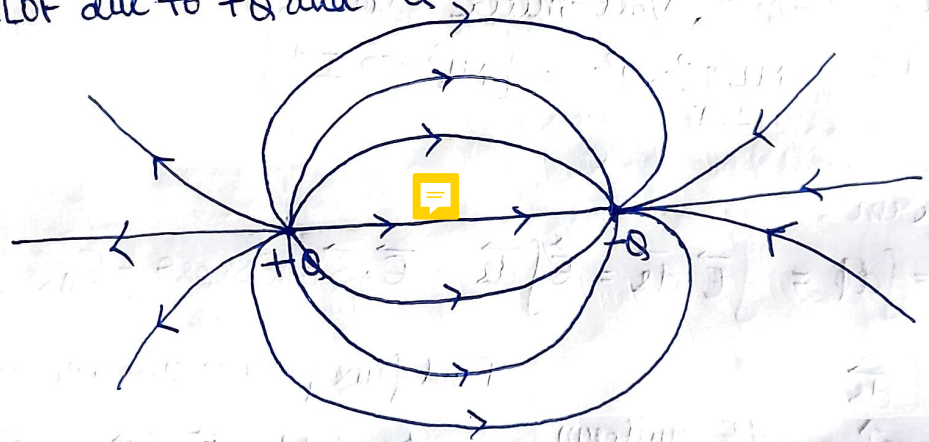
Ex



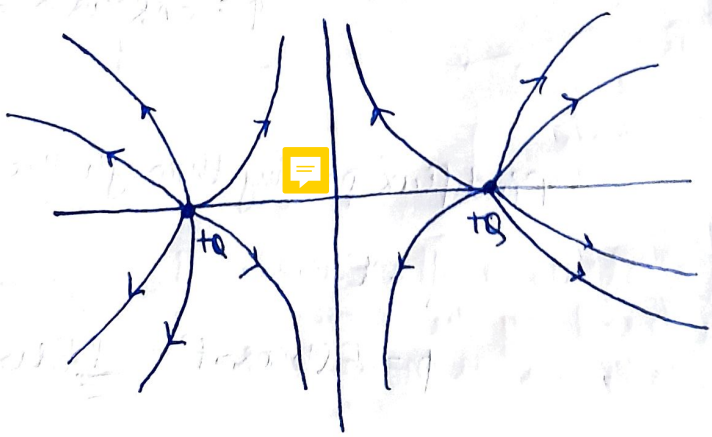
$\boxed{E_A > E_B}$ by comparing line density

$\boxed{V_A < V_B}$ since EF is in the direction decreasing potential

Q Draw ELOF due to +Q and −Q



Q Draw ELOF due to +Q and +Q

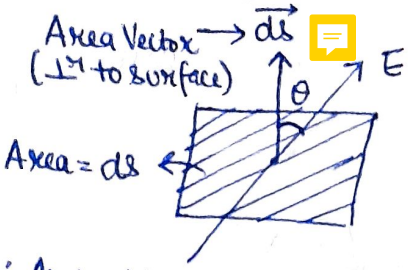


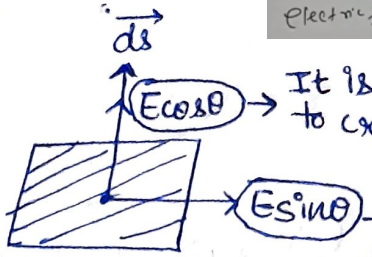
Electric Flux :

→ It represents number of ELOF passing through a surface.

Electric flux ↑ ⇒ ELOF no. ↑
— — — — ↓ ⇒ ELOF no. ↓
Electric flux = 0 ⇒ no. of ELOF = 0

★ **Electric flux ∝ No. of ELOF**





→ It is responsible for El to cross the surface

→ It is not respons for any ELOF to cr the surface

θ : Angle b/w Electric Field and area vector

$d\phi$ = Electric flux passing through surface ds.

$$d\phi = (E\cos\theta)\,ds = E_{\perp} ds = E\,ds\cos\theta = \vec{E}\cdot\vec{ds}$$

$$\phi = \int d\phi = \int \vec{E}\cdot\vec{ds}$$

★ $\int \vec{E}\cdot\vec{ds}$ is called surface integral of electric field.

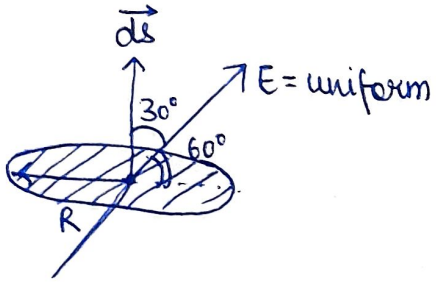
thus, electric flux is the surface integral of electric field

→ Electric flux is scalar quantity.
→ Can be +ve, −ve or zero
→ Unit = $\frac{N}{C}-m^2$, Volt-meter = Gauss
→ Dimension : $\frac{MLT^{-2}}{IT} \times L^2 = [ML^3T^{-3}I^{-1}]$

If $\vec{E}$ = uniform :

$$\phi = \int d\phi = \int \vec{E}\cdot\vec{ds} = \vec{E}\cdot\int\vec{ds} = \vec{E}\cdot\vec{S} = ES\cos\theta = E_{\perp}S$$

Q



Find flux passing through the surface.

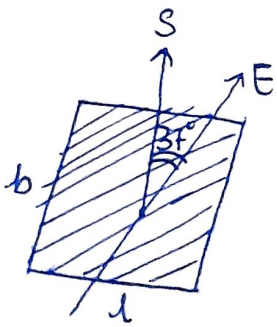
$S = \pi R^2$

$$\phi = \int d\phi = \vec{E}\cdot\int\vec{ds} = \vec{E}\cdot\vec{S} = ES\cos$$

$$\phi = E\pi R^2\cos 30^\circ = \frac{\sqrt{3}}{2}E\pi R^2$$

Note:

Q



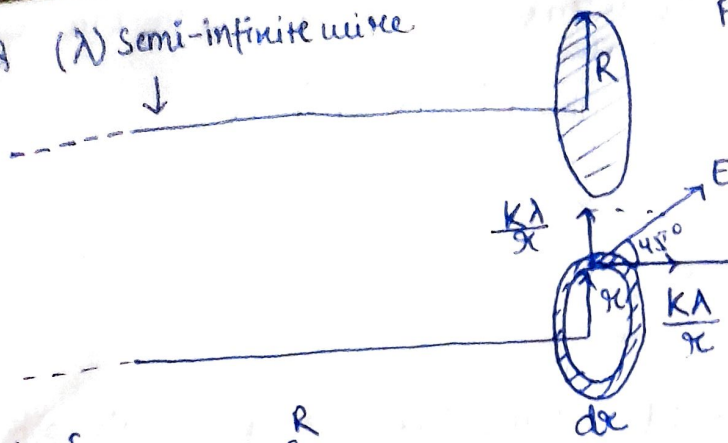
Find flux passing through the surface

$$\phi = ES\cos\theta$$

$$\phi = Elb\cos 37^\circ = \frac{4}{5}Elb$$

Q (λ) Semi-infinite wire

Find electric flux passing through the circular cross section.



$d\phi = \vec{E}\cdot\vec{ds}$

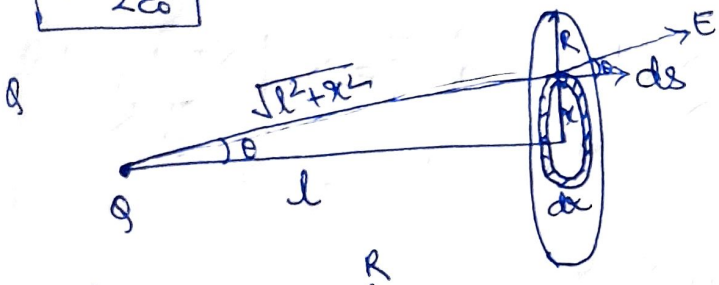
$= \frac{\sqrt{2}K\lambda}{x}(2\pi x\,dx)\cos 45°$

$\boxed{d\phi = K\lambda 2\pi\,dx}$

$\phi = \int d\phi = K\lambda 2\pi \int_0^R dx$

$= \frac{\lambda}{2\varepsilon_0}(x)_0^R = \frac{\lambda R}{2\varepsilon_0}$

$\boxed{\phi = \frac{\lambda R}{2\varepsilon_0}}$



Find electric flux passing through the circular cross section.

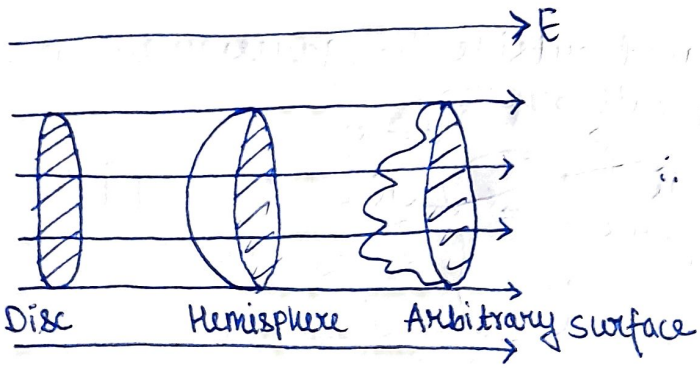
$d\phi = E\,ds\cos\theta$

$= \frac{KQ}{l^2+x^2}(2\pi x\,dx)\frac{l}{\sqrt{l^2+x^2}}$

$\phi = \int d\phi = KQ2\pi l \int_0^R \frac{x\,dx}{(l^2+x^2)^{3/2}} = \frac{Ql}{2\varepsilon_0}\left(\frac{-1}{\sqrt{l^2+x^2}}\right)_0^R = \frac{Ql}{2\varepsilon_0}\left\{\frac{1}{l} - \frac{1}{\sqrt{l^2+R^2}}\right\}$

$\boxed{\phi = \frac{Q}{2\varepsilon_0}\left\{1 - \frac{l}{\sqrt{l^2+R^2}}\right\}}$

Note:



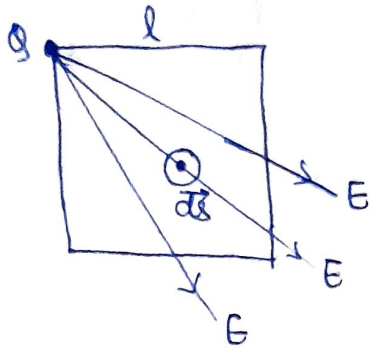
$\boxed{\phi_{Disc} = E\pi R^2}$

$\therefore$ $\boxed{\phi_{Hemisphere} = E\pi R^2}$

$\boxed{\phi_{Arbi.} = E\pi R^2}$

As no. of ELOF passing through all the surface is same, flux through them is equal irrespective of their shape, size or geometry.

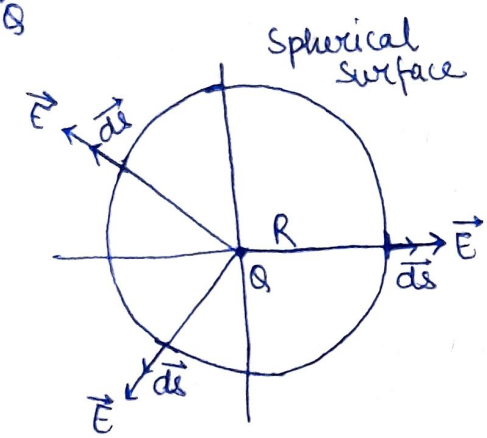
Q



Find flux passing through the square plate.

$\boxed{\phi = 0}$ as $\vec{E} \perp d\vec{s}$

Q



Find electric flux passing through spherical surface.

$\vec{E} \neq$ uniform

$\phi = \oint \vec{E} \cdot d\vec{s}$ } Surface integral of electric field on closed surface

$\phi = \oint E ds \cos 0^\circ = \oint E ds = E \oint ds$

$\phi = \frac{KQ}{R^2} \times 4\pi R^2 \Rightarrow \boxed{\phi = \frac{Q}{\varepsilon_0}}$ → independent of R

Q

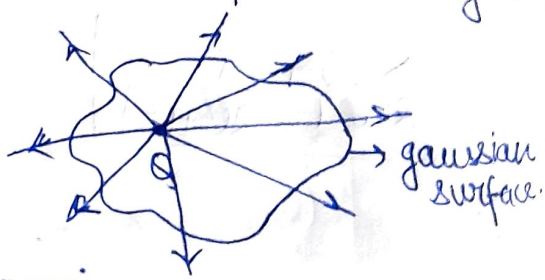
Q

Q

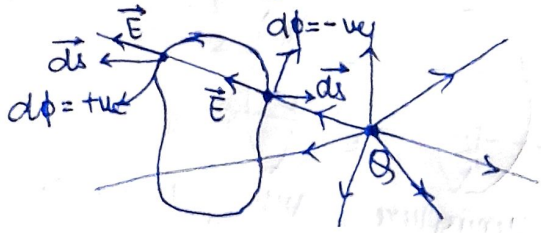
Gauss Theorem:

① A charge will emit electric flux of $\frac{Q}{\varepsilon_0}$.
(Q)

② A surface enclosing the charge Q, will have electric flux crossing its surface of magnitude $\phi = \frac{Q}{\varepsilon_0}$.



③ If a charge is placed outside the gaussian surface, flux by it through the surface will be zero.

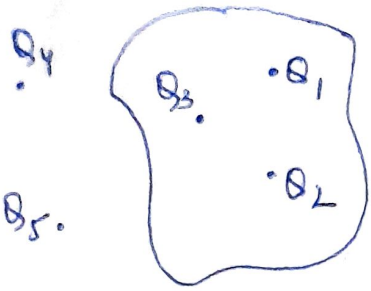


$\boxed{\phi = 0}$

flux entering the surface will be -ve
flux exiting the surface will be +ve
we cannot mathematically calculate the flux by integration here
however, since the no. of ELOF entering and exiting are equal, hence the flux will be zero

④ Surface integral of electric field on a closed surface is always equal to charge enclosed by the surface per unit permitivitty of free space. This is known as gauss law.

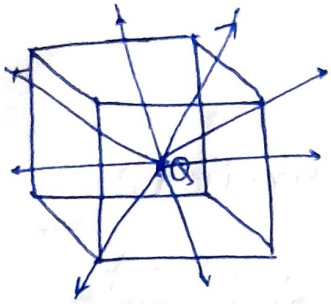
i.e electric flux



$$\phi = \oint \vec{E}_{net} \cdot \vec{ds} = \frac{Q_{in}}{\varepsilon_0} = \frac{Q_1 + Q_2 + Q_3}{\varepsilon_0}$$

{ Put charges with sign }

Q



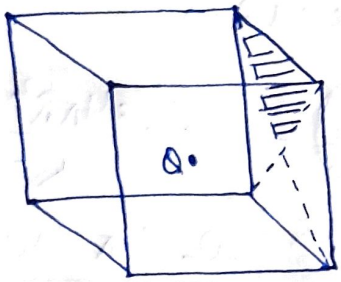
Find flux through one face of the cube.

Net flux passing through all 6 faces $= \frac{Q}{\varepsilon_0} = \phi_{net}$

Flux passing through one of the face $= \phi$

$$\phi_{net} = \frac{Q}{\varepsilon_0} = 6\phi \Rightarrow \boxed{\phi = \frac{Q}{6\varepsilon_0}}$$

Q



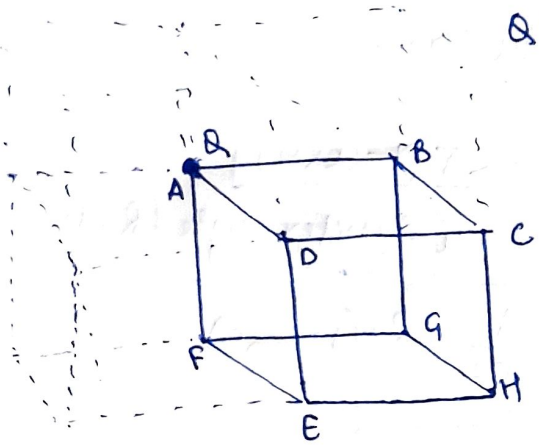
Find flux through the shaded area.

Net flux through cube $= \frac{Q}{\varepsilon_0}$

Flux through one face $= \frac{Q}{6\varepsilon_0}$

Flux through shaded area $= \frac{Q}{24\varepsilon_0}$

Q

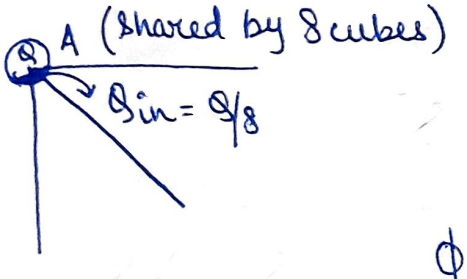


Q {sphere, radius → 0
centre at vertex.

Find electric flux passing through the surfaces of cube.

Flux through cube ABCDEFGH :

$$\phi = \frac{Q_{in}}{\varepsilon_0} = \frac{Q/8}{\varepsilon_0} = \frac{Q}{8\varepsilon_0}$$



$$\boxed{\phi_{ABCD} = \phi_{ADEF} = \phi_{ABGF} = 0}$$

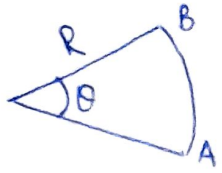
↳ as Q is at vertex of square.

$$\phi_{net} = 3\phi_1 + 3\times 0 \Rightarrow 3\phi_1 = \frac{Q}{8\varepsilon_0} \Rightarrow \phi_1 = \frac{Q}{24\varepsilon_0}$$

$$\boxed{\phi_{BCGH} = \phi_{EFGH} = \phi_{CDEH} = \frac{Q}{24\varepsilon_0}}$$

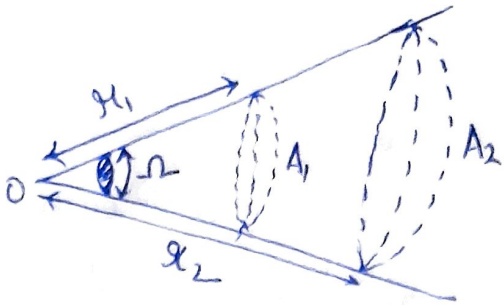
Solid Angle (Steradian): Angle in 3-D

In 2-D:



$$\theta(\text{Radian}) = \frac{\widehat{AB}}{R}$$

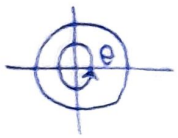
In 3-D:



$$\Omega = \frac{A_1}{r_1^2} = \frac{A_2}{r_2^2} \text{ (steradian)}$$

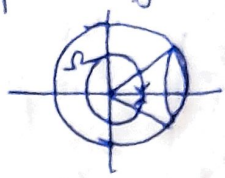
$$\boxed{A_1 = \Omega r_1^2} \qquad \therefore \boxed{\Omega = \frac{A}{r^2}}$$

Complete angle in 2D:



$$\theta = \frac{2\pi R}{R} = 2\pi \text{ Radian}$$

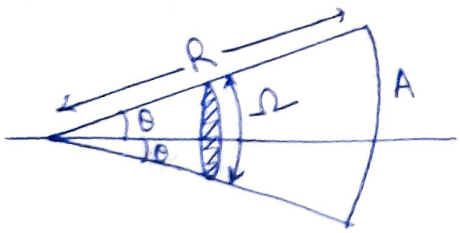
Complete angle in 3-D:



$$\Omega = \frac{4\pi R^2}{R^2} = 4\pi$$

$$\boxed{\Omega = 4\pi \text{ Steradian}}$$

Relation between solid angle and angle in 2D:-

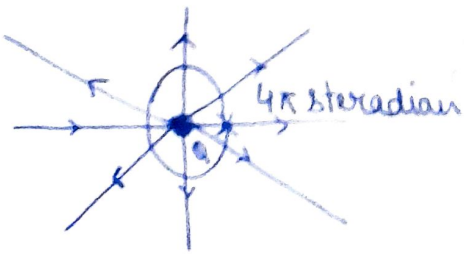


$$\boxed{\Omega = 2\pi(1-\cos\theta)}$$

θ : semi-vertex angle (Radian)

$$\boxed{\Omega = \frac{A}{R^2} = 2\pi(1-\cos\theta)}$$

Electric flux per unit solid angle:



$$\phi = \frac{Q}{\varepsilon_0}$$

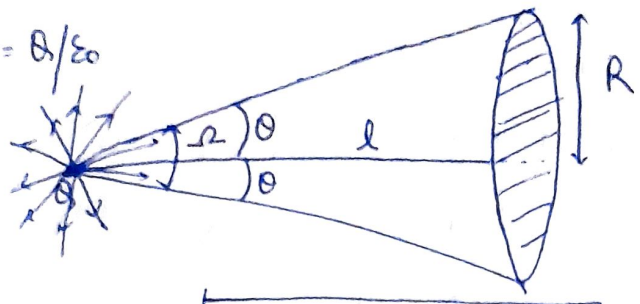
$$\frac{\phi}{\Omega} = \frac{Q/\varepsilon_0}{4\pi} = \frac{Q}{4\pi\varepsilon_0} = KQ$$

$$\therefore \boxed{\frac{\phi}{\Omega} = \frac{Q}{4\pi\varepsilon_0} = KQ}$$

Unit = $\frac{\text{Gauss}}{\text{Steradian}}$

Q

$\phi = Q/\varepsilon_0$



Find electric flux crossing this circular cross section

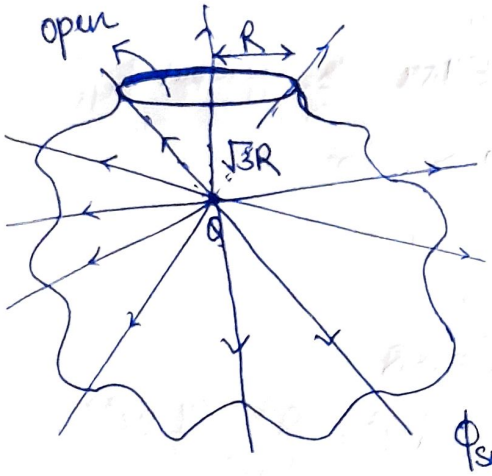
$4\pi \longrightarrow \frac{Q}{\varepsilon_0}$

$1 \longrightarrow \frac{Q}{4\pi\varepsilon_0}$

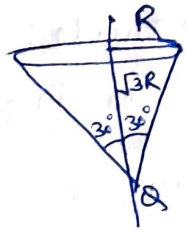
$\Omega = \frac{Q}{4\pi\varepsilon_0}\Omega = \frac{Q}{4\pi\varepsilon_0}2\pi(1-\cos\theta)$

$$\boxed{\phi = \frac{Q}{2\varepsilon_0}\left\{1 - \frac{l}{\sqrt{l^2+R^2}}\right\}}$$

Q open



Find electric flux passing through the random surface



$\phi_{open} = \frac{Q}{4\pi\varepsilon_0}\Omega$

$= \frac{Q}{4\pi\varepsilon_0}2\pi(1-\cos 30^\circ)$

$= \frac{Q}{2\varepsilon_0}\left(1-\frac{\sqrt{3}}{2}\right)$

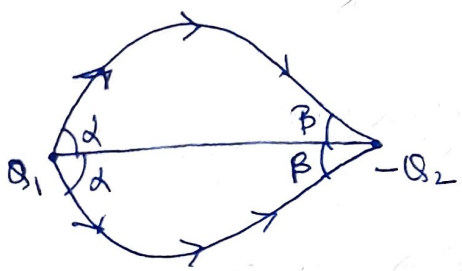
$\phi_{surface} = \phi_{Total} - \phi_{open}$

$= \frac{Q}{\varepsilon_0} - \frac{Q}{2\varepsilon_0}\left(1-\frac{\sqrt{3}}{2}\right)$

$$\therefore \boxed{\phi_{surface} = \frac{Q}{2\varepsilon_0}\left\{1+\frac{\sqrt{3}}{2}\right\}}$$

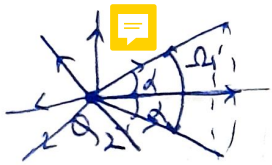
***

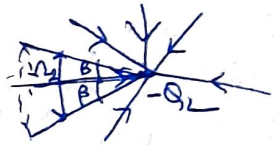
Q



Find Q_1/Q_2

Flux through Q_1 in the region should be equal to flux entering $-Q_2$ as no line can intersect each other so they must originate from Q_1 and terminate at $-Q_2$.





$\frac{Q_1}{4\pi\varepsilon_0}\Omega_1 = \frac{Q_2}{4\pi\varepsilon_0}\Omega_2$

$Q_1\{2\pi(1-\cos\alpha)\} = Q_2\{2\pi(1-\cos\beta)\}$

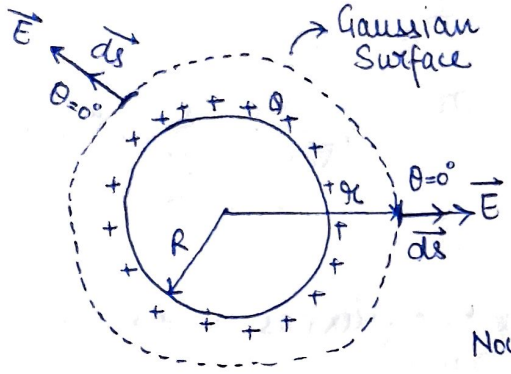
$\frac{Q_1}{Q_2} = \frac{1-\cos\beta}{1-\cos\alpha} = \frac{2\sin^2\beta/2}{2\sin^2\alpha/2}$

$$\therefore \boxed{\frac{Q_1}{Q_2} = \frac{\sin^2\beta/2}{\sin^2\alpha/2}}$$

Gauss law is very useful in finding Electric field due to symmetric systems.

① Electric Field due to Uniform Hollow sphere:

(i) Outside



By definition: $\phi = \oint \vec{E}\cdot\vec{ds}$

$= \oint E ds\cos 0^\circ = \oint E ds = E\oint ds$

$\phi = E 4\pi r^2$ — ①

By Gauss law: $\phi = \frac{Q_{in}}{\varepsilon_0}$

$\phi = \frac{Q}{\varepsilon_0}$ — ②

Now, ① = ② $\Rightarrow E 4\pi r^2 = \frac{Q}{\varepsilon_0} \Rightarrow E_{out} = \frac{Q}{4\pi\varepsilon_0 r^2}$

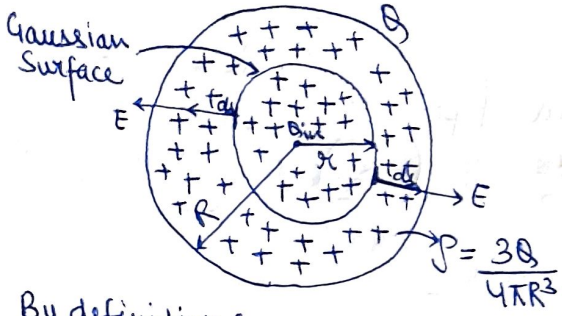
$$\boxed{E_{out} = \frac{KQ}{r^2}}$$

(ii) Inside: $Q_{in} = 0 \Rightarrow \phi = 0$

$$\therefore \boxed{E_{in} = 0}$$

② Electric Field due to Uniform Solid sphere:

Inside:



$\frac{4}{3}\pi R^3 \longrightarrow Q$

$\Rightarrow \frac{4}{3}\pi r^3 = \frac{Q}{\frac{4}{3}\pi R^3}\left(\frac{4}{3}\pi r^3\right)$

$$\therefore \boxed{Q_{in} = \frac{Q r^3}{R^3}}$$

By definition:

$\phi = \oint \vec{E}\cdot\vec{ds} = \oint E ds\cos 0^\circ = \oint E ds$

$= E\oint ds = E 4\pi r^2$ — ①

By Gauss law

$\phi = \frac{Q_{in}}{\varepsilon_0} = \frac{Q r^3}{R^3 \varepsilon_0}$

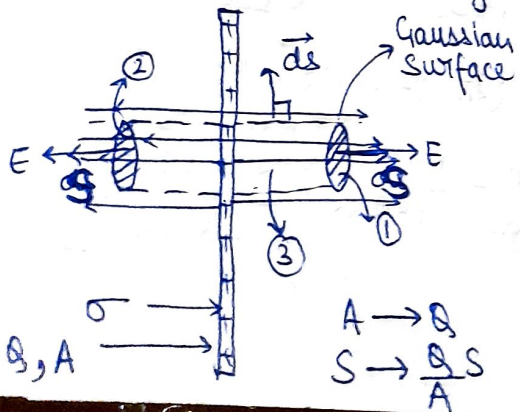
Now, ① = ② $\Rightarrow E 4\pi r^2 = \frac{Q r^3}{R^3 \varepsilon_0}$

(ii) outside:

$$\boxed{E_{out} = \frac{KQ}{r^2}}$$

$$\boxed{E_{in} = \frac{KQr}{R^3} = \frac{\rho r}{3\varepsilon_0}}$$

③ Electric Field due to large Non-Conducting sheet:



By definition: $\phi = \phi_1 + \phi_2 + \phi_3$

$\phi = E S\cos 0^\circ + E S\cos 0^\circ + 0$

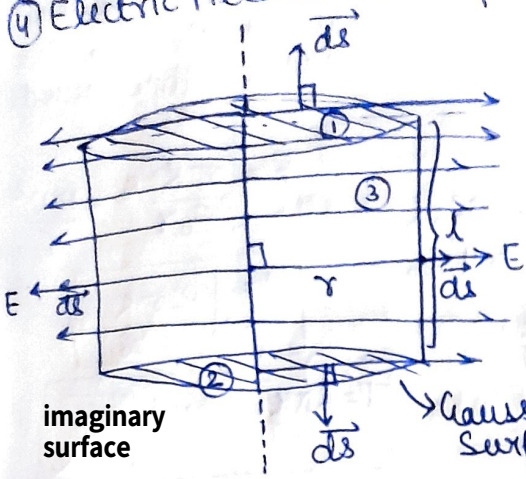
$\phi = 2ES$ — ①

By Gauss law: $\phi = \frac{\sigma S}{\varepsilon_0} = \frac{QS}{A\varepsilon_0}$ — ②

Now ① = ② $\Rightarrow$

$$\boxed{E = \frac{\sigma}{2\varepsilon_0} = \frac{Q}{2A\varepsilon_0}}$$

④ Electric field due to uniform infinitely long wire:



By definition: $\phi = \phi_1 + \phi_2 + \phi_3$

$\phi = \int \vec{E} \cdot \vec{ds} = \int E\,ds\cos 0^\circ = E\int ds$

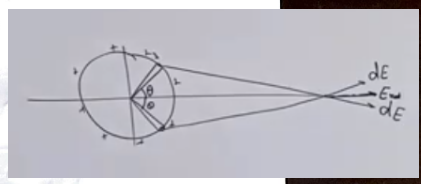
$\phi = E\,2\pi r l$ — (1)

By Gauss law:

$\phi = \frac{Q_{in}}{\varepsilon_0} = \frac{\lambda l}{\varepsilon_0}$ — (2)

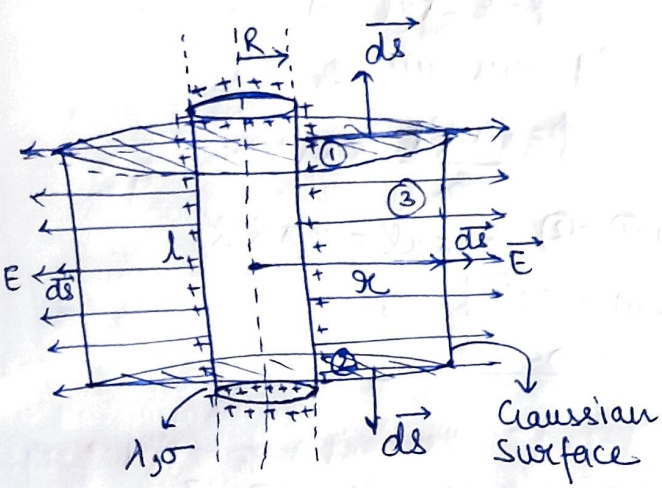
Now, (1) = (2) $\Rightarrow E\,2\pi r l = \frac{\lambda l}{\varepsilon_0}$

$$E = \frac{\lambda}{2\pi\varepsilon_0 r} = \frac{2k\lambda}{r}$$



⑤ Electric field due to long uniform hollow cylinder:

(i) Outside:



By definition:

$\phi = \phi_1 + \phi_2 + \phi_3$

$\phi = \int \vec{E} \cdot \vec{ds} = \int E\,ds\cos 0^\circ = E\int ds$

$\phi = E\,2\pi r l$ — (1)

By Gauss law:

$\phi = \frac{Q_{in}}{\varepsilon_0} = \frac{\lambda l}{\varepsilon_0} = \frac{\sigma\,2\pi R l}{\varepsilon_0}$ — (2)

Now, (1) = (2) $\Rightarrow E\,2\pi r l = \frac{\lambda l}{\varepsilon_0}$

$q = \lambda l = \sigma\,2\pi R l$

$$\lambda = \sigma\,2\pi R$$

$$E_{out} = \frac{\lambda}{2\pi\varepsilon_0 r}$$

$$E_{out} = \frac{2k\lambda}{r}$$

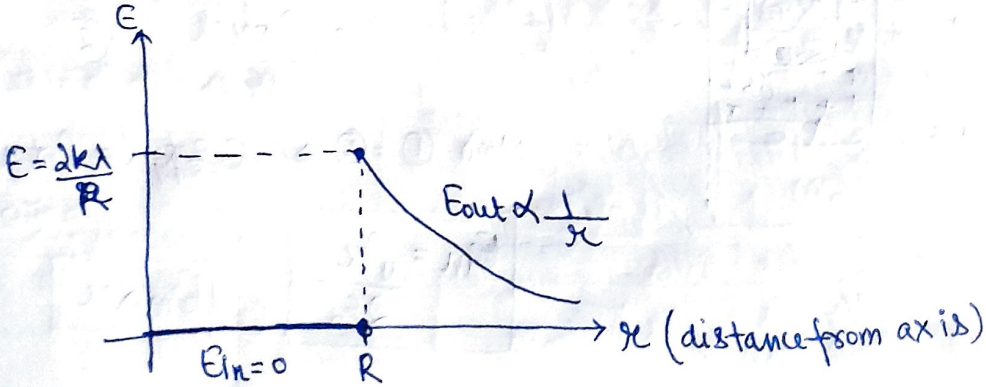
$$E_{out} = \frac{\sigma R}{\varepsilon_0 r}$$

$$E_{surface} = \frac{2k\lambda}{R}$$

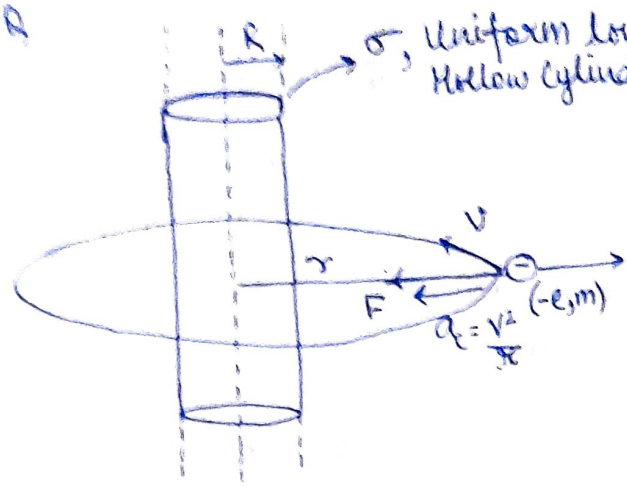
(ii) Inside:

$E_{in} = 0$ As charge inside is zero

Graph:



Q



$$E = \frac{\sigma R}{\varepsilon_0 r}$$

Find v such that e⊖ perfor... UCM around cylinder. ... find its time perio...

$$F = \frac{\sigma R}{\varepsilon_0 r}\, e = \frac{mv^2}{r}$$

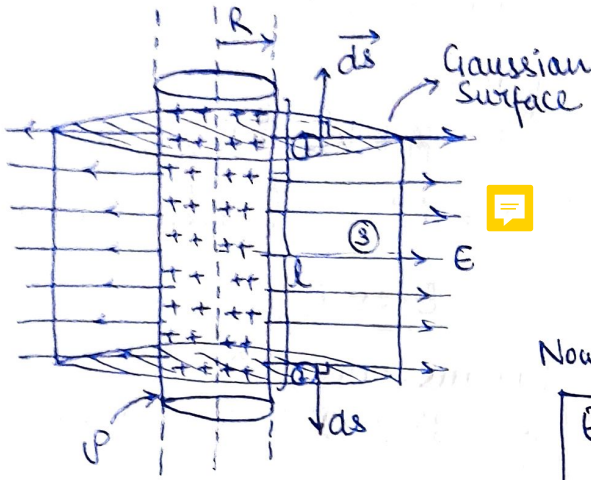
$$v = \sqrt{\frac{e\sigma R}{m\varepsilon_0}}$$

$$T = \frac{2\pi r}{v} = 2\pi r\sqrt{\frac{m\varepsilon_0}{\sigma R e}}$$

Q

Electric Field due to long uniform solid cylinder:

(i) Outside:



By definition: $\phi = \phi_1 + \phi_2 + \phi_3$ ($\phi_1 = 0$, $\phi_2 = 0$)

$$\phi = \int \vec{E}\cdot\vec{ds} = \int E\,ds\cos 0^\circ = E\int ds$$

$$\phi = E\,2\pi r l \quad -(1)$$

By Gauss law: $\phi = \frac{Q_{in}}{\varepsilon_0}$

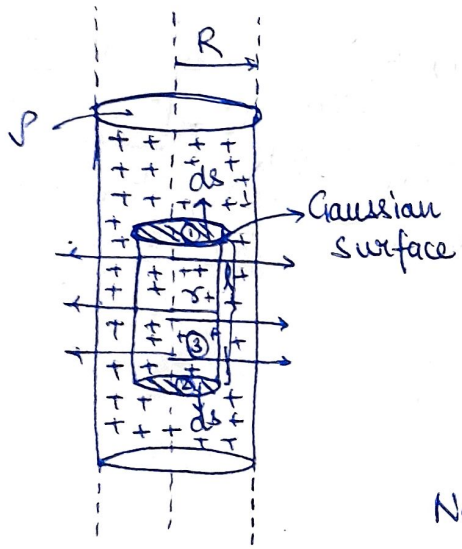
$$\phi = \frac{\rho\pi R^2 l}{\varepsilon_0}$$

Now (1) = (2) $\quad E\,2\pi r l = \frac{\rho\pi R^2 l}{\varepsilon_0}$

$$E_{out} = \frac{\rho R^2}{2\varepsilon_0 r}$$

$$E_{out} \propto \frac{1}{r}$$

(ii) Inside:



By definition: $\phi = \phi_1 + \phi_2 + \phi_3$ ($\phi_1 = 0$, $\phi_2 = 0$)

$$\phi = \int \vec{E}\cdot\vec{ds} = \int E\,ds\cos 0^\circ = E\int ds$$

$$\phi = E\,2\pi r l \quad -(1)$$

By Gauss law: $\phi = \frac{Q_{in}}{\varepsilon_0}$

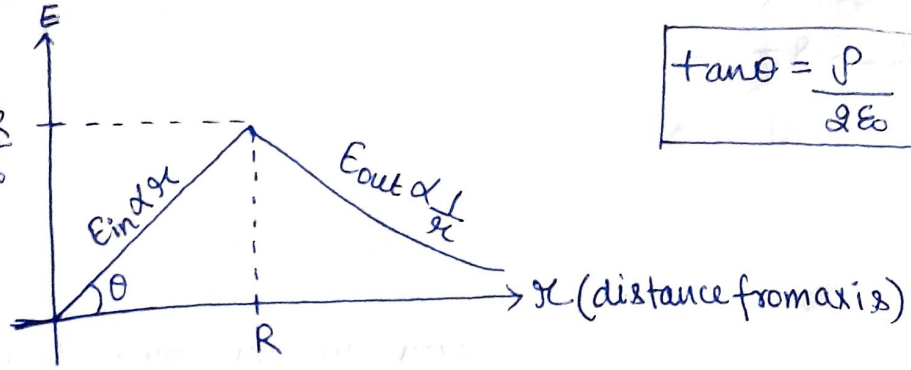
$$\phi = \frac{\rho\pi r^2 l}{\varepsilon_0} \quad -(2)$$

Now (1) = (2) $\Rightarrow \quad E\,2\pi r l = \frac{\rho\pi r^2 l}{\varepsilon_0}$

$$E_{in} = \frac{\rho r}{2\varepsilon_0}$$

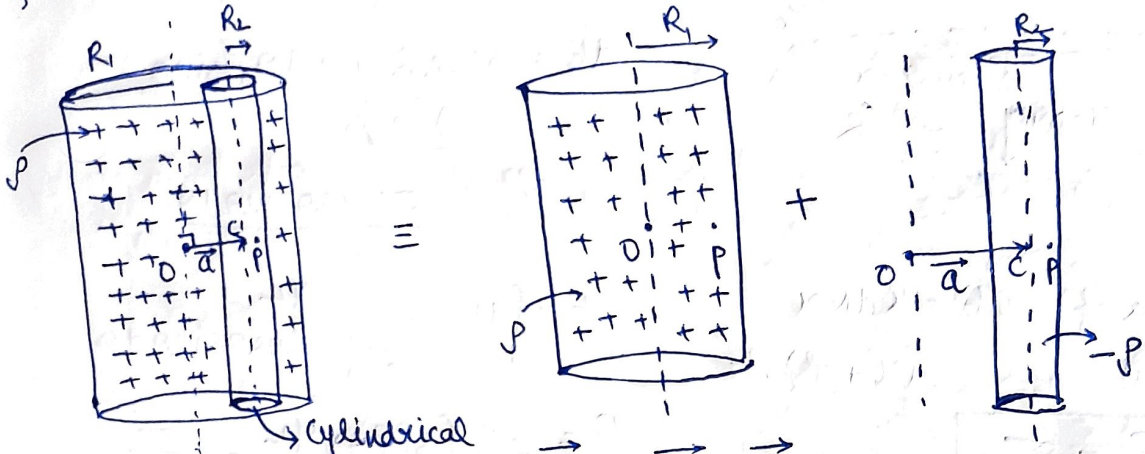
$$E_{in} \propto r$$

Graph:



$$\tan\theta = \frac{\rho}{2\varepsilon_0}$$

Q Find $\vec{E}$ at P.



$0 = \rho - \rho$

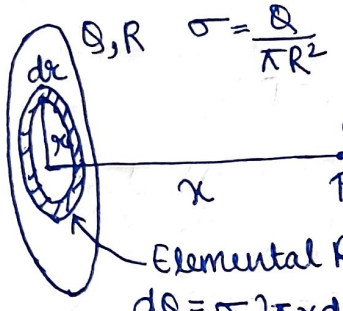
$$\vec{E}_P = \vec{E}_\rho + \vec{E}_{-\rho}$$

$$= \frac{\rho(\vec{OP})}{2\varepsilon_0} + \frac{(-\rho)(\vec{CP})}{2\varepsilon_0} = \frac{\rho}{2\varepsilon_0}(\vec{OP} - \vec{CP})$$

$$= \frac{\rho}{2\varepsilon_0}(\vec{OP} + \vec{PC}) = \frac{\rho}{2\varepsilon_0}(\vec{OC}) = \frac{\rho}{2\varepsilon_0}(\vec{a})$$

$$\boxed{\vec{E}_P = \frac{\rho}{2\varepsilon_0}(\vec{a})}$$

Electric Potential due to Uniform Disc:



$$dV = \frac{K dQ}{\sqrt{r^2+x^2}}$$

$$dV = \frac{K dQ}{\sqrt{r^2+x^2}} = \frac{\sigma 2\pi r dr}{4\pi\varepsilon_0\sqrt{r^2+x^2}}$$

$$dV = \frac{\sigma r dr}{2\varepsilon_0\sqrt{r^2+x^2}}$$

$$V = \int dV = \frac{\sigma}{2\varepsilon_0}\int_0^R \frac{r dr}{\sqrt{r^2+x^2}} = \frac{\sigma}{2\varepsilon_0}\left(\sqrt{r^2+x^2}\right)_0^R$$

$$\boxed{V = \frac{\sigma}{2\varepsilon_0}\left(\sqrt{R^2+x^2} - x\right)}$$

$$V = \frac{\sigma}{2\varepsilon_0}\left(\sqrt{R^2+x^2} - x\right)\frac{\left(\sqrt{R^2+x^2}+x\right)}{\left(\sqrt{R^2+x^2}+x\right)}$$

remember

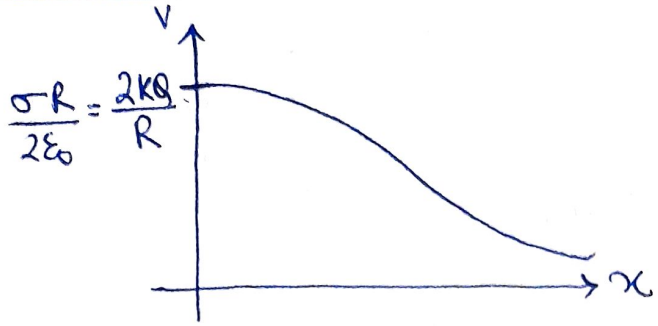
Very near point $\Rightarrow x = 0$

$$\boxed{V = \frac{\sigma R^2}{2\varepsilon_0\left(\sqrt{R^2+x^2}+x\right)} = \frac{2KQ}{\left(\sqrt{R^2+x^2}+x\right)}}$$

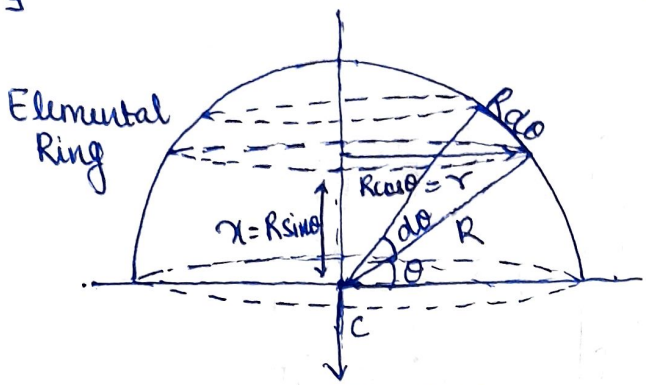
$$\boxed{V = \frac{2KQ}{R} = \frac{\sigma R}{2\varepsilon_0}}$$

As $x \uparrow$, $V \downarrow$

Graph:



Q



(At axis of) ring $dE = \frac{K\,dQ\,x}{(x^2+x^2)^{3/2}}$

$\therefore \boxed{E_C = \frac{\sigma}{4\varepsilon_0}}$

Given uniform Hemi-spherical shell (σ, R)

Find Electric field at point C.

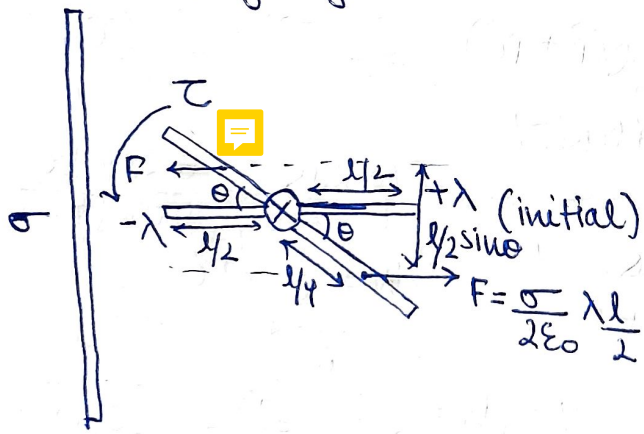
$$dE = \frac{\sigma\, 2\pi R^2 \cos\theta d\theta R\sin\theta}{4\pi\varepsilon_0 R^3}$$

$$dE = \frac{\sigma \cos\theta \sin\theta d\theta}{2\varepsilon_0}$$

$$E = \int dE = \frac{\sigma}{2\varepsilon_0}\int_0^{\pi/2} \cos\theta\sin\theta d\theta$$

$$E = \frac{\sigma}{2\varepsilon_0}\left(\frac{\sin^2\theta}{2}\right)_0^{\pi/2} = \frac{\sigma}{4\varepsilon_0}$$

Q Non-conducting large sheet



Now, $\tau = I\alpha \Rightarrow \frac{\sigma\lambda l^2}{8\varepsilon_0}\theta = \left(\frac{ml^2}{12}\right)\alpha$

$$\boxed{\alpha = \frac{3\sigma\lambda}{2\varepsilon_0 m}\theta}$$

$$\boxed{\omega = \sqrt{\frac{3\sigma\lambda}{2\varepsilon_0 m}}}$$

$$\boxed{T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{2\varepsilon_0 m}{3\sigma\lambda}}}$$

Prove that if rod is given small angular displacement it will perform angular SHM. Also find time period of SHM

$$\tau = F l_\perp$$

$$= \frac{\sigma\lambda l}{4\varepsilon_0}\frac{l}{2}\sin\theta$$

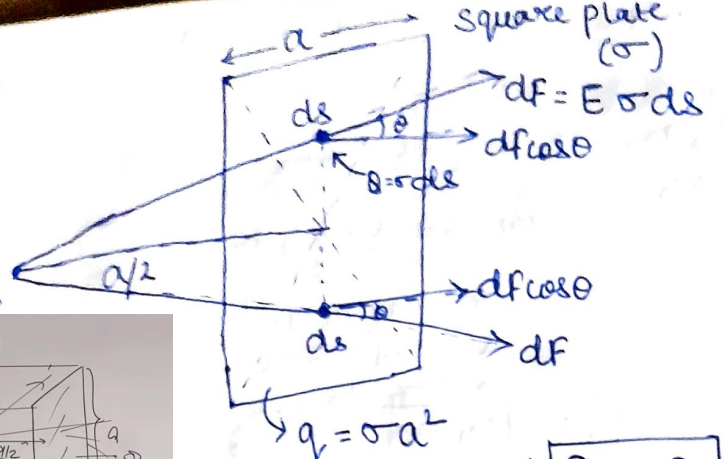
$$\tau = \frac{\sigma\lambda l^2}{8\varepsilon_0}\sin\theta$$

As $\theta \to$ small, $\sin\theta \approx \theta$

$$\boxed{\tau = \frac{\sigma\lambda l^2}{8\varepsilon_0}\theta}$$

SHM proved

Q



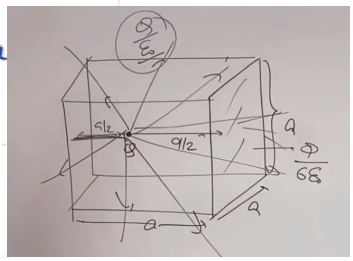
Find electrostatic force b/w Q and charged square plate.

$$F = \int df\cos\theta = \int (E\sigma\, ds)\cos\theta$$

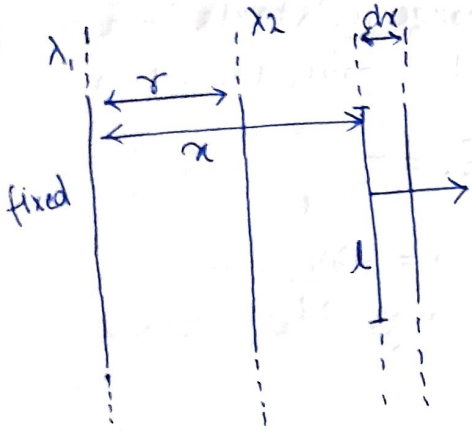
$$F = \sigma\int E\, ds\cos\theta = \sigma\int \vec{E}\cdot d\vec{s}$$

$$F = \sigma\phi = \frac{\sigma Q}{6\varepsilon_0}$$

$$\boxed{F = \frac{\sigma Q}{6\varepsilon_0}}$$



Q



Find W.d per unit length by e.f. on second wire, if separation increases from r to 2r.

$$F = \frac{2k\lambda_1}{x}(\lambda_2 l)$$

Method1:

$$dW = \frac{2k\lambda_1\lambda_2 l\, dx\cos 0^\circ}{x}$$

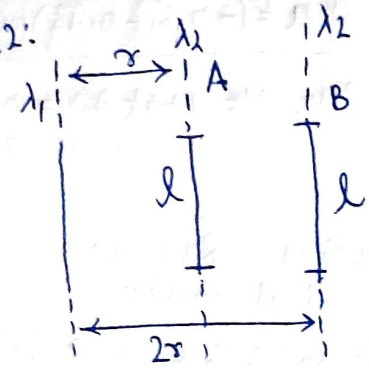
$$W = \int dW = 2k\lambda_1\lambda_2 l\int_r^{2r}\frac{dx}{x}$$

$$W = 2k\lambda_1\lambda_2 l\,(\ln x)_r^{2r} = 2k\lambda_1\lambda_2 l\ln\frac{2r}{r}$$

$$W = 2k\lambda_1\lambda_2 l\ln 2 \Rightarrow \boxed{\frac{W}{l} = 2k\lambda_1\lambda_2\ln 2}$$

2 infinitely long uniform wires.

Method 2:

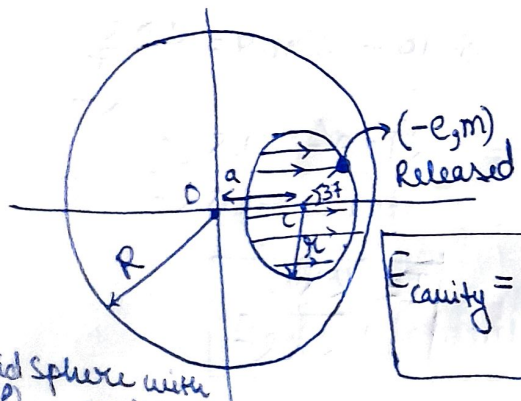


$$W_{elec} = U_i - U_f = \lambda_2 l V_A - \lambda_2 l V_B$$

$$W = \lambda_2 l(V_A - V_B) = \lambda_2 l\, 2k\lambda_1 \ln\frac{2r}{r}$$

$$\boxed{\frac{W}{l} = 2k\lambda_1\lambda_2\ln 2}$$

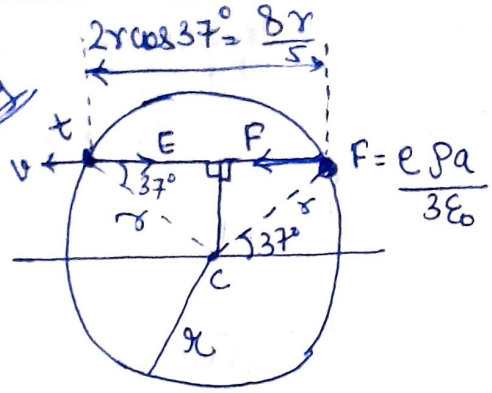
Q



After how much time electron will touch the sphere again. Also find velocity at this instant

$$E_{cavity} = \frac{\rho a}{3\varepsilon_0}$$

Inside cavity



$F = ma \Rightarrow \frac{e\rho a}{3\varepsilon_0} = m\frac{dv}{dt}$

$\frac{dv}{dt} = \frac{e\rho a}{3m\varepsilon_0} = \text{const.}$

Now, $s = ut + \frac{1}{2}at^2$

$\frac{8r}{5} = \frac{1}{2}\frac{e\rho a}{3m\varepsilon_0}t^2$

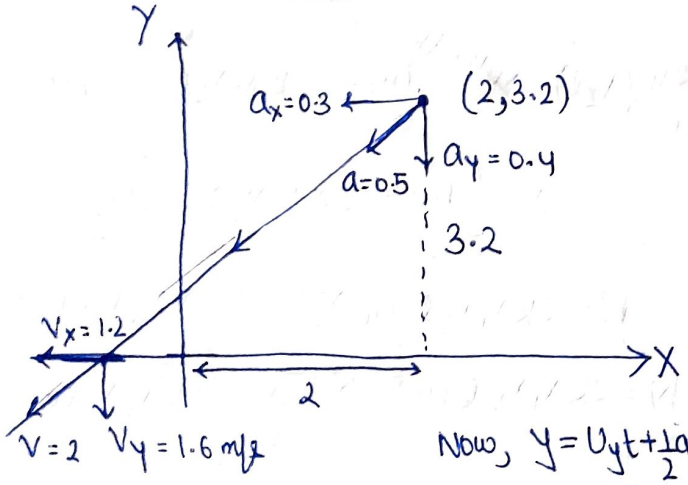
$\boxed{t = \sqrt{\frac{48mr\varepsilon_0}{5e\rho a}}}$

Also, $v^2 = u^2 + 2as$

$v^2 = 2\left(\frac{e\rho a}{3m\varepsilon_0}\right)\frac{8r}{5}$

$\boxed{v = \sqrt{\frac{16e\rho ar}{15m\varepsilon_0}}}$

Q $V = 3x + 4y$. A particle of mass 10kg, charge 1C starts from rest from point (2, 3.2)m. Find the speed of particle when it crosses X-axis.



$E_x = -\frac{\partial V}{\partial x} = -3\ N/C$

$E_y = -\frac{\partial V}{\partial y} = -4\ N/C$

$\boxed{\vec{E} = (-3\hat{i} - 4\hat{j})\ N/C}$

$F = q\vec{E} = 1(-3\hat{i} - 4\hat{j})\ N$

$F = m\vec{a} \Rightarrow -3\hat{i} - 4\hat{j} = 10\vec{a}$

$\boxed{\vec{a} = (-0.3\hat{i} - 0.4\hat{j})\ m/s^2}$

Now, $y = u_y t + \frac{1}{2}a_y t^2$

$-3.2 = \frac{1}{2}(-0.4)t^2$

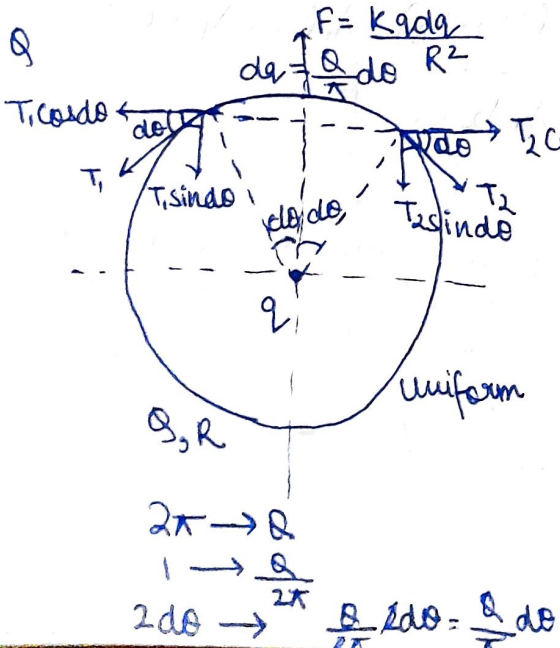
$\boxed{t = 4s}$

Also, $V = u + at \Rightarrow V = 0.5 \times 4$

$\boxed{V = 2m/s}$

$\boxed{\vec{V} = (-1.2\hat{i} - 1.6\hat{j})\ m/s}$

Q



① Find tension in Ring when q charge is placed at center.

$T_1\cos d\theta = T_2\cos d\theta \Rightarrow \boxed{T_1 = T_2 = T}$

$T\sin d\theta + T\sin d\theta = kq\frac{\frac{Q}{\pi}d\theta}{R^2}$

$2Td\theta = \frac{kqQd\theta}{\pi R^2}$

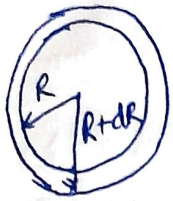
$\boxed{T = \frac{kQq}{2\pi R^2}}$

$2\pi \rightarrow Q$

$1 \rightarrow \frac{Q}{2\pi}$

$2d\theta \rightarrow \frac{Q}{2\pi}2d\theta = \frac{Q}{\pi}d\theta$

② Find change in radius when q is placed at center. Given Area = A, Young's Modulus = Y

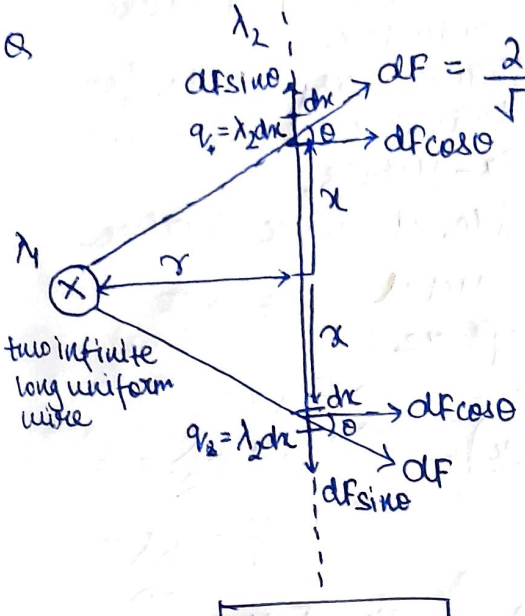


$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{T/A}{\Delta l/l} = \frac{KQq \times 2\pi R}{2\pi R^2 A \, 2\pi dR}$$

$$\boxed{dR = \frac{KQq}{2\pi RAY}}$$

$$\left\{ \begin{array}{l} l \to 2\pi R \\ l+\Delta l = 2\pi(R+dR) \\ \Delta l = 2\pi dR \end{array} \right\}$$

Q



Find electric force b/w these wires.

$$F_{net} = \int_{-\infty}^{\infty} dF\cos\theta = \int_{0}^{\infty} 2\,dF\cos\theta$$

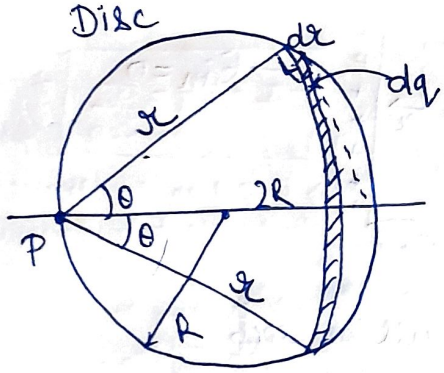
$$F_{net} = \int_0^\infty 2 \cdot \frac{2k\lambda_1\lambda_2}{\sqrt{r^2+x^2}}\,dx \times \frac{r}{\sqrt{r^2+x^2}}$$

$$F_{net} = 4k\lambda_1\lambda_2 r \int_0^\infty \frac{dx}{(r^2+x^2)} = 4k\lambda_1\lambda_2 r \left[\frac{1}{r}\left(\tan^{-1}\frac{x}{r}\right)\right]_0^\infty$$

$$F_{net} = 4k\lambda_1\lambda_2 (\tan^{-1}\infty - \tan^{-1}0) = 2\pi k\lambda_1\lambda_2 = \frac{2\pi\lambda_1\lambda_2}{4\pi\varepsilon_0}$$

$$\therefore \boxed{F = \frac{\lambda_1\lambda_2}{2\varepsilon_0}}$$

Q



Find electric potential at point P.

$$dq = \sigma(2r\theta)dx \qquad dV = \frac{k\,dq}{r}$$

$$dV = \frac{2\sigma r\,dx\,d\theta}{4\pi\varepsilon_0 r} = \frac{\sigma\,dx\,d\theta}{2\pi\varepsilon_0}$$

$$V = \int dV = \frac{\sigma}{2\pi\varepsilon_0}\int \theta\,dx = \frac{-\sigma 2R}{2\pi\varepsilon_0}\int_{\pi/2}^{0} (\sin\theta)\,\theta\,d\theta$$

$$V = \frac{-\sigma R}{\pi\varepsilon_0}\int_{\pi/2}^{0} \theta\sin\theta\,d\theta = \frac{-\sigma R}{\pi\varepsilon_0}\left(-\theta\cos\theta+\sin\theta\right)_{\pi/2}^{0}$$

$$V = \frac{-\sigma R}{\pi\varepsilon_0}\{0-(1)\}$$

$$\boxed{V = \frac{\sigma R}{\pi\varepsilon_0}}$$

$\cos\theta = \frac{r}{2R}$

$r = 2R\cos\theta$

$dx = 2R(-\sin\theta\,d\theta)$

$r \to 0 \Rightarrow \theta \to \pi/2$

$r \to 2R \Rightarrow \theta \to 0$